

Card column reference

All **711** columns across the five cards, grouped by card and then by block, with blocks ordered by **measured importance**. Each entry leads with what the column *is*; the caveats fold underneath.

Generated 2026-08-20 by `analyses/ops/gen_column_reference.py` from `card_glossary.tsv` and `card_registry.tsv`. **Do not hand-edit**.

Vintage mix “ read before quoting a fitted column

The delivered edge card is **16 days newer** than the base it layers over. Its **84 base-owned columns** (`coupling_`, `beta_`, `dose_`, `share_`, `rank_`) carry the older vintage; the other 102 are current.

The one rule behind this layout

The rung is a property of (CARD, COLUMN) “ never of the column name. `beta` is edge-rung on the edge card and family-rung on the `gene_family` card: same estimator, different unit. That is why this is organised card-first, and why one name can appear twice with different text.

How blocks are ordered

By three measured signals, not by size: how often a block's columns are named in the **discovery registry** (what the findings rest on, weighted highest), how many **modules** read them, and whether the block is **base-owned** (fitted) rather than an annotation join. Names under 6 characters count only backtick-quoted registry hits “ otherwise `n` and `z` match ordinary prose.

One correction the measurement needed: mention-frequency is not importance “ a *settled* block gets discussed less precisely because nobody argues about it. `adm_` scored 27th of 44 on the raw signals while MH-216 calls `adm_has_site` the single most load-bearing conditioning variable, on 4 registry mentions. Five blocks therefore carry a **curated floor**, each citing the finding that justifies it (see `CURATED_FLOOR`). That is the only hand-set input here.

Cards

- edge “ 186 columns, 47 blocks
- gene “ 135 columns, 29 blocks
- gene_family “ 61 columns, 12 blocks
- arm “ 296 columns, 44 blocks
- seed_family “ 33 columns, 2 blocks

edge

One row per one (miRNA arm, gene) pair. The widest card and the one most claims are made from. 84 of its columns come from the BASE card and 102 from the annotation layer “ check the vintage banner before quoting a fitted one.

186 columns 5,649 rows 47 blocks median coverage 72%

Contents “ 47 blocks, most important first

1. (bare) 6 “ CORE

2. seed_ 1 “ CORE

3. coupling_ 13 “ CORE

4. adm_ 4 “ CORE

5. arm_ 15 “ CORE

6. beta_ 8 “ CORE

7. share_ 4 “ CORE

8. n_ 5 “ CORE

9. rank_ 4 “ CORE

10. cal_ 7 “ CORE

11. echim_ 3 “ CORE

12. ctx_ 18 “ CORE

13. shift_ 1 “ CORE

14. healthy_ 3 “ SUPPORTING

15. dose_ 7 “ SUPPORTING

16. identity_ 5 “ SUPPORTING

17. d_rank_ 4 “ SUPPORTING

18. gene_ 6 “ SUPPORTING

19. ago_ 2 “ SUPPORTING

20. pip_ 2 “ REFERENCE

21. oof_ 3 “ REFERENCE

22. retention_ 3 “ REFERENCE

23. grank_ 3 “ REFERENCE
24. dGlobal_ 3 “ REFERENCE

25. own_ 1 “ REFERENCE

26. composition_ 1 “ REFERENCE

27. w_ 1 “ REFERENCE

28. m_ 1 “ REFERENCE

29. term_ 3 “ REFERENCE

30. prior_ 1 “ REFERENCE

31. edge_ 1 “ REFERENCE

32. esub_ 8 “ REFERENCE

33. mean_ 2 “ REFERENCE

34. surrogate_ 2 “ REFERENCE

35. cptac_ 20 “ REFERENCE

36. realization_ 2 “ REFERENCE

37. kd_ 2 “ REFERENCE

38. wiring_ 1 “ REFERENCE

39. sd_ 1 “ REFERENCE

40. cell_ 1 “ REFERENCE

41. z_ 1 “ REFERENCE

42. acquisition_ 1 “ REFERENCE

43. dShare_ 2 “ REFERENCE

44. heur_ 1 “ REFERENCE

45. family_ 1 “ REFERENCE

46. hly_ 1 “ REFERENCE

47. is_ 1 “ REFERENCE

(BARE) “ 6

core “ the columns findings rest on

6 columns “ spans edge, key “ this prefix groups columns living on DIFFERENT units “ coverage 96%“100% (median 100%)
2× identifier “ 2× real “ 0 “ 1× boolean “ 1× real, signed

arm

KEY IDENTIFIER 100%

KEY “ mature miRNA arm (e.g. hsa-miR-21-5p). Normalised through `_arm_key()` , which gates against the canonical universe and LOGS drops rather than silently dropping. AUTHORED
example: CDH1 / miR-25-3p/32-5p/92-3p/363-3p/367-3p “ hsa-miR-25-3p

beta

EDGE REAL “ 0 100%

EDGE-rung coupling coefficient “ `readouts.run(level='arm')` , the same-seed collapse REMOVED. AUTHORED
example: CDH1 / hsa-miR-25-3p “ 0.2203

gene KEY IDENTIFIER 100%	KEY “ HGNC gene symbol. AUTHORED example: hsa-miR-21-5p / miR-21-5p/590-5p “ ACAT1
identified EDGE BOOLEAN 100%	$ z > 2$ on the UNCALIBRATED posterior SD, where $z = \text{beta} / \text{beta_sd}$ (MH-83 records precision 0.86 / recall 0.89 against a held-out standard). True on 24.8% of edges. AUTHORED values: False (4,269) · True (1,375) example: CDH1 / hsa-miR-25-3p “ True
identity EDGE REAL, SIGNED 96%	Shapley/LMG share of the gene's R^2 credited to this edge, with bagged-NNLS weights. AUTHORED example: CDH1 / hsa-miR-25-3p “ 0.3367
z EDGE REAL “ 0 100%	$\text{beta} / \text{beta_sd}$ for the EDGE-level fit (<code>readouts.run(level='arm')</code>), the whole gene's arms as separate predictors), on the UNCALIBRATED posterior SD. AUTHORED example: CDH1 / hsa-miR-25-3p “ 4.542

SEED_ “ 1

core “ the columns findings rest on

1 columns · family · coverage 100%“100% (median 100%) 1x identifier	
seed_family FAMILY IDENTIFIER 100%	KEY “ the seed family itself, one row per family. AUTHORED example: CDH1 / hsa-miR-25-3p “ miR-25-3p/32-5p/92-3p/363-3p/367-3p

COUPLING_ “ 13

core “ the columns findings rest on

13 columns · spans edge, family “ this prefix groups columns living on DIFFERENT units · coverage 3%“100% (median 62%) 5x signed “1“1 · 5x p-value · 3x real, signed	
coupling_fam FAMILY SIGNED “1“1 100%	Family-level coupling for the edge's (gene, seed_family) cell. AUTHORED example: CDH1 / hsa-miR-25-3p “ -0.2577
coupling_hly EDGE SIGNED “1“1 56%	Partial-Spearman coupling of arm dose against target mRNA in the GTEx-HEALTHY state, conditioned on C (composition + target CN + proliferation). AUTHORED example: CDH1 / hsa-miR-25-3p “ 0.007
coupling_hly_surrogate EDGE SIGNED “1“1 3%	Healthy coupling via a same-seed SURROGATE arm, used where the arm itself has no GTEx signal. AUTHORED example: CDH1 / hsa-miR-30a-5p “ -0.037
coupling_nat EDGE SIGNED “1“1 62%	Partial-Spearman coupling of arm dose against target mRNA in matched NORMAL (NAT), conditioned on C (composition + target CN + proliferation). AUTHORED example: CDH1 / hsa-miR-25-3p “ -0.116
coupling_p_hly EDGE P-VALUE 56%	The site-free-null p for that coupling in the GTEx-HEALTHY state, conditioned on C (composition + target CN + proliferation). AUTHORED example: CDH1 / hsa-miR-25-3p “ 0.6225

coupling_p_hly_resolved EDGE P-VALUE 59%	The site-free-null p for that coupling in the GTEx-HEALTHY state, conditioned on C (composition + target CN + proliferation). AUTHORED example: <code>CDH1 / hsa-miR-25-3p</code> $\hat{+}$ 0.6225
coupling_p_hly_surrogate EDGE P-VALUE 3%	The site-free-null p for that coupling in the GTEx-HEALTHY state, conditioned on C (composition + target CN + proliferation). AUTHORED example: <code>CDH1 / hsa-miR-30a-5p</code> $\hat{+}$ 0.4457
coupling_p_nat EDGE P-VALUE 62%	The site-free-null p for that coupling in matched NORMAL (NAT), conditioned on C (composition + target CN + proliferation). AUTHORED example: <code>CDH1 / hsa-miR-25-3p</code> $\hat{+}$ 0.1658
coupling_p_tum EDGE P-VALUE 69%	p against the CALIBRATED site-free null (not a theoretical t-null). Median 0.375; <0.05 on 16.2%. AUTHORED example: <code>CDH1 / hsa-miR-25-3p</code> $\hat{+}$ 0.0012
coupling_tum EDGE SIGNED $\hat{+}$1$\hat{+}$1 72%	Partial-Spearman coupling of arm dose against target mRNA in TUMOUR, conditioned on C (composition + target CN + proliferation). AUTHORED example: <code>CDH1 / hsa-miR-25-3p</code> $\hat{+}$ -0.271
coupling_z_hly EDGE REAL, SIGNED 59%	That coupling expressed in NULL SDs $\hat{+}$ the honest effect size in the GTEx-HEALTHY state, conditioned on C (composition + target CN + proliferation). AUTHORED example: <code>CDH1 / hsa-miR-25-3p</code> $\hat{+}$ 0.31
coupling_z_nat EDGE REAL, SIGNED 62%	That coupling expressed in NULL SDs $\hat{+}$ the honest effect size in matched NORMAL (NAT), conditioned on C (composition + target CN + proliferation). AUTHORED example: <code>CDH1 / hsa-miR-25-3p</code> $\hat{+}$ -0.97
coupling_z_tum EDGE REAL, SIGNED 69%	The tumour coupling expressed in NULL SDs $\hat{+}$ the CALIBRATED effect size, and the scale to quote. Median $\hat{+}$0.32; z >2 on 15.5%, z >3 on 6.4%. AUTHORED example: <code>CDH1 / hsa-miR-25-3p</code> $\hat{+}$ -3.04

ADM_ 4

core $\hat{+}$ the columns findings rest on

4 columns $\hat{+}$ rung edge $\hat{+}$ per-edge FLAGS; the arm card carries their rollup $\hat{+}$ coverage 87%$\hat{+}$87% (median 87%) 4x boolean	
adm_admissible EDGE BOOLEAN 87%	The admissibility verdict: site + evidence + expression. AUTHORED values: True (2,981) $\hat{+}$ False (1,956) example: <code>CDH1 / hsa-miR-25-3p</code> $\hat{+}$ True
adm_expr_gtex EDGE BOOLEAN 87%	ADMISSIBILITY $\hat{+}$ could this edge repress in breast AT ALL? <code>has_site</code> (seed match in the 3'UTR) AND <code>expressed</code> (arm present in TCGA tumour OR GTEx healthy breast). AUTHORED values: False (3,927) $\hat{+}$ True (1,010) example: <code>CDH1 / hsa-miR-25-3p</code> $\hat{+}$ True
adm_expr_tcga EDGE BOOLEAN 87%	Is the arm expressed above floor in TCGA? AUTHORED values: True (3,154) $\hat{+}$ False (1,783) example: <code>CDH1 / hsa-miR-25-3p</code> $\hat{+}$ True
adm_has_site EDGE BOOLEAN 87%	Does the arm have a seed match in this target's 3'UTR. AUTHORED values: True (4,436) $\hat{+}$ False (501) example: <code>CDH1 / hsa-miR-25-3p</code> $\hat{+}$ True

ARM_ 15

core the columns findings rest on

15 columns spans arm, arm-in-family, family this prefix groups columns living on DIFFERENT units coverage 20%100% (median 69%)
4x real 0 4x real, signed 3x fraction 1 2x categorical 1x percent 1x boolean

arm_credit_share ARM-IN-FAMILY FRACTION 1 20%	This arm's share of its (gene, seed_family) cell's credit. Verified to sum to 1.0000 within every one of the 467 multi-arm cells, and it genuinely varies PER ARM inside the cell (467/467). AUTHORED example: CDH1 / hsa-miR-25-3p 0.7094 Defined on: multi-arm cells only (n_arm_in_cell > 1) 20.3% of edges
arm_dbeta FAMILY REAL 0 20%	max^min I^2 across the cell's member arms. AUTHORED example: CDH1 / hsa-miR-25-3p 0.03
arm_detection ARM FRACTION 1 95%	Fraction of patients in which this ARM is measured at all (readouts._detection). AUTHORED example: CDH1 / hsa-miR-25-3p 1
arm_id_status ARM-IN-FAMILY CATEGORICAL 100%	How the arm's identity within its cell was resolved: resolved_by_dose 2,344 singleton 2,260 unvalidated_balanced 1,045. AUTHORED values: resolved_by_dose (2,344) singleton (2,260) unvalidated_balanced (1,045) example: CDH1 / hsa-miR-25-3p unvalidated_balanced
arm_iqr ARM REAL 0 69%	Interquartile range of the arm's abundance its DYNAMIC RANGE. AUTHORED example: CDH1 / hsa-miR-25-3p 0.9
arm_lfc_HLY_NAT_raw ARM REAL, SIGNED 63%	log-FC GTEx-healthy TCGA-NAT, raw. AUTHORED example: CDH1 / hsa-miR-25-3p 9.49
arm_lfc_HLY_TUM_QN ARM REAL, SIGNED 64%	log-FC GTEx-healthy TCGA-tumour, on quantile-normalised values. AUTHORED example: CDH1 / hsa-miR-25-3p 4.79
arm_lfc_HLY_TUM_raw ARM REAL, SIGNED 63%	As arm_lfc_HLY_TUM_QN but against the RAW GTEx median. AUTHORED example: CDH1 / hsa-miR-25-3p 9.46
arm_lfc_NAT_TUM ARM REAL, SIGNED 68%	log-FC of the arm's median abundance, matched NAT tumour. AUTHORED example: CDH1 / hsa-miR-25-3p -0.03
arm_med_rpm ARM REAL 0 69%	Median linear RPM of this ARM across the cohort. Arm-rung, so it REPEATS across the arm's genes. Fill 0.686 the gap is arms with no expression record, not zeros. AUTHORED example: CDH1 / hsa-miR-25-3p 13.1 Defined on: edges present in the progression panel (arm expressed + state measured)
arm_pct_above_floor ARM PERCENT 69%	Percent of samples in which the arm sits above the detection floor (median 99). Feeds arm_spiker together with arm_iqr. AUTHORED example: CDH1 / hsa-miR-25-3p 100 Defined on: edges present in the progression panel (arm expressed + state measured)
arm_role_in_family ARM-IN-FAMILY CATEGORICAL 72%	The ARM's DOSE role inside its seed family NOT the family's role in the gene, which is what the name suggests. AUTHORED values: sole (2,260) co (1,045) minor (395) dominant (379) example: CDH1 / hsa-miR-25-3p co
arm_sep_z FAMILY REAL 0 20%	arm_dbeta in units of the cell's own pooled posterior SD can these same-seed arms be told apart at all. AUTHORED example: CDH1 / hsa-miR-25-3p 3.196

arm_share_of_family_dose

ARM-IN-FAMILY

FRACTION 0-1

69%

The arm's share of its seed family's TOTAL dose – denominator is the **full family**, not the design (median 0.69). The quantity `family_role` is cut from.

AUTHORED

example: CDH1 / hsa-miR-25-3p - 0.56

Defined on: edges present in the progression panel (arm expressed + state measured)

arm_spiker

ARM

BOOLEAN

69%

arm_pct_floor < 40 AND arm_iqr > 1.5 – the arm sits at/below the detection floor in most samples yet swings widely when present.

AUTHORED

values: False (3,706) - True (171)

example: CDH1 / hsa-miR-25-3p - False

BETA_ - 8

core – the columns findings rest on

8 columns - rung edge - coverage 20-100% (median 100%)
4x real - 4x fraction 0-1

beta_arm

EDGE

REAL - 0

20%

The arm's OWN $\hat{\tau}^2$ from the within-cell fit (`arm_rung.py`), defined only in multi-arm cells (fill 0.203).

AUTHORED

example: CDH1 / hsa-miR-25-3p - 0.04066

beta_deconv

EDGE

REAL - 0

100%

$\hat{\tau}^2$ refit on the DECONV design (core C + the 8 non-malignant Wu-major lineages, Cancer-Epithelial held out).

AUTHORED

example: CDH1 / hsa-miR-25-3p - 0.1737

beta_frac

EDGE

FRACTION 0-1

100%

MAGNITUDE share (`beta_f` / sum beta).

AUTHORED

example: CDH1 / hsa-miR-25-3p - 0.1575

beta_frac_abs

EDGE

FRACTION 0-1

100%

$|E[\hat{\tau}_f^2]| / \hat{\tau}^2 |E[\hat{\tau}^2]|$ – the share computed from the POINT ESTIMATES.

AUTHORED

example: CDH1 / hsa-miR-25-3p - 0.1565

beta_frac_deconv

EDGE

FRACTION 0-1

100%

`beta_frac` recomputed on the deconv design – the composition-adjusted share of the gene's budget.

AUTHORED

example: CDH1 / hsa-miR-25-3p - 0.1411

beta_frac_sd

EDGE

FRACTION 0-1

100%

Posterior SD of the per-draw fraction $\hat{\tau}_f^2 / \hat{\tau}^2$.

AUTHORED

example: CDH1 / hsa-miR-25-3p - 0.03573

beta_sd

EDGE

REAL - 0

100%

Posterior summaries of the coupling coefficient from the dense Gibbs fit (`learned/readouts.py`). `_sd` is the posterior SD, `_frac` the share of the gene's total beta.

AUTHORED

example: CDH1 / hsa-miR-25-3p - 0.04851

beta_sd_deconv

EDGE

REAL - 0

100%

Posterior SD on the deconv design.

AUTHORED

example: CDH1 / hsa-miR-25-3p - 0.04548

SHARE_ - 4

core – the columns findings rest on

4 columns - rung edge - coverage 46-72% (median 72%)
4x fraction 0-1

share_HLY

EDGE

FRACTION 0-1

72%

The arm's share of the gene's healthy (GTEx) pressure budget, median **0.129**.

AUTHORED

example: CDH1 / hsa-miR-25-3p - 0.15

share_HLY_meas EDGE FRACTION 0.1 46%	share_HLY over ONLY the arms GTEx genuinely measures “invisible arms ABSTAIN (NaN) instead of being zeroed (MH-210). Median 0.244 against the imputed 0.129. AUTHORED example: CDH1 / hsa-miR-25-3p † 0.297
share_NAT EDGE FRACTION 0.1 72%	The arm's share of its gene's pressure budget in matched normal (NAT). AUTHORED example: CDH1 / hsa-miR-25-3p † 0.219
share_TUM EDGE FRACTION 0.1 72%	The arm's share of its gene's pressure budget in tumour. AUTHORED example: CDH1 / hsa-miR-25-3p † 0.243

N_ 5
core “the columns findings rest on

5 columns spans edge, family, gene “this prefix groups columns living on DIFFERENT units coverage 20%100% (median 100%)
4x count 1x constant

n_HLY_meas EDGE COUNT 72%	How many of this gene's arms have a MEASURED healthy (GTEx) level “as opposed to one manufactured by the multi-mapping collapse. AUTHORED example: CDH1 / hsa-miR-25-3p † 8
n_arm_in_cell FAMILY COUNT 20%	How many arms sit in this edge's (gene, seed_family) CELL “i.e. how many of that family's members appear in THIS gene's design. AUTHORED values: 2.0 (670) 3.0 (258) 4.0 (100) 5.0 (55) 6.0 (36) 7.0 (28) example: CDH1 / hsa-miR-25-3p † 2
n_fam GENE COUNT 100%	How many families compete at this edge's GENE. Gene-rung “it repeats across the gene's edges by construction. beta is mathematically inert where it is 1. AUTHORED example: CDH1 / hsa-miR-25-3p † 25
n_family_in_design FAMILY COUNT 100%	Members of this arm's seed family THAT ARE IN THIS GENE'S DESIGN “1 on 4,497 of 5,644 edges (79.7%). AUTHORED values: 1.0 (4,497) 2.0 (670) 3.0 (258) 4.0 (100) 5.0 (55) 6.0 (36) 7.0 (28) example: CDH1 / hsa-miR-25-3p † 2
n_samples GENE CONSTANT 100%	Samples the edge was fit on “CONSTANT (1,040) across the card. AUTHORED values: 1040.0 (5,644) example: CDH1 / hsa-miR-25-3p † 1040

RANK_ 4
core “the columns findings rest on

4 columns rung edge coverage 46%72% (median 72%)
4x count

rank_HLY EDGE COUNT 72%	The arm's RANK within its gene's pressure budget in the GTEx-healthy state (1 = largest). AUTHORED example: CDH1 / hsa-miR-25-3p † 4
rank_HLY_meas EDGE COUNT 46%	The arm's WITHIN-GENE rank by dose in one state. AUTHORED example: CDH1 / hsa-miR-25-3p † 2

rank_NAT

EDGE

COUNT

72%

The arm's RANK within its gene's pressure budget in matched normal (NAT) (1 = largest).

AUTHORED

example: CDH1 / hsa-miR-25-3p → 2

rank_TUM

EDGE

COUNT

72%

The arm's RANK within its gene's pressure budget in tumour (1 = largest).

AUTHORED

example: CDH1 / hsa-miR-25-3p → 2

CAL_ → 7

core → the columns findings rest on

7 columns → rung edge → coverage 100%→100% (median 100%)

4× real → 0 → 3× boolean

cal_beta_sd

EDGE

REAL → 0

100%

The calibrated posterior SD of $\hat{\Gamma}^2$ (central estimate). Range 0.00101 → 3.94, 100% filled.

AUTHORED

example: CDH1 / hsa-miR-25-3p → 0.05952

cal_beta_sd_hi

EDGE

REAL → 0

100%

Upper end of the calibrated SD band. Range 0.00112 → 4.39, 100% filled.

AUTHORED

example: CDH1 / hsa-miR-25-3p → 0.06636

cal_beta_sd_lo

EDGE

REAL → 0

100%

Lower end of the calibrated SD band. Range 0.000912 → 3.57, 100% filled.

AUTHORED

example: CDH1 / hsa-miR-25-3p → 0.05396

cal_identified

EDGE

BOOLEAN

100%

Is the edge identified ($|cal_z| > 2$) under the CALIBRATED posterior width → the honest version of identified : 1,110 vs 1,375 edges, i.e. 19.9% [18.1→22.4] against 24.8%.

AUTHORED

values: False (4,534) → True (1,110)

example: CDH1 / hsa-miR-25-3p → True

cal_identified_hi

EDGE

BOOLEAN

100%

The UPPER end of the calibrated identification band → $|cal_z| > 2$ under the optimistic calibration (990 edges vs 1,110 central).

AUTHORED

values: False (4,654) → True (990)

example: CDH1 / hsa-miR-25-3p → True

cal_identified_lo

EDGE

BOOLEAN

100%

The LOWER end of the calibrated identification band (1,234 edges vs 1,110 central). See cal_identified_hi → the pair is the uncertainty, and reporting one alone hides it.

AUTHORED

values: False (4,410) → True (1,234)

example: CDH1 / hsa-miR-25-3p → True

cal_z

EDGE

REAL → 0

100%

beta / cal_beta_sd → the CALIBRATED z.

AUTHORED

example: CDH1 / hsa-miR-25-3p → 3.702

ECHIM_ → 3

core → the columns findings rest on

3 columns → rung edge → coverage 8%→26% (median 22%)

2× count → 1× real → 0

echim_manakov_w

EDGE

REAL → 0

22%

Manakov chimeric-eCLIP support weight for this edge (median 3, max 1,712).

AUTHORED

example: ESR1 / hsa-miR-20b-5p → 0.3333

echim_n_sources EDGECOUNT26%	How many chimeric-eCLIP sources support this edge (1 on 1,151 Â· 2 on 262 Â· 3 on 34). AUTHORED values: 1.0 (1,151) Â· 2.0 (262) Â· 3.0 (34) Â· 4.0 (2) example: CDH1 / hsa-miR-92a-3p â†’ 1
echim_tarbase_w EDGECOUNT8%	TarBase chimeric support weight (median 2, max 28) â€” a much shorter tail than the Manakov twin, and defined on only 8.4% of edges. AUTHORED example: CDH1 / hsa-miR-92a-3p â†’ 4 Defined on: edges with a chimeric (AGO-ligation) duplex â€” âš per-source, NEVER pooled; cross-tissue, not breast
CTX_ Â· 18 core â€” the columns findings rest on	
18 columns Â· spans arm, gene â€” this prefix groups columns living on DIFFERENT units Â· coverage 84%â€”100% (median 99%) 4× real â‰¥ 0 Â· 4× signed âˆ’1â‰¥1 Â· 3× real, signed Â· 3× count Â· 2× boolean Â· 1× categorical Â· 1× fraction 0â‰¥1	
ctx_apriori_class GENECATEGORICAL100%	The a-priori measurability class assigned to the gene's design before any fit (gene_atlas via card_context). Range 4 levels, 100% filled. AUTHORED values: A_COMPETENT (3,989) Â· D_UNDEFINED (654) Â· B_PARTIAL (609) Â· C_WEAK (397) example: CDH1 / hsa-miR-25-3p â†’ A_COMPETENT
ctx_arm_abundant ARMBOOLEAN95%	ctx_arm_dose >= log2(11) â€” the arm-abundance gate, ARM-rung on an edge card. See the arm card's entry. AUTHORED values: True (3,384) Â· False (1,981) example: CDH1 / hsa-miR-25-3p â†’ True Defined on: arms present on the EDGE card (729 of 2,450) â€” lifted arm-rung columns, names preserved
ctx_arm_dose ARMREAL â‰¥ 095%	This arm's median log2(RPM+1) â€” see the arm card's entry. AUTHORED example: CDH1 / hsa-miR-25-3p â†’ 13.05
ctx_ceiling GENESIGNED âˆ’1â‰¥199%	The UPPER BOUND on what ANY estimator can achieve for this gene â€” the 5-fold out-of-fold RÂ² of an UNRESTRICTED OLS on the gene's real design (card_context.py:15). AUTHORED example: CDH1 / hsa-miR-25-3p â†’ 0.1535
ctx_collapse GENEREAL â‰¥ 0100%	n_arm / n_fam for the gene â€” how much curation BUNDLES same-seed mates onto it (1â‰¥5). AUTHORED example: CDH1 / hsa-miR-25-3p â†’ 1.091
ctx_d_collin GENESIGNED âˆ’1â‰¥184%	Within-design COLLINEARITY asymmetry, real minus decoy. The second design-match axis MH-167 tested; also bounded and not moving the gap. AUTHORED example: CDH1 / hsa-miR-25-3p â†’ -0.02497 Defined on: genes with >=2 seed families (a collinearity measure needs 2 columns)
ctx_d_fam GENEREAL, SIGNED99%	Design-WIDTH asymmetry between the real design and its matched decoy (real minus fake families). AUTHORED example: CDH1 / hsa-miR-25-3p â†’ 0
ctx_dose_max GENEREAL â‰¥ 0100%	The LARGEST regulator dose in the gene's design (gene_atlas via card_context). Range 0.215 â€” 17.8, 100% filled. AUTHORED example: CDH1 / hsa-miR-25-3p â†’ 15.38

ctx_dose_med GENE REAL 0 100%	The MEDIAN regulator dose in the gene's design (<code>gene_atlas</code> via <code>card_context</code>). Range 0.215 to 17.8, 100% filled. AUTHORED example: <code>CDH1 / hsa-miR-25-3p</code> at <code>6.318</code>
ctx_frac_abund GENE FRACTION 0 to 1 100%	FRACTION of the gene's regulators clearing the abundance gate (<code>gene_atlas</code> via <code>card_context</code>). Range 0 to 1, 100% filled. AUTHORED example: <code>CDH1 / hsa-miR-25-3p</code> at <code>0.5833</code>
ctx_gap_core GENE SIGNED -1 to 1 99%	The $\hat{I}^2$-vs-decoy MARGIN under the CORE confounder block (<code>gene_atlas</code> via <code>card_context</code>). Range -0.476 to +0.541, 99% filled. AUTHORED example: <code>CDH1 / hsa-miR-25-3p</code> at <code>-0.2714</code>
ctx_gap_d_dose GENE REAL, SIGNED 99%	Dose asymmetry between the real and decoy designs (median $\hat{I}^2$ 0.43). AUTHORED example: <code>CDH1 / hsa-miR-25-3p</code> at <code>0.1238</code>
ctx_gap_deconv GENE SIGNED -1 to 1 99%	The same margin under the DECONVOLUTION block (<code>gene_atlas</code> via <code>card_context</code>). Range -0.33 to +0.374, 99% filled. AUTHORED example: <code>CDH1 / hsa-miR-25-3p</code> at <code>-0.1864</code>
ctx_gap_retention GENE REAL, SIGNED 95%	<code>gap_deconv / gap_core</code> to how much of the decoy margin survives adjustment (<code>gene_atlas</code> via <code>card_context</code>). Range -16.5 to 17, 95% filled. AUTHORED example: <code>CDH1 / hsa-miR-25-3p</code> at <code>0.6867</code>
ctx_measurable GENE BOOLEAN 99%	Is the gene's <code>ctx_ceiling</code> above <code>CEILING_ROBUST</code> (0.02, not 0 to MH-144)? AUTHORED values: True (4,189) and False (1,428) example: <code>CDH1 / hsa-miR-25-3p</code> at <code>True</code>
ctx_n_abund GENE COUNT 100%	How many of the gene's regulators clear the abundance gate (<code>gene_atlas</code> via <code>card_context</code>). Range 0 to 53, 100% filled. AUTHORED example: <code>CDH1 / hsa-miR-25-3p</code> at <code>7</code>
ctx_n_fam_abund GENE COUNT 100%	Gene-design context from the decoy/atlas layer (<code>learned/analyses/gene_atlas.py</code> via <code>card_context</code>): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it. AUTHORED example: <code>CDH1 / hsa-miR-25-3p</code> at <code>6</code>
ctx_n_fam_multi GENE COUNT 100%	How many of the gene's families hold MORE THAN ONE arm (<code>gene_atlas</code> via <code>card_context</code>). Range 0 to 14, 100% filled. AUTHORED example: <code>CDH1 / hsa-miR-25-3p</code> at <code>1</code>

SHIFT_ 1

core to the columns findings rest on

1 columns and 1 rung edge and coverage 72% to 72% (median 72%) 1x categorical	shift_class EDGE CATEGORICAL 72%	Cross-state class for the edge. AUTHORED example: <code>CDH1 / hsa-miR-25-3p</code> at <code>tumour_realized</code>
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HEALTHY_ 3

supporting read these when the core raises a question

3 columns 1x categorical 1x real 1x boolean 1x categorical 1x real 1x boolean	
healthy_leg ARM CATEGORICAL 72%	Provenance of this arm's GTEx healthy leg measured , collapse_blind or true_absent. AUTHORED values: measured (2,573) collapse_blind (854) true_absent (652) example: CDH1 / hsa-miR-25-3p measured
healthy_potential ARM REAL 64%	The arm's IMPUTED healthy abundance a repression-POTENTIAL proxy, collapse-repaired (0.27 17.9). AUTHORED example: CDH1 / hsa-miR-25-3p 8.33
healthy_uninformative ARM BOOLEAN 64%	Is the healthy leg both collapse_blind AND high-potential i.e. the one combination where an "acquired" call is genuinely unsafe? True on 143 of 3,490 edges (4%). AUTHORED values: False (3,490) True (143) example: CDH1 / hsa-miR-25-3p False

DOSE_ 7

supporting read these when the core raises a question

7 columns 3x count 2x real 1x categorical 1x constant (median 72%)	
dose_class EDGE CATEGORICAL 72%	Direction of the arm's NAT-tumour dose change gain (1,620) / flat (1,286) / loss (1,173). Range 3 levels, 72% filled. AUTHORED values: gain (1,620) flat (1,286) loss (1,173) example: CDH1 / hsa-miR-25-3p flat Defined on: edges present in the progression panel (arm expressed + state measured)
dose_comp_retention ARM REAL 55%	The arm's dose signal after conditioning on COMPOSITION, over its raw dose. 1.00 for real doses. Range 0.51 1.66, 55% filled. AUTHORED example: CDH1 / hsa-miR-30a-5p 1.02 Defined on: edges with a matched NAT leg (n~104 paired participants)
dose_confounded ARM CONSTANT 72%	Is this edge's NAT-tumour dose a composition/proliferation artifact (either gated retention < 0.4)? AUTHORED values: False (4,079) example: CDH1 / hsa-miR-25-3p False
dose_prolif_retention ARM REAL 55%	The same against PROLIFERATION. 1.00 for real doses the NAT-tumour acquisitions are cell-intrinsic. Range 0.69 1.25, 55% filled. AUTHORED example: CDH1 / hsa-miR-30a-5p 1.02 Defined on: edges with a matched NAT leg (n~104 paired participants)
dose_rank_HLY EDGE COUNT 72%	The arm's within-gene DOSE rank (abundance only, no I²) in healthy. Range 1 53, 72% filled. AUTHORED example: CDH1 / hsa-miR-25-3p 3 Defined on: arms with a GTEx-HEALTHY leg (the thinnest state many arms are multi-mapping-collapsed, MH-166)

dose_rank_NAT EDGECOUNT72%	Its within-gene dose rank in NAT. Range 1 – 57, 72% filled. AUTHORED example: CDH1 / hsa-miR-25-3p – 2 Defined on: edges with a matched NAT leg (n~104 paired participants)
dose_rank_TUM EDGECOUNT72%	Its within-gene dose rank in tumour. Range 1 – 58, 72% filled. AUTHORED example: CDH1 / hsa-miR-25-3p – 2 Defined on: edges present in the progression panel (arm expressed + state measured)

IDENTITY_ – 5

supporting – read these when the core raises a question

5 columns – rung edge – coverage 96%–98% (median 96%) 2× fraction 0–1 – 2× real, signed – 1× boolean	
identity_abs EDGEFRACTION 0–196%	identity as a fraction of the gene's total identity , clipped to [0,1]. AUTHORED example: CDH1 / hsa-miR-25-3p – 0.3305
identity_allocated EDGEREAL, SIGNED96%	Family identity with phantom minor arms forced to 0 – the allocation actually used downstream. AUTHORED example: CDH1 / hsa-miR-25-3p – 0.3367
identity_coherence EDGEFRACTION 0–196%	1 / Î identity over the gene, clipped to (0,1] – how much the gene's identity shares CANCEL. AUTHORED example: CDH1 / hsa-miR-25-3p – 0.9817
identity_deconv EDGEREAL, SIGNED98%	Shapley identity recomputed under the composition-adjusted design. AUTHORED example: CDH1 / hsa-miR-25-3p – 0.2929
identity_reliable EDGEBOOLEAN96%	Is this edge's Shapley identity trustworthy – identity_coherence ≥ 0.5 AND identity ≥ 1? AUTHORED values: True (5,272) – False (159) example: CDH1 / hsa-miR-25-3p – True

D_RANK_ – 4

supporting – read these when the core raises a question

4 columns – rung edge – coverage 46%–72% (median 72%) 4× real, signed	
d_rank_HLY_NAT EDGEREAL, SIGNED72%	Change in the arm's WITHIN-GENE budget rank from the GTEx-healthy state to matched normal (NAT). AUTHORED example: CDH1 / hsa-miR-25-3p – 2
d_rank_HLY_TUM EDGEREAL, SIGNED72%	Change in the arm's WITHIN-GENE budget rank from the GTEx-healthy state to tumour. AUTHORED example: CDH1 / hsa-miR-25-3p – 2
d_rank_HLY_TUM_meas EDGEREAL, SIGNED46%	d_rank_HLY_TUM recomputed over arms with a MEASURED healthy level only – the collapse-safe version. Prefer it over the bare column. AUTHORED example: CDH1 / hsa-miR-25-3p – 0 Defined on: edges whose gene has ≥ 1 GTEx-MEASURED regulator – NaN = GTEx cannot see it (abstention), NOT 0

d_rank_NAT_TUM

EDGE

REAL, SIGNED

72%

Change in the arm's WITHIN-GENE budget rank from matched normal (NAT) to tumour.

AUTHORED

example: CDH1 / hsa-miR-25-3p → 0

GENE_ → 6

supporting → read these when the core raises a question

6 columns → rung gene → coverage 72%→100% (median 99%)
2× categorical → 2× boolean → 1× real → 0 → 1× real, signed

gene_cancer_role

GENE

CATEGORICAL

100%

→ NOT the arm's role → the GENE's cancer role (oncogene / tsg / oncogene/tsg / unknown), from gene_roles.load_gene_roles.

AUTHORED

values: unknown (3,667) → oncogene (1,189) → tsg (661) → oncogene/tsg (132)

example: CDH1 / hsa-miR-25-3p → tsg

gene_dominated

GENE

BOOLEAN

97%

Does one family carry more than 60% of this gene's tumour pressure budget (2,641 of 5,495)?

AUTHORED

values: False (2,854) → True (2,641)

example: CDH1 / hsa-miR-25-3p → False

gene_iqr

GENE

REAL → 0

72%

Interquartile range of the TARGET GENE's expression → its dynamic range. Range 0 → 6.27, 72% filled.

AUTHORED

example: CDH1 / hsa-miR-25-3p → 1.72

Defined on: genes with a NAT-TUM gene-level leg

gene_lfc_NAT_TUM

GENE

REAL, SIGNED

72%

The target gene's own NAT→tumour log fold-change. Range -5.85 → 5.38, 72% filled.

AUTHORED

example: CDH1 / hsa-miR-25-3p → 0.73

Defined on: genes with a NAT-TUM gene-level leg

gene_net_repr

GENE

BOOLEAN

100%

Is this gene NET-REPPRESSED in tumour by the →6b-RETIRED heuristic lane (1,376 of 5,634 edges)?

AUTHORED

values: False (4,258) → True (1,376)

example: CDH1 / hsa-miR-25-3p → False

gene_repr_class

GENE

CATEGORICAL

100%

The gene's repression class from the retired heuristic lane. Range 5 levels, 100% filled.

AUTHORED

values: never_repressed (2,939) → lost_repression (1,318) → constitutive_repressed (770) → gained_repression (606) → na (1)

example: CDH1 / hsa-miR-25-3p → never_repressed

AGO_ → 2

supporting → read these when the core raises a question

2 columns → rung arm → → ago_loading is rank-identical to the arm card's ago_dom → one quantity, two names → coverage 95%→100% (median 97%)
1× fraction →1 → 1× boolean

ago_loading

ARM

FRACTION →1

95%

AGO/RISC loading discount.

AUTHORED

example: CDH1 / hsa-miR-25-3p → 0.996

ago_loading_measured

ARMBOOLEAN100%

Was ago_loading actually MEASURED for this edge (vs defaulted to 1.0). True for 3,820/5,648 edges (67.6%).

AUTHORED

values: True (3,820) · False (1,829)

example: CDH1 / hsa-miR-25-3p â†’ True

Defined on: arms with a Manakov AGO-dominance measurement â€”â€™ NOT loading(), which fabricates 1.0

PIP_ · 2

reference â€” context, provenance and rarely-quoted detail

2 columns · rung edge · coverage 100%â€”100% (median 100%)
1× constant · 1× fraction 0â€”1

pip_dense

EDGECONSTANT100%

â€” CONSTANT by construction â€” the coupling readout is called with ĨĈ%ı1, so every edge has the same value.

AUTHORED

values: 1.0 (5,644)

example: CDH1 / hsa-miR-25-3p â†’ 1

pip_discovery

EDGEFRACTION 0â€”1100%

Posterior inclusion probability. pip_dense comes from the pi=1 coupling readout; pip_discovery from the evidence-pi readout.

AUTHORED

example: CDH1 / hsa-miR-25-3p â†’ 1

OOF_ · 3

reference â€” context, provenance and rarely-quoted detail

3 columns · rung family · coverage 20%â€”20% (median 20%)
3× signed âˆ’1â€”1

oof_drho

FAMILYSIGNED âˆ’1â€”120%

Arm-level OOF rho MINUS family-level.

AUTHORED

example: CDH1 / hsa-miR-25-3p â†’ -0.004735

oof_rho_arm

FAMILYSIGNED âˆ’1â€”120%

Out-of-fold coupling from the ARM-grain fit. Range -0.542 â€” +0.324, 20% filled.

AUTHORED

example: CDH1 / hsa-miR-25-3p â†’ -0.2411

Defined on: multi-arm cells only (n_arm_in_cell > 1) â€” 20.3% of edges

oof_rho_fam

FAMILYSIGNED âˆ’1â€”120%

Out-of-fold coupling from the FAMILY-grain fit â€” their difference is oof_drho, the arm-resolution gain. Range -0.545 â€” +0.309, 20% filled.

AUTHORED

example: CDH1 / hsa-miR-25-3p â†’ -0.2364

Defined on: multi-arm cells only (n_arm_in_cell > 1) â€” 20.3% of edges

RETENTION_ · 3

reference â€” context, provenance and rarely-quoted detail

3 columns · rung edge · coverage 72%â€”100% (median 100%)
1× real â‰¥ 0 · 1× boolean · 1× real, signed

retention_beta

EDGEREAL â‰¥ 0100%

Fraction of an effect surviving covariate adjustment (adjusted / raw).

AUTHORED

example: CDH1 / hsa-miR-25-3p â†’ 0.7885

retention_reliable

EDGE

BOOLEAN

100%

Is this edge's `retention` ratio trustworthy â€” i.e. does it have a real denominator?

AUTHORED

values: False (4,269) â€” True (1,375)

example: `CDH1 / hsa-miR-25-3p` â†’ True

retention_rho

EDGE

REAL, SIGNED

72%

`İ_deconv / İ_core` â€”

AUTHORED

example: `CDH1 / hsa-miR-25-3p` â†’ 0.65

GRANK_ 3

reference â€” context, provenance and rarely-quoted detail

3 columns

â€”

rung

arm

â€”

coverage

63%â€”69%

(median 68%)

3x count

grank_HLY

ARM

COUNT

63%

The arm's GLOBAL abundance percentile across the whole cohort in the GTEx-healthy state.

AUTHORED

example: `CDH1 / hsa-miR-25-3p` â†’ 96

grank_NAT

ARM

COUNT

68%

The arm's GLOBAL abundance percentile across the whole cohort in matched normal (NAT).

AUTHORED

example: `CDH1 / hsa-miR-25-3p` â†’ 99

grank_TUM

ARM

COUNT

69%

The arm's GLOBAL abundance percentile across the whole cohort in tumour.

AUTHORED

example: `CDH1 / hsa-miR-25-3p` â†’ 99

DGLOBAL_ 3

reference â€” context, provenance and rarely-quoted detail

3 columns

â€”

rung

arm

â€”

coverage

63%â€”68%

(median 63%)

3x real, signed

dGlobal_HLY_NAT

ARM

REAL, SIGNED

63%

Change in the arm's GLOBAL (cohort-wide) abundance percentile from the GTEx-healthy state to matched normal (NAT).

AUTHORED

example: `CDH1 / hsa-miR-25-3p` â†’ 3

dGlobal_HLY_TUM

ARM

REAL, SIGNED

63%

Change in the arm's GLOBAL (cohort-wide) abundance percentile from the GTEx-healthy state to tumour.

AUTHORED

example: `CDH1 / hsa-miR-25-3p` â†’ 3

dGlobal_NAT_TUM

ARM

REAL, SIGNED

68%

Change in the arm's GLOBAL (cohort-wide) abundance percentile from matched normal (NAT) to tumour.

AUTHORED

example: `CDH1 / hsa-miR-25-3p` â†’ 0

OWN_ 1

reference â€” context, provenance and rarely-quoted detail

1 columns

â€”

rung

arm

â€”

coverage

88%â€”88%

(median 88%)

1x real

â‰¥ 0

own_specific_frac

ARM

REAL

â‰¥ 0

88%

Fraction of the arm's variance that is target-specific.

AUTHORED

example: `CDH1 / hsa-miR-25-3p` â†’ 0.55

COMPOSITION_ 1

reference " context, provenance and rarely-quoted detail

1 columns rung edge coverage 100%100% (median 100%)
1x categorical

composition_class

EDGE CATEGORICAL 100%

Composition-adjustment verdict for this row (e.g. composition_explained when the effect does not survive the deconvolution block). AUTHORED
values: cell_intrinsic (3,764) partial (1,167) composition_explained (713)
example: CDH1 / hsa-miR-25-3p cell_intrinsic

W_ 1

reference " context, provenance and rarely-quoted detail

1 columns rung gene coverage 100%100% (median 100%)
1x real 0

w_max

GENE REAL 0 100%

Maximum curated EVIDENCE weight over the gene's regulators. Gene-rung repeat. AUTHORED
example: CDH1 / hsa-miR-25-3p 13.24

M_ 1

reference " context, provenance and rarely-quoted detail

1 columns rung edge coverage 100%100% (median 100%)
1x real 0

m_nnl

EDGE REAL 0 100%

Bagged-NNLS weight. Exactly 0 in 31.7% of families " this is what lets identity say a regulator contributes nothing, which beta structurally cannot. AUTHORED
example: CDH1 / hsa-miR-25-3p 0.2293

TERM_ 3

reference " context, provenance and rarely-quoted detail

3 columns rung edge coverage 68%68% (median 68%)
2x signed 1 1 1x real, signed

term_ABUND

EDGE SIGNED 1 1 68%

Decomposition of the acquisition score into ABUNDANCE, WIRING and INTERACTION terms. AUTHORED
example: CDH1 / hsa-miR-25-3p -0.004
Defined on: edges present in the progression panel (arm expressed + state measured)

term_INTERACT

EDGE SIGNED 1 1 68%

The INTERACTION term " the part of the NAT'tumour change attributable to abundance and wiring moving TOGETHER. AUTHORED
example: CDH1 / hsa-miR-25-3p -0.003

term_WIRING

EDGE REAL, SIGNED 68%

The WIRING term of the dose-vs-wiring decomposition: how much of the NAT'tumour change comes from the arm's M-weight moving rather than its abundance. AUTHORED
example: CDH1 / hsa-miR-25-3p 1.376

PRIOR_ 1

reference " context, provenance and rarely-quoted detail

1 columns 1 rung edge 1 coverage 100%100% (median 100%)

1x constant

prior_pi

EDGE CONSTANT 100%

The Gibbs sampler's inclusion prior " CONSTANT 0.05 across the card. AUTHORED

values: 0.05 (5,644)

example: CDH1 / hsa-miR-25-3p + 0.05

EDGE_ 1

reference " context, provenance and rarely-quoted detail

1 columns 1 rung edge 1 coverage 73%73% (median 73%)

1x signed 11

edge_rho_adj

EDGE SIGNED 11 73%

An edge-rung quantity carried onto this row. AUTHORED

example: CDH1 / hsa-miR-25-3p + -0.198

| Defined on: the WITHIN-PATIENT paired subset (n=103 matched tumour/NAT pairs, MH-158)

ESUB_ 8

reference " context, provenance and rarely-quoted detail

8 columns 1 rung edge 1 coverage 98%100% (median 99%)

6x signed 11 1x categorical 1x p-value

esub_best_sub

EDGE CATEGORICAL 100%

WHICH subtype gave the strongest coupling. AUTHORED

values: Her2 (1,782) Basal (1,679) LumB (1,178) LumA (1,005)

example: CDH1 / hsa-miR-25-3p + Her2

esub_q_adj

EDGE P-VALUE 98%

BH-adjusted q for the best-subtype contrast " AUTHORED

example: CDH1 / hsa-miR-25-3p + 0.3849

esub_rho_Basal

EDGE SIGNED 11 99%

This edge's coupling computed WITHIN the Basal PAM50 subtype. AUTHORED

example: CDH1 / hsa-miR-25-3p + -0.1683

esub_rho_Her2

EDGE SIGNED 11 98%

This edge's coupling computed WITHIN the Her2 PAM50 subtype. AUTHORED

example: CDH1 / hsa-miR-25-3p + -0.3531

esub_rho_LumA

EDGE SIGNED 11 99%

This edge's coupling computed WITHIN the LumA PAM50 subtype. AUTHORED

example: CDH1 / hsa-miR-25-3p + -0.174

esub_rho_LumB

EDGE SIGNED 11 99%

This edge's coupling computed WITHIN the LumB PAM50 subtype. AUTHORED

example: CDH1 / hsa-miR-25-3p + -0.1127

esub_rho_best

EDGE SIGNED 11 100%

Coupling in the arm's BEST PAM50 subtype. AUTHORED

example: CDH1 / hsa-miR-25-3p + -0.3531

esub_rho_pool

EDGE SIGNED 11 100%

Coupling pooled across subtypes " the unselected reference for esub_rho_best (median 0.0135, indistinguishable from a random-subtype draw at 0.0123). AUTHORED

example: CDH1 / hsa-miR-25-3p + -0.1909

MEAN_ 2

reference " context, provenance and rarely-quoted detail

2 columns rung arm coverage 90%90% (median 90%) 2x real, signed	
mean_dGlobalRank ARMREAL, SIGNED90%	Mean change in the arm's GLOBAL abundance percentile across the paired patients. AUTHORED example: CDH1 / hsa-miR-25-3p 0.01
mean_own_shift ARMREAL, SIGNED90%	A per-row mean of a lower-rung quantity; check agg_of for which rung it summarises. AUTHORED example: CDH1 / hsa-miR-25-3p -0.194 Defined on: the WITHIN-PATIENT paired subset (n=103 matched tumour/NAT pairs, MH-158)

SURROGATE_ 2

reference " context, provenance and rarely-quoted detail

2 columns rung arm coverage 3%3% (median 3%) 1x fraction 01 1x categorical	
surrogate_corr ARMFRACTION 013%	How well the surrogate tracks the arm where both are observed (0.54 0.89). AUTHORED values: 0.874 (43) 0.892 (35) 0.621 (29) 0.842 (26) 0.541 (23) 0.606 (18) 0.728 (16) example: CDH1 / hsa-miR-30a-5p 0.606
surrogate_instrument ARMCATEGORICAL3%	Which SURROGATE stands in for a collapsed GTEx healthy leg (7 levels; defined on 3% of edges). AUTHORED values: hsa-miR-20a-5p (43) hsa-miR-181b-5p (35) hsa-miR-93-5p (29) hsa-miR-429 (26) hsa-let-7b-5p (23) hsa-miR-30c-5p (18) hsa-miR-221-3p (16) example: CDH1 / hsa-miR-30a-5p hsa-miR-30c-5p

CPTAC_ 20

reference " context, provenance and rarely-quoted detail

20 columns rung edge per-edge; the gene card aggregates it coverage 45%79% (median 65%) 12x signed 11 4x real, signed 2x constant 2x categorical	
cptac_prosp_comp_class_prot EDGECATEGORICAL59%	Composition class of this edge's PROTEIN coupling (cell_intrinsic / partial / composition_explained) in the prospective cohort. AUTHORED values: composition_explained (1,427) cell_intrinsic (1,278) partial (647) example: CDH1 / hsa-miR-25-3p composition_explained
cptac_prosp_n EDGECONSTANT79%	Patients in the prospective CPTAC cohort " CONSTANT 101. AUTHORED values: 101.0 (4,486) example: CDH1 / hsa-miR-25-3p 101
cptac_prosp_ret_prot EDGEREAL, SIGNED59%	RETENTION at EDGE rung " this edge's composition-ADJUSTED protein coupling divided by its UNADJUSTED one, in the CPTAC prospective cohort. AUTHORED example: CDH1 / hsa-miR-25-3p 0.2189

cptac_prosp_ret_rna EDGE REAL, SIGNED 61%	As cptac_prosp_ret_prot but against mRNA rather than protein: adjusted $\hat{A}$ · unadjusted mRNA coupling for this edge, prospective cohort. AUTHORED example: <code>CDH1 / hsa-miR-25-3p</code> $\hat{A}$ †' 0.3 <i>Defined on: genes/edges present in the CPTAC cohort</i>
cptac_prosp_rho_disc EDGE SIGNED $\hat{A}^{\sim}1\hat{A}\epsilon;1$ 78%	Spearman coupling of this single edge's arm against the discovery/decoy layer in the PROSPECTIVE cohort (independent patients, TMT-11, $n\hat{\%}^{\sim}101$), adjusted for the C block (composition + target CN + proliferation). AUTHORED example: <code>CDH1 / hsa-miR-25-3p</code> $\hat{A}$ †' 0.08776
cptac_prosp_rho_disc_raw EDGE SIGNED $\hat{A}^{\sim}1\hat{A}\epsilon;1$ 78%	Spearman coupling of this single edge's arm against the discovery/decoy layer in the PROSPECTIVE cohort (independent patients, TMT-11, $n\hat{\%}^{\sim}101$), UNADJUSTED $\hat{A}\epsilon$ " no confounder block. AUTHORED example: <code>CDH1 / hsa-miR-25-3p</code> $\hat{A}$ †' 0.014
cptac_prosp_rho_prot EDGE SIGNED $\hat{A}^{\sim}1\hat{A}\epsilon;1$ 78%	Spearman coupling of this single edge's arm against PROTEIN in the PROSPECTIVE cohort (independent patients, TMT-11, $n\hat{\%}^{\sim}101$), adjusted for the C block (composition + target CN + proliferation). AUTHORED example: <code>CDH1 / hsa-miR-25-3p</code> $\hat{A}$ †' 0.03748
cptac_prosp_rho_prot_raw EDGE SIGNED $\hat{A}^{\sim}1\hat{A}\epsilon;1$ 79%	UNADJUSTED protein coupling, prospective. AUTHORED example: <code>CDH1 / hsa-miR-25-3p</code> $\hat{A}$ †' 0.1712
cptac_prosp_rho_rna EDGE SIGNED $\hat{A}^{\sim}1\hat{A}\epsilon;1$ 79%	Spearman coupling of this single edge's arm against mRNA in the PROSPECTIVE cohort (independent patients, TMT-11, $n\hat{\%}^{\sim}101$), adjusted for the C block (composition + target CN + proliferation). AUTHORED example: <code>CDH1 / hsa-miR-25-3p</code> $\hat{A}$ †' 0.06188
cptac_prosp_rho_rna_raw EDGE SIGNED $\hat{A}^{\sim}1\hat{A}\epsilon;1$ 79%	Spearman coupling of this single edge's arm against mRNA in the PROSPECTIVE cohort (independent patients, TMT-11, $n\hat{\%}^{\sim}101$), UNADJUSTED $\hat{A}\epsilon$ " no confounder block. Range -0.578 $\hat{A}\epsilon$" +0.692 , 79% filled. AUTHORED example: <code>CDH1 / hsa-miR-25-3p</code> $\hat{A}$ †' 0.2063
cptac_t105_comp_class_prot EDGE CATEGORICAL 45%	Composition class of this row's PROTEIN coupling in the TCGA-OVERLAP cohort (n=105) $\hat{A}\epsilon$ " cell_intrinsic , partial or composition_explained . AUTHORED values: cell_intrinsic (1,447) $\hat{A}$ · composition_explained (632) $\hat{A}$ · partial (463) example: <code>CDH1 / hsa-miR-25-3p</code> $\hat{A}$ †' composition_explained
cptac_t105_n EDGE CONSTANT 70%	Patients in the TCGA-overlap CPTAC cohort $\hat{A}\epsilon$ " CONSTANT 105 . See the prospective twin. AUTHORED values: 105.0 (3,968) example: <code>CDH1 / hsa-miR-25-3p</code> $\hat{A}$ †' 105
cptac_t105_ret_prot EDGE REAL, SIGNED 45%	Retention at edge rung on the TCGA-overlap cohort $\hat{A}\epsilon$ " see the prospective twin. AUTHORED example: <code>CDH1 / hsa-miR-25-3p</code> $\hat{A}$ †' -0.5722
cptac_t105_ret_rna EDGE REAL, SIGNED 49%	As the prospective mRNA twin, on the TCGA-overlap cohort. AUTHORED example: <code>CDH1 / hsa-miR-25-3p</code> $\hat{A}$ †' 0.6976 <i>Defined on: genes/edges present in the CPTAC cohort</i>
cptac_t105_rho_disc EDGE SIGNED $\hat{A}^{\sim}1\hat{A}\epsilon;1$ 64%	Spearman coupling of this single edge's arm against the discovery/decoy layer in the TCGA-OVERLAP cohort (n=105), adjusted for the C block (composition + target CN + proliferation). AUTHORED example: <code>CDH1 / hsa-miR-25-3p</code> $\hat{A}$ †' 0.05624

cptac_t105_rho_disc_raw EDGE SIGNED $\hat{A}^{*1}\hat{A}\epsilon 1$ 65%	Spearman coupling of this single edge's arm against the discovery/decoy layer in the TCGA-OVERLAP cohort (n=105), UNADJUSTED $\hat{A}\epsilon$ no confounder block. Range -0.643 $\hat{A}\epsilon$ +0.727, 65% filled. AUTHORED example: <code>CDH1 / hsa-miR-25-3p $\hat{A}^+$ -0.03572</code>
cptac_t105_rho_prot EDGE SIGNED $\hat{A}^{*1}\hat{A}\epsilon 1$ 64%	Spearman coupling of this single edge's arm against PROTEIN in the TCGA-OVERLAP cohort (n=105), adjusted for the C block (composition + target CN + proliferation). AUTHORED example: <code>CDH1 / hsa-miR-25-3p $\hat{A}^+$ 0.0313</code>
cptac_t105_rho_prot_raw EDGE SIGNED $\hat{A}^{*1}\hat{A}\epsilon 1$ 65%	Spearman coupling of this single edge's arm against PROTEIN in the TCGA-OVERLAP cohort (n=105), UNADJUSTED $\hat{A}\epsilon$ no confounder block. Range -0.714 $\hat{A}\epsilon$ +0.736, 65% filled. AUTHORED example: <code>CDH1 / hsa-miR-25-3p $\hat{A}^+$ -0.0547</code>
cptac_t105_rho_rna EDGE SIGNED $\hat{A}^{*1}\hat{A}\epsilon 1$ 64%	Spearman coupling of this single edge's arm against mRNA in the TCGA-OVERLAP cohort (n=105), adjusted for the C block (composition + target CN + proliferation). Range -0.593 $\hat{A}\epsilon$ +0.684, 64% filled. AUTHORED example: <code>CDH1 / hsa-miR-25-3p $\hat{A}^+$ -0.06411</code>
cptac_t105_rho_rna_raw EDGE SIGNED $\hat{A}^{*1}\hat{A}\epsilon 1$ 65%	Spearman coupling of this single edge's arm against mRNA in the TCGA-OVERLAP cohort (n=105), UNADJUSTED $\hat{A}\epsilon$ no confounder block. Range -0.782 $\hat{A}\epsilon$ +0.695, 65% filled. AUTHORED example: <code>CDH1 / hsa-miR-25-3p $\hat{A}^+$ -0.0919</code>

REALIZATION_ $\hat{A}$ 2

reference $\hat{A}\epsilon$ context, provenance and rarely-quoted detail

2 columns $\hat{A}$ rung edge $\hat{A}$ coverage 58$\hat{A}\epsilon$69% (median 63%) 1$\times$ fraction $0\hat{A}\epsilon$1 $\hat{A}$ 1$\times$ real, signed	
realization_identity EDGE FRACTION $0\hat{A}\epsilon$1 58%	Within-patient realization quantities $\hat{A}\epsilon$ the score, its identity allocation, and who owns it. AUTHORED example: <code>CDH1 / hsa-miR-25-3p $\hat{A}^+$ 0.849</code> <i>Defined on: genes with a family-level realization fit</i>
realization_score EDGE REAL, SIGNED 69%	$\hat{A}^{*}$coupling_z_tum $\hat{A}\epsilon$ the calibrated tumour repression strength as a positive score, so bigger = more repressed. AUTHORED example: <code>CDH1 / hsa-miR-25-3p $\hat{A}^+$ 3.04</code>

KD_ $\hat{A}$ 2

reference $\hat{A}\epsilon$ context, provenance and rarely-quoted detail

2 columns $\hat{A}$ rung edge $\hat{A}$ coverage 85$\hat{A}\epsilon$85% (median 85%) 1$\times$ percent $\hat{A}$ 1$\times$ real, signed	
kd_affinity_pct EDGE PERCENT 85%	Is this gene among the arm's strongest predicted targets $\hat{A}\epsilon$ a WITHIN-ARM percentile, higher = stronger target. AUTHORED example: <code>CDH1 / hsa-miR-25-3p $\hat{A}^+$ 0.8387</code>
kd_repression EDGE REAL, SIGNED 85%	scanMiR RBNS binding affinity. AUTHORED example: <code>CDH1 / hsa-miR-25-3p $\hat{A}^+$ -0.8171</code>

WIRING_ 1

reference context, provenance and rarely-quoted detail

1 columns 1 rung edge 1 coverage 68%68% (median 68%)

1x fraction 01

wiring_frac

EDGE FRACTION 01 68%

Fraction of the acquisition score attributable to WIRING rather than abundance. AUTHORED

example: CDH1 / hsa-miR-25-3p 1

| Defined on: edges present in the progression panel (arm expressed + state measured)

SD_ 1

reference context, provenance and rarely-quoted detail

1 columns 1 rung edge 1 coverage 20%20% (median 20%)

1x real 0

sd_arm

EDGE REAL 0 20%

Posterior SD of the arm-level $\hat{\mu}$. AUTHORED

example: CDH1 / hsa-miR-25-3p 0.01097

CELL_ 1

reference context, provenance and rarely-quoted detail

1 columns 1 rung family 1 coverage 20%20% (median 20%)

1x boolean

cell_arms_resolvable

FAMILY BOOLEAN 20%

Are the cell's arms separable AND does splitting them help out-of-fold. True for 31.7% of edges in multi-arm cells. AUTHORED

values: False (783) 1 True (364)

example: CDH1 / hsa-miR-25-3p 1 True

Z_ 1

reference context, provenance and rarely-quoted detail

1 columns 1 rung edge 1 coverage 20%20% (median 20%)

1x real 0

z_arm

EDGE REAL 0 20%

beta_arm / sd_arm

 from the WITHIN-CELL fit (`arm_rung.py`) the arm's z inside its (gene, seed_family) cell only, defined on the 20.3% of edges in multi-arm cells. AUTHORED

example: CDH1 / hsa-miR-25-3p 3.706

ACQUISITION_ 1

reference context, provenance and rarely-quoted detail

1 columns 1 rung edge 1 coverage 59%59% (median 59%)

1x real, signed

acquisition_score

EDGE

REAL, SIGNED

59%

Composite acquisition score across states.

AUTHORED

example: CDH1 / hsa-miR-25-3p → 3.35

Defined on: arms with a GTEx-HEALTHY leg (the thinnest state → many arms are multi-mapping-collapsed, MH-166)

DSHARE_ → 2

reference → context, provenance and rarely-quoted detail

2 columns → rung edge → coverage 65%→65% (median 65%)

2x signed →1→1

dShare_M_own

EDGE

SIGNED →1→1

65%

Mean over the 103 paired patients of this arm's change in M-WEIGHTED share of its gene's total regulator dose, tumour minus that patient's OWN NAT.

AUTHORED

example: CDH1 / hsa-miR-25-3p → 0.045

dShare_raw_own

EDGE

SIGNED →1→1

65%

As dShare_M_own but on RAW abundance share, with the model's M weights removed.

AUTHORED

example: CDH1 / hsa-miR-25-3p → 0.04

HEUR_ → 1

reference → context, provenance and rarely-quoted detail

1 columns → rung gene → coverage 100%→100% (median 100%)

1x count

heur_gene_nreg

GENE

COUNT

100%

Regulator count for this gene from the →6b-RETIRED heuristic lane.

AUTHORED

example: CDH1 / hsa-miR-25-3p → 12

FAMILY_ → 1

reference → context, provenance and rarely-quoted detail

1 columns → rung family → coverage 69%→69% (median 69%)

1x signed →1→1

family_rho_adj

FAMILY

SIGNED →1→1

69%

Family-level properties carried onto this row (size, the arm's dose share of it, and its role).

AUTHORED

example: CDH1 / hsa-miR-25-3p → -0.332

HLY_ → 1

reference → context, provenance and rarely-quoted detail

1 columns → rung edge → coverage 61%→61% (median 61%)

1x boolean

hly_leg_concordant

EDGE

BOOLEAN

61%

Do the two HLY→TUM legs (_QN and _raw) at least AGREE ON SIGN. **Concordant 76.1%, DISCORDANT 23.9%** of the 3,433 edges carrying both.

AUTHORED

values: True (2,611) → False (822)

example: CDH1 / hsa-miR-25-3p → True

IS_ 1

reference “ context, provenance and rarely-quoted detail

1	columns	1	rung	edge	1	coverage	58%58% (median 58%)
1x	boolean						

is_realization_owner

EDGE

BOOLEAN

58%

A boolean flag; read the suffix.

AUTHORED

values: False (2,498) 1 True (759)

example: CDH1 / hsa-miR-25-3p 1 True

| Defined on: genes with a family-level realization fit

realization/gene_card.tsv

gene

One row per the gene's total incoming regulation. One row per gene, so nothing here is checkable by invariance – `agg_of` carries the meaning instead.

135 columns 1,549 rows 29 blocks median coverage 83%

Contents – 29 blocks, most important first

1. n_7

2. (bare) 2

3. comp_16

4. ctx_16

5. top_8

6. realized_5

7. w_1

8. lit_7

9. cptac_24

10. frac_3

11. dominant_4

12. static_1

13. total_1

14. owner_1
15. median_1

16. armres_7

17. identity_1

18. greal_10

19. heur_5

20. acquired_vs_2

21. pressure_1

22. tcga_4

23. realization_1

24. disc_2

25. max_1

26. regulatory_1

27. mean_1

28. concentration_1

29. fam_1
- CORE

• CORE

• CORE

• CORE

• SUPPORTING

• SUPPORTING

• SUPPORTING

• REFERENCE

• REFERENCE

• REFERENCE

• REFERENCE

• REFERENCE

• REFERENCE

• REFERENCE

N – 7

core – the columns findings rest on

7 columns – rung gene – coverage 91%–100% (median 100%)
7x count

n_arms

GENE COUNT AGG ARM 100%

Arms in the LEARNED MODEL's design – `X.shape[1]` from `assemble_gene`, carried via `gene_atlas`. **This is the width the Gibbs actually fit** (median 2, max 90). AUTHORED
example: `CDH1` – 26

n_cell_intrinsic

GENE COUNT 100%

FAMILIES whose coupling SURVIVES the composition block (retention ≥ 0.7). AUTHORED
example: `CDH1` – 22

n_composition_explained

GENE COUNT 100%

FAMILIES whose coupling does NOT survive the composition block (retention < 0.4). AUTHORED
values: 0 (1,133) – 1 (286) – 2 (80) – 3 (26) – 4 (13) – 6 (7) – 5 (2) – 8 (2)
example: `CDH1` – 0

n_fam

GENE COUNT AGG FAMILY
100%

How many seed families regulate this gene. AUTHORED
example: `CDH1` – 25

n_hallmark_sets

GENE COUNT 91%

How many MSigDB Hallmark programs contain this gene (1–10, median 2). Membership only – it says nothing about regulation. AUTHORED
example: `CDH1` – 3

n_identified
GENE COUNT 100%
Families reaching $|z| > 2$ on the UNCALIBRATED SD. **Zero for 691 genes (44.6%)** meaning the model cannot distinguish ANY of that gene's families from zero. **AUTHORED**
example: CDH1 4

n_pip_disc_gt50
GENE COUNT 100%
Count of the gene's families with `pip_discovery > 0.5` under the evidence-readout. **AUTHORED**
example: CDH1 4

(BARE) 2
core the columns findings rest on

2 columns spans gene, key this prefix groups columns living on DIFFERENT units coverage 100%100% (median 100%)
1x fraction 1 1x identifier

concentration
GENE FRACTION 1 100%
`beta.max() / beta.sum()` the TOP FAMILY's share of the gene's total $\hat{\mu}^2$; median **0.954**, i.e. most genes are dominated by one family. **AUTHORED**
example: CDH1 0.191

gene
KEY IDENTIFIER 100%
KEY HGNC gene symbol. **AUTHORED**
example: JUN

COMP_ 16
core the columns findings rest on

16 columns rung gene coverage 59%90% (median 73%)
6x categorical 6x signed 1 3x real, signed 1x real 0

comp_cptac_prot_driver
GENE CATEGORICAL 59%
WHICH deconvolved compartment most explains this gene's PROTEIN coupling prolifer CAFs 173 purity 140 PVL 77 Endothelial 67. **AUTHORED**
example: CDH1 T-cells

comp_cptac_prot_driver_ret
GENE REAL, SIGNED 59%
`1 - comp_cptac_prot_driver_share`, exactly. Median +0.398 raw, **+0.469** gated to $|I_raw| \geq 0.10$ quote the gated value. See the mRNA twin. **AUTHORED**
example: CDH1 0.5085
Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)

comp_cptac_prot_driver_share
GENE REAL, SIGNED 59%
Protein-layer twin of `comp_tcga_mrna_driver_share`. Same exact complementarity with `_driver_ret` (8.9e-16 on 918 rows). **AUTHORED**
example: CDH1 0.4915

comp_cptac_prot_rho_raw
GENE SIGNED 1 73%
Unadjusted aggregate PROTEIN coupling denominator of `comp_cptac_prot_driver_ret`. **AUTHORED**
example: CDH1 -0.1416

comp_cptac_prot_top_budget
GENE CATEGORICAL 73%
The compartment the gene's miRNA BUDGET loads on most, on CPTAC protein. Range **10 levels**, 73% filled. **AUTHORED**
example: CDH1 Normal Epithelial
Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)

comp_cptac_prot_top_budget_r
GENE SIGNED 1 73%
As the TCGA mRNA twin, on CPTAC protein the budget-vs-compartment correlation. Median **0.118**, i.e. it runs the OPPOSITE way to mRNA (+0.121). **AUTHORED**
example: CDH1 0.3126
Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)

comp_cptac_prot_top_target

GENE CATEGORICAL 73%

The compartment the TARGET's expression loads on most. Range **10 levels**, 73% filled. AUTHORED

example: CDH1 ↑ T-cells

Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)

comp_cptac_prot_top_target_r

GENE SIGNED $\hat{r} \in [-1, 1]$ 73%

As the TCGA mRNA twin, on CPTAC protein the target-vs-compartment correlation. Median **+0.203**. AUTHORED

example: CDH1 ↑ -0.2385

Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)

comp_tcga_mrna_driver

GENE CATEGORICAL 66%

WHICH compartment drives the gene's mRNA coupling the one whose removal moves it most (CAFs 279 purity 174 prolifer 148 Endothelial 122 Myeloid 109). AUTHORED

example: CDH1 ↑ T-cells

comp_tcga_mrna_driver_ret

GENE REAL, SIGNED 66%

1 the comp_tcga_mrna_driver_share, exactly (see there). The RETENTION orientation: what fraction of the coupling survives removing the driver compartment. AUTHORED

example: CDH1 ↑ 0.8681

comp_tcga_mrna_driver_share

GENE REAL $\in [0, 1]$ 66%

Share of the gene's mRNA coupling attributable to the single compartment that most changes it the **axiom 8's composition GATE**, the axis that separates the harm tails at p=0.00999 where the ensemble axis gives p=0.10. AUTHORED

example: CDH1 ↑ 0.1319

comp_tcga_mrna_rho_raw

GENE SIGNED $\hat{r} \in [-1, 1]$ 90%

The gene's UNADJUSTED aggregate mRNA coupling the DENOMINATOR of comp_tcga_mrna_driver_ret. AUTHORED

example: CDH1 ↑ -0.1921

comp_tcga_mrna_top_budget

GENE CATEGORICAL 90%

The compartment the gene's miRNA BUDGET loads on most a property of the REGULATORS. AUTHORED

example: CDH1 ↑ Normal Epithelial

comp_tcga_mrna_top_budget_r

GENE SIGNED $\hat{r} \in [-1, 1]$ 90%

Correlation between the gene's **miRNA PRESSURE BUDGET** and the fraction of the compartment that budget loads on most (comp_tcga_mrna_top_budget), in TCGA mRNA. AUTHORED

example: CDH1 ↑ 0.2545

comp_tcga_mrna_top_target

GENE CATEGORICAL 90%

The compartment the TARGET GENE's expression loads on most. AUTHORED

example: CDH1 ↑ prolifer

comp_tcga_mrna_top_target_r

GENE SIGNED $\hat{r} \in [-1, 1]$ 90%

Correlation between the **TARGET GENE's own expression** and the fraction of the compartment it loads on most. AUTHORED

example: CDH1 ↑ 0.3289

CTX_ 16

core the columns findings rest on

16 columns the rung gene SUBSET of the edge block, identical rungs coverage **45%100%** (median 90%)
4x signed $\hat{r} \in [-1, 1]$ 3x real $\in [0, 1]$ 3x real, signed 3x count 1x categorical 1x fraction 01 1x boolean

ctx_apriori_class

GENE CATEGORICAL 100%

A-priori competence class from the gene atlas (A_COMPETENT / B_PARTIAL / C_WEAK / D_UNDEFINED). AUTHORED

values: D_UNDEFINED (730) A_COMPETENT (428) B_PARTIAL (295) C_WEAK (96)

example: CDH1 ↑ A_COMPETENT

ctx_ceiling	GENE SIGNED $\hat{A}^{\sim 1} \hat{A} \in \{1\}$ 98%	The UPPER BOUND on what ANY estimator can achieve for this gene $\hat{A}$ the 5-fold out-of-fold $\hat{R}^2$ of an UNRESTRICTED OLS on the gene's real design (<code>card_context.py:15</code>). AUTHORED example: <code>CDH1</code> $\hat{A} \dagger$ 0.1535
ctx_collapse	GENE REAL $\hat{A} \in \mathbb{Y} \emptyset$ 90%	<code>n_arm / n_fam</code> for the gene $\hat{A}$ how much curation BUNDLES same-seed mates onto it ($\hat{A} \in \{5\}$). AUTHORED example: <code>CDH1</code> $\hat{A} \dagger$ 1.091
ctx_d_collin	GENE SIGNED $\hat{A}^{\sim 1} \hat{A} \in \{1\}$ 45%	Mean pairwise COLLINEARITY among the gene's regulators $\hat{A}$ how much they move together (<code>gene_atlas</code> via <code>card_context</code>). Range -0.699 $\hat{A}$ +0.495 , 45% filled. AUTHORED example: <code>CDH1</code> $\hat{A} \dagger$ -0.02497
ctx_d_fam	GENE REAL, SIGNED 87%	Design width at FAMILY grain (<code>gene_atlas</code> via <code>card_context</code>). Range -2 $\hat{A}$ 25 , 87% filled. AUTHORED example: <code>CDH1</code> $\hat{A} \dagger$ 0
ctx_dose_max	GENE REAL $\hat{A} \in \mathbb{Y} \emptyset$ 90%	The LARGEST regulator dose in the gene's design (<code>gene_atlas</code> via <code>card_context</code>). Range 0.215 $\hat{A}$ 17.8 , 90% filled. AUTHORED example: <code>CDH1</code> $\hat{A} \dagger$ 15.38
ctx_dose_med	GENE REAL $\hat{A} \in \mathbb{Y} \emptyset$ 90%	The MEDIAN regulator dose in the gene's design (<code>gene_atlas</code> via <code>card_context</code>). Range 0.215 $\hat{A}$ 17.8 , 90% filled. AUTHORED example: <code>CDH1</code> $\hat{A} \dagger$ 6.318
ctx_frac_abund	GENE FRACTION $\emptyset \hat{A} \in \{1\}$ 90%	FRACTION of the gene's regulators clearing the abundance gate (<code>gene_atlas</code> via <code>card_context</code>). Range 0 $\hat{A}$ 1 , 90% filled. AUTHORED example: <code>CDH1</code> $\hat{A} \dagger$ 0.5833
ctx_gap_core	GENE SIGNED $\hat{A}^{\sim 1} \hat{A} \in \{1\}$ 87%	Real-minus-decoy coupling gap for this gene. Pooled it is ~ -0.012 , but the per-gene IQR STRADDLES ZERO $\hat{A}$ this is a set-level distributional shift, not a per-gene effect. Do not rank genes by it. AUTHORED example: <code>CDH1</code> $\hat{A} \dagger$ -0.2714
ctx_gap_d_dose	GENE REAL, SIGNED 87%	Difference in dose between the real design and its decoy. Range -5.41 $\hat{A}$ 1.74 , 87% filled. AUTHORED example: <code>CDH1</code> $\hat{A} \dagger$ 0.1238
ctx_gap_deconv	GENE SIGNED $\hat{A}^{\sim 1} \hat{A} \in \{1\}$ 87%	The same margin under the DECONVOLUTION block (<code>gene_atlas</code> via <code>card_context</code>). Range -0.33 $\hat{A}$ +0.374 , 87% filled. AUTHORED example: <code>CDH1</code> $\hat{A} \dagger$ -0.1864
ctx_gap_retention	GENE REAL, SIGNED 83%	<code>gap_deconv / gap_core</code> $\hat{A}$ how much of the decoy margin survives adjustment (<code>gene_atlas</code> via <code>card_context</code>). Range -16.5 $\hat{A}$ 17 , 83% filled. AUTHORED example: <code>CDH1</code> $\hat{A} \dagger$ 0.6867
ctx_measurable	GENE BOOLEAN 98%	Is the gene's <code>ctx_ceiling</code> above <code>CEILING_ROBUST</code> (0.02 , not 0 $\hat{A}$ MH-144)? AUTHORED values: False (845) $\hat{A}$ True (672) example: <code>CDH1</code> $\hat{A} \dagger$ True
ctx_n_abund	GENE COUNT 90%	How many of the gene's regulators clear the abundance gate (<code>gene_atlas</code> via <code>card_context</code>). Range 0 $\hat{A}$ 53 , 90% filled. AUTHORED example: <code>CDH1</code> $\hat{A} \dagger$ 7
ctx_n_fam_abund	GENE COUNT 90%	How many of the gene's families clear the abundance gate. Range 0 $\hat{A}$ 34 , 90% filled. AUTHORED example: <code>CDH1</code> $\hat{A} \dagger$ 6

ctx_n_fam_multi

GENE

COUNT

90%

How many of the gene's families hold MORE THAN ONE arm (`gene_atlas` via `card_context`). Range **0** – **14**, 90% filled.

AUTHORED

example: `CDH1` – `1`

TOP_ – 8

supporting – read these when the core raises a question

8 columns – rung **gene** – coverage **78%–100%** (median 89%)

3× identifier – 2× real – 0 – 1× fraction – 1 – 1× signed – 1 – 1× count

top_beta

GENE

REAL – 0

AGG FAMILY

100%

The largest $\hat{I}^2$ among the gene's families. Range **0.00096** – **1.52**, 100% filled.

AUTHORED

example: `CDH1` – `0.2584`

top_beta_frac

GENE

FRACTION – 1

AGG FAMILY

100%

That family's SHARE of the gene's total $\hat{I}^2$. Range **0.076** – **1**, 100% filled.

AUTHORED

example: `CDH1` – `0.1919`

top_family_identity

GENE

IDENTIFIER

AGG FAMILY

87%

The family Shapley/LMG credits with the gene's regulation. NaN for 197 genes (12.7%) where NNLS zeroed everything – an honest 'undefined', which the magnitude answer cannot express.

AUTHORED

example: `CDH1` – `miR-25-3p/32-5p/92-3p/363-3p/367-3p`

top_family_magnitude

GENE

IDENTIFIER

AGG FAMILY

100%

The family with the largest beta.

AUTHORED

example: `CDH1` – `miR-25-3p/32-5p/92-3p/363-3p/367-3p`

top_gaining_arms

GENE

IDENTIFIER

91%

Which arms gained the most repression healthy – tumour. Range **1,279 distinct**, 91% filled.

AUTHORED

example: `CDH1` – `hsa-miR-30a-5p(+0.25);hsa-miR-9-5p(+0.24);hsa-miR-544a-3p(+0.02);hsa-miR-34c-3p(+0.01);hsa-miR-25-3p(+0.01);hsa-miR-199a-5p(-0.00)`

top_identity

GENE

REAL – 0

AGG FAMILY

87%

The largest `identity` share among the gene's families.

AUTHORED

example: `CDH1` – `0.4477`

top_identity_gated

GENE

SIGNED – 1

78%

`top_identity` recomputed over `identity_reliable` edges only – **max 1.0000, 0 genes above 1** (vs +740.007 ungated). Defined on 1,203 genes.

AUTHORED

example: `CDH1` – `0.3367`

top_identity_n_reliable

GENE

COUNT

78%

How many of the gene's families survive the identity gate – the DENOMINATOR behind `top_identity_gated`. **Median 2**.

AUTHORED

example: `CDH1` – `26`

REALIZED_ – 5

supporting – read these when the core raises a question

5 columns – rung **gene** – coverage **63%–82%** (median 81%)

2× signed – 1 – 1× boolean – 1× count – 1× real, signed

realized_composition_explained

GENE

BOOLEAN

82%

Is the gene's REALIZED coupling composition-explained (retention < 0.4)?

AUTHORED

values: False (785) – True (490)

example: `CDH1` – `True`

realized_n_reg GENE COUNT 82%	How many regulators the realized coupling was computed over. Range 1 – 63 , 82% filled. AUTHORED example: <code>CDH1</code> $\hat{+}$ <code>17</code>
realized_retention GENE REAL, SIGNED 63%	Realized-coupling retention, $\hat{I}_{\text{adjusted}} / \hat{I}_{\text{raw}}$. AUTHORED example: <code>CDH1</code> $\hat{+}$ <code>-0.16</code>
realized_rho_adj GENE SIGNED $\hat{A}^{\sim 1} \hat{A} \{ 1$ 81%	The same, composition-adjusted. Range -0.479 – +0.509 , 81% filled. AUTHORED example: <code>CDH1</code> $\hat{+}$ <code>-0.035</code>
realized_rho_raw GENE SIGNED $\hat{A}^{\sim 1} \hat{A} \{ 1$ 81%	The gene's realized coupling, UNADJUSTED. Range -0.67 – +0.62 , 81% filled. AUTHORED example: <code>CDH1</code> $\hat{+}$ <code>0.225</code>

W_ $\hat{A}$ **1**
supporting $\hat{A}$ ” read these when the core raises a question

1 columns $\hat{A}$ rung gene $\hat{A}$ coverage 100% – 100% (median 100%) 1x real $\hat{A} \{ 0$
--

w_max GENE REAL $\hat{A} \{ 0$ 100%	The MAXIMUM curated evidence weight w over the gene's regulators (<code>gene_atlas : nanmax(w)</code>), where w comes from the same PMID-deduped ledger that feeds <code>lit_</code> and <code>fame_</code> . AUTHORED example: <code>CDH1</code> $\hat{+}$ <code>13.24</code>
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LIT_ $\hat{A}$ **7**
reference $\hat{A}$ ” context, provenance and rarely-quoted detail

7 columns $\hat{A}$ rung gene $\hat{A}$ coverage 21% – 21% (median 21%) 4x count $\hat{A}$ 2x boolean $\hat{A}$ 1x identifier
--

lit_agrees_identity GENE BOOLEAN 21%	Does <code>identity</code> name the same family as the most-published one for this gene $\hat{A}$ ” True on 100 of 323 (31%) . AUTHORED values: False (223) $\hat{A}$ True (100) example: <code>CDH1</code> $\hat{+}$ <code>False</code>
lit_agrees_magnitude GENE BOOLEAN 21%	Does the model's largest- $\hat{I}^2$ family match the most-published one (96 of 329 genes, 29%)? AUTHORED values: False (233) $\hat{A}$ True (96) example: <code>CDH1</code> $\hat{+}$ <code>False</code>
lit_family GENE IDENTIFIER 21%	The most-published seed family for this gene by distinct low-throughput functional PMIDs. AUTHORED example: <code>CDH1</code> $\hat{+}$ <code>miR-9-5p</code>
lit_margin GENE COUNT 21%	Gap between the top and second literature family, in PMIDs $\hat{A}$ ” median 1.00 , i.e. the literature's 'winner' is usually decided by a SINGLE paper. $\hat{A}$ ” gate on this before reading lit_agrees_* : a 1-PMID margin is a coin flip dressed as ground truth. AUTHORED example: <code>CDH1</code> $\hat{+}$ <code>3</code> Defined on: genes with a versioned literature canonical family (329 of 1,549, sha256-stamped, MH-196)

lit_n_lit_families

GENE COUNT 21%

” A STUDY-COUNT AXIS, NOT A BIOLOGICAL GROUND TRUTH ” read the name with suspicion. eval/lit_ground_truth.py defines the 'canonical' family of a gene as the **argmax of DISTINCT PubMed IDs** carrying low-throughput functional evidence (reporter / western / proteomics; miRTarBase-Weak and TarBase-Positive excluded). AUTHORED
example: CDH1 25

lit_n_pmid

GENE COUNT 21%

Distinct low-throughput functional PMIDs behind lit_family . Median 3 ” the 'ground truth' rests on a handful of papers per gene. AUTHORED
example: CDH1 6
Defined on: genes with a versioned literature canonical family (329 of 1,549, sha256-stamped, MH-196)

lit_tier

GENE COUNT 21%

Literature-evidence tier for the gene (levels 2, 3 and 4; 4 is the most populated at 132 genes). AUTHORED
values: 4.0 (132) 2.0 (129) 3.0 (68)
example: CDH1 4

CPTAC_ 24

reference ” context, provenance and rarely-quoted detail

24 columns rung gene AGGREGATE of the edge block via an agg_/abund_ infix ” NOT derivable (MH-262) coverage 42%89% (median 66%)
18x signed 1 2x boolean 2x real, signed 2x count

cptac_prosp_abund_rho_disc

GENE SIGNED 11
AGG ARM 72%

Spearman coupling of an **UNWEIGHTED ABUNDANCE SUM** ” the reference the aggregate is judged against against the discovery/decoy layer in the PROSPECTIVE cohort (independent patients, TMT-11, 101), adjusted for the C block (composition + target CN + proliferation). AUTHORED
example: CDH1 -0.08033

cptac_prosp_abund_rho_prot

GENE SIGNED 11
AGG ARM 73%

Spearman coupling of an **UNWEIGHTED ABUNDANCE SUM** ” the reference the aggregate is judged against against PROTEIN in the PROSPECTIVE cohort (independent patients, TMT-11, 101), adjusted for the C block (composition + target CN + proliferation). AUTHORED
example: CDH1 0.109

cptac_prosp_abund_rho_rna

GENE SIGNED 11
AGG ARM 74%

Spearman coupling of an **UNWEIGHTED ABUNDANCE SUM** ” the reference the aggregate is judged against against mRNA in the PROSPECTIVE cohort (independent patients, TMT-11, 101), adjusted for the C block (composition + target CN + proliferation). AUTHORED
example: CDH1 0.2848

cptac_prosp_agg_beats_abund_prot

GENE BOOLEAN 73%

Does the learned $\hat{I}^2$ -weighted aggregate couple more negatively to PROTEIN than a plain abundance sum, in the CPTAC prospective cohort? AUTHORED
values: False (770) True (357)
example: CDH1 True

cptac_prosp_agg_ret_prot

GENE REAL, SIGNED 59%

rho_prot(composition-adjusted) / rho_prot_raw ” the ADJUSTMENT sense of retention, on the prospective cohort. æ... **The denominator is ALREADY GATED** at RHO_GATE in cptac_card.py:322 , so values outside [0,1] are NOT the vanishing-denominator artifact of axiom 5 (verified: 0 rows with |raw| < 0.02). AUTHORED
example: CDH1 0.855

cptac_prosp_agg_rho_disc

GENE SIGNED 11
AGG ARM 72%

Spearman coupling of the $\hat{I}^2$ -WEIGHTED AGGREGATE of the gene's arms against the discovery/decoy layer in the PROSPECTIVE cohort (independent patients, TMT-11, 101), adjusted for the C block (composition + target CN + proliferation). AUTHORED
example: CDH1 -0.1539

cptac_prosp_agg_rho_disc_raw GENE SIGNED $\hat{A}^{*1} \hat{A} \in \{1\}$ 72%	Spearman coupling of the $\hat{I}^2$-WEIGHTED AGGREGATE of the gene's arms against the discovery/decoy layer in the PROSPECTIVE cohort (independent patients, TMT-11, $n \hat{=} 101$), UNADJUSTED $\hat{A} \in$ no confounder block. AUTHORED example: $CDH1 \hat{A} \dagger' -0.1971$
cptac_prosp_agg_rho_prot GENE SIGNED $\hat{A}^{*1} \hat{A} \in \{1\}$ AGG ARM 73%	Spearman coupling of the $\hat{I}^2$-WEIGHTED AGGREGATE of the gene's arms against PROTEIN in the PROSPECTIVE cohort (independent patients, TMT-11, $n \hat{=} 101$), adjusted for the C block (composition + target CN + proliferation). AUTHORED example: $CDH1 \hat{A} \dagger' -0.1211$
cptac_prosp_agg_rho_prot_raw GENE SIGNED $\hat{A}^{*1} \hat{A} \in \{1\}$ 73%	Spearman coupling of the $\hat{I}^2$-WEIGHTED AGGREGATE of the gene's arms against PROTEIN in the PROSPECTIVE cohort (independent patients, TMT-11, $n \hat{=} 101$), UNADJUSTED $\hat{A} \in$ no confounder block. AUTHORED example: $CDH1 \hat{A} \dagger' -0.1416$
cptac_prosp_agg_rho_rna GENE SIGNED $\hat{A}^{*1} \hat{A} \in \{1\}$ AGG ARM 74%	Spearman coupling of the $\hat{I}^2$-WEIGHTED AGGREGATE of the gene's arms against mRNA in the PROSPECTIVE cohort (independent patients, TMT-11, $n \hat{=} 101$), adjusted for the C block (composition + target CN + proliferation). AUTHORED example: $CDH1 \hat{A} \dagger' 0.04264$
cptac_prosp_agg_rho_rna_raw GENE SIGNED $\hat{A}^{*1} \hat{A} \in \{1\}$ 74%	Spearman coupling of the $\hat{I}^2$-WEIGHTED AGGREGATE of the gene's arms against mRNA in the PROSPECTIVE cohort (independent patients, TMT-11, $n \hat{=} 101$), UNADJUSTED $\hat{A} \in$ no confounder block. AUTHORED example: $CDH1 \hat{A} \dagger' 0.06539$
cptac_prosp_n_arms GENE COUNT 89%	Arms contributing to this gene's prospective-cohort aggregate (median 2, max 61). AUTHORED example: $CDH1 \hat{A} \dagger' 23$
cptac_t105_abund_rho_disc GENE SIGNED $\hat{A}^{*1} \hat{A} \in \{1\}$ 60%	Spearman coupling of an UNWEIGHTED ABUNDANCE SUM $\hat{A} \in$ the reference the aggregate is judged against against the discovery/decoy layer in the TCGA-OVERLAP cohort (n=105), adjusted for the C block (composition + target CN + proliferation). AUTHORED example: $CDH1 \hat{A} \dagger' -0.0005806$
cptac_t105_abund_rho_prot GENE SIGNED $\hat{A}^{*1} \hat{A} \in \{1\}$ 60%	Spearman coupling of an UNWEIGHTED ABUNDANCE SUM $\hat{A} \in$ the reference the aggregate is judged against against PROTEIN in the TCGA-OVERLAP cohort (n=105), adjusted for the C block (composition + target CN + proliferation). AUTHORED example: $CDH1 \hat{A} \dagger' 0.01071$
cptac_t105_abund_rho_rna GENE SIGNED $\hat{A}^{*1} \hat{A} \in \{1\}$ 60%	Spearman coupling of an UNWEIGHTED ABUNDANCE SUM $\hat{A} \in$ the reference the aggregate is judged against against mRNA in the TCGA-OVERLAP cohort (n=105), adjusted for the C block (composition + target CN + proliferation). AUTHORED example: $CDH1 \hat{A} \dagger' 0.06946$
cptac_t105_agg_beats_abund_prot GENE BOOLEAN 60%	As the prospective twin, on the TCGA-105 overlap cohort. AUTHORED values: False (633) $\hat{A}$ True (299) example: $CDH1 \hat{A} \dagger' True$
cptac_t105_agg_ret_prot GENE REAL, SIGNED 42%	As the prospective twin, TCGA-105. AUTHORED example: $CDH1 \hat{A} \dagger' 0.2276$
cptac_t105_agg_rho_disc GENE SIGNED $\hat{A}^{*1} \hat{A} \in \{1\}$ AGG ARM 60%	Spearman coupling of the $\hat{I}^2$-WEIGHTED AGGREGATE of the gene's arms against the discovery/decoy layer in the TCGA-OVERLAP cohort (n=105), adjusted for the C block (composition + target CN + proliferation). AUTHORED example: $CDH1 \hat{A} \dagger' 0.04722$

cptac_t105_agg_rho_disc_raw GENE SIGNED $\hat{A}^{\prime 1} \hat{A} \leq 1$ 60%	Spearman coupling of the $\hat{I}^2$-WEIGHTED AGGREGATE of the gene's arms against the discovery/decoy layer in the TCGA-OVERLAP cohort (n=105), UNADJUSTED $\hat{A}$ no confounder block. AUTHORED example: $CDH1 \hat{A} \dagger' -0.03473$
cptac_t105_agg_rho_prot GENE SIGNED $\hat{A}^{\prime 1} \hat{A} \leq 1$ AGG ARM 60%	Spearman coupling of the $\hat{I}^2$-WEIGHTED AGGREGATE of the gene's arms against PROTEIN in the TCGA-OVERLAP cohort (n=105), adjusted for the C block (composition + target CN + proliferation). AUTHORED example: $CDH1 \hat{A} \dagger' -0.02019$
cptac_t105_agg_rho_prot_raw GENE SIGNED $\hat{A}^{\prime 1} \hat{A} \leq 1$ 60%	Spearman coupling of the $\hat{I}^2$-WEIGHTED AGGREGATE of the gene's arms against PROTEIN in the TCGA-OVERLAP cohort (n=105), UNADJUSTED $\hat{A}$ no confounder block. Range -0.484 $\hat{A}$ +0.446, 60% filled. AUTHORED example: $CDH1 \hat{A} \dagger' -0.08874$
cptac_t105_agg_rho_rna GENE SIGNED $\hat{A}^{\prime 1} \hat{A} \leq 1$ AGG ARM 60%	Spearman coupling of the $\hat{I}^2$-WEIGHTED AGGREGATE of the gene's arms against mRNA in the TCGA-OVERLAP cohort (n=105), adjusted for the C block (composition + target CN + proliferation). AUTHORED example: $CDH1 \hat{A} \dagger' -0.07787$
cptac_t105_agg_rho_rna_raw GENE SIGNED $\hat{A}^{\prime 1} \hat{A} \leq 1$ 60%	Spearman coupling of the $\hat{I}^2$-WEIGHTED AGGREGATE of the gene's arms against mRNA in the TCGA-OVERLAP cohort (n=105), UNADJUSTED $\hat{A}$ no confounder block. Range -0.636 $\hat{A}$ +0.54, 60% filled. AUTHORED example: $CDH1 \hat{A} \dagger' -0.1085$
cptac_t105_n_arms GENE COUNT 85%	Arms behind the TCGA-105 aggregate (median 2, max 57). See the prospective twin. AUTHORED example: $CDH1 \hat{A} \dagger' 21$ <i>Defined on: genes/edges present in the CPTAC cohort</i>

FRAC_ $\hat{A}$ 3

reference $\hat{A}$ context, provenance and rarely-quoted detail

3 columns $\hat{A}$ rung gene $\hat{A}$ coverage 60$\hat{A}$100% (median 82%) 3x fraction $0 \hat{A} 1$	
frac_edges_realized GENE FRACTION $0 \hat{A} 1$ 82%	Fraction of the gene's scored edges that realize a coupling below the threshold. AUTHORED example: $CDH1 \hat{A} \dagger' 0.7059$
frac_gain_true_acquired GENE FRACTION $0 \hat{A} 1$ 60%	Fraction of the gene's acquired-repression calls that survive the healthy-leg triage $\hat{A}$ i.e. are not <code>collapse_blind</code> artifacts. AUTHORED example: $CDH1 \hat{A} \dagger' 0$
frac_identified GENE FRACTION $0 \hat{A} 1$ 100%	Fraction of this gene's families reaching $z > 2$ on the UNCALIBRATED SD. AUTHORED example: $CDH1 \hat{A} \dagger' 0.16$

DOMINANT_ $\hat{A}$ 4

reference $\hat{A}$ context, provenance and rarely-quoted detail

4 columns $\hat{A}$ rung gene $\hat{A}$ coverage 77$\hat{A}$83% (median 82%) 3x identifier $\hat{A}$ 1x categorical	
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dominant_HLY GENE IDENTIFIER 77%	The arm carrying the largest share of the gene's HEALTHY budget. Range 280 distinct, 77% filled. AUTHORED example: <code>CDH1</code> <code>hsa-miR-92a-3p</code>
dominant_NAT GENE IDENTIFIER 81%	The arm dominating its NAT budget. Range 309 distinct, 81% filled. AUTHORED example: <code>CDH1</code> <code>hsa-miR-30a-5p</code>
dominant_TUM GENE IDENTIFIER 82%	The arm carrying the largest share of the gene's tumour pressure budget (miR-21-5p on 56 genes, miR-200c-3p next). AUTHORED example: <code>CDH1</code> <code>hsa-miR-30a-5p</code>
dominant_edge_shift_class GENE CATEGORICAL 83%	How the gene's dominant edge moved healthy→tumour. Range 10 levels, 83% filled. AUTHORED example: <code>CDH1</code> <code>uncoupled</code>

STATIC_ 1

reference context, provenance and rarely-quoted detail

1 columns <code>rung</code> <code>gene</code> coverage 87%87% (median 87%) 1x identifier

static_owner_family GENE IDENTIFIER 87%	The static (dose-based) owner of the gene, for comparison against the realization owner. AUTHORED example: <code>CDH1</code> <code>miR-25-3p/32-5p/92-3p/363-3p/367-3p</code> <i>Defined on: MULTI-FAMILY genes only (an owner needs >1 family to choose between)</i>
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TOTAL_ 1

reference context, provenance and rarely-quoted detail

1 columns <code>rung</code> <code>gene</code> coverage 100%100% (median 100%) 1x real 0
--

total_pressure GENE REAL 0 100%	Sum over the gene's regulators. AUTHORED example: <code>CDH1</code> 1.353
--	--

OWNER_ 1

reference context, provenance and rarely-quoted detail

1 columns <code>rung</code> <code>gene</code> coverage 41%41% (median 41%) 1x boolean
--

owner_agrees GENE BOOLEAN 41%	Whether the realization owner and the static owner agree. Defined only where both exist. AUTHORED values: False (385) True (253) example: <code>CDH1</code> <code>True</code> <i>Defined on: MULTI-FAMILY genes only (an owner needs >1 family to choose between)</i>
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MEDIAN_ 1

reference " context, provenance and rarely-quoted detail

1 columns 1 rung gene coverage 100%100% (median 100%)
1x real 0

median_retention

GENE REAL 0 100%

A per-row median of a lower-rung quantity; check agg_of . AUTHORED
example: CDH1 0.9148

ARMRES_ 7

reference " context, provenance and rarely-quoted detail

7 columns 1 rung gene coverage 18%18% (median 18%)
2x signed 1 2x count 1x categorical 1x fraction 1 1x real 0

armres_best_drho

GENE SIGNED 1 18%

The largest arm-resolution gain over the family fit. Range -0.206 +0.0421, 18% filled. AUTHORED
example: CDH1 -0.004735

armres_class

GENE CATEGORICAL 18%

arm_modelable (every multi-arm cell resolvable, 68 genes) partial (37) family_only (no cell resolvable, 168). Absent = the gene has no multi-arm cell at all, so the question is undefined. AUTHORED
values: family_only (168) arm_modelable (68) partial (37)
example: CDH1 arm_modelable

armres_frac_resolvable

GENE FRACTION 1 18%

Fraction of this gene's multi-arm cells where the arms are separable. AUTHORED
example: CDH1 1

armres_med_drho

GENE SIGNED 1 18%

Median of_drho over the gene's multi-arm cells " negative means arm resolution predicts repression better out of fold. AUTHORED
example: CDH1 -0.004735

armres_med_sep_z

GENE REAL 0 18%

Median separation z between same-cell arms. Range 0.0536 18.5, 18% filled. AUTHORED
example: CDH1 3.196

armres_n_arms_split

GENE COUNT 18%

How many arms sit in those multi-arm cells. Range 2 40, 18% filled. AUTHORED
example: CDH1 2

armres_n_multi_cells

GENE COUNT 18%

How many of the gene's family cells hold >1 arm. Range 1 14, 18% filled. AUTHORED
example: CDH1 1

IDENTITY_ 1

reference " context, provenance and rarely-quoted detail

1 columns 1 rung gene coverage 87%87% (median 87%)
1x boolean

identity_eq_magnitude

GENE BOOLEAN AGG FAMILY
87%

Do the identity and magnitude answers name the SAME top family. Exactly 100% agreement at n_fam==1 by construction; 76% at n_fam>=2. Read it only on multi-family genes. AUTHORED
values: True (1,161) False (191)
example: CDH1 True

GREAL_ 5 10

reference “ context, provenance and rarely-quoted detail

10 columns 5 rung gene 5 coverage 29%83% (median 83%)

7x count 5 2x signed 51 5 1x fraction 01

greal_best_coupling

GENE

SIGNED 51 51

83%

The STRONGEST (most negative) realized coupling over this gene's scored regulators.

AUTHORED

example: CDH1 5 0.271

greal_frac_c10

GENE

FRACTION 01

29%

Gene-rung twin of real_frac_c10 “ defined on 446 genes against greal_n_c10's 1,289; gated-out genes have a median greal_n_scored of 1. Exact on 100%.

AUTHORED

example: CDH1 5 0.4

Defined on: genes with 51 calibrated-coupling-scored regulator (1,260) “ the GENE-side ladder; rates gated n53

greal_med_coupling

GENE

SIGNED 51 51

83%

The MEDIAN realized coupling over this gene's scored regulators. Range -0.503 “ +0.298, 83% filled.

AUTHORED

example: CDH1 5 0.083

Defined on: genes with 51 calibrated-coupling-scored regulator (1,260) “ the GENE-side ladder; rates gated n53

greal_n_c05

GENE

COUNT

83%

How many of this gene's scored regulators couple below 50.05.

AUTHORED

example: CDH1 5 7

greal_n_c10

GENE

COUNT

83%

How many of this gene's scored regulators couple below 50.10.

AUTHORED

example: CDH1 5 4

greal_n_c15

GENE

COUNT

83%

How many of this gene's scored regulators couple below 50.15.

AUTHORED

example: CDH1 5 2

greal_n_c20

GENE

COUNT

83%

How many of this gene's scored regulators couple below 50.20.

AUTHORED

example: CDH1 5 2

greal_n_c30

GENE

COUNT

83%

How many of this gene's scored regulators couple below 50.30.

AUTHORED

example: CDH1 5 0

greal_n_scored

GENE

COUNT

83%

Genes/edges scored for realized coupling (median 2, max 59) “ the DENOMINATOR of every greal_frac_*. “... The greal_n_c05 “ n_c30 ladder is properly nested: 0 monotonicity violations and 0 rows where a threshold count exceeds n_scored (verified).

AUTHORED

example: CDH1 5 10

Defined on: genes with 51 calibrated-coupling-scored regulator (1,260) “ the GENE-side ladder; rates gated n53

greal_n_sig

GENE

COUNT

83%

How many of this gene's scored regulators reach significance.

AUTHORED

example: CDH1 5 2

HEUR_ 5 5

reference “ context, provenance and rarely-quoted detail

5 columns 5 rung gene 5 coverage 89%91% (median 91%)

2x signed 51 51 5 1x count 5 1x boolean 5 1x categorical

heur_delta_tumor_nat

GENE

SIGNED 51 51

89%

Tumour-minus-NAT change in the heuristic pressure correlation.

AUTHORED

example: CDH1 5 0.03725

heur_n_regulators GENE COUNT 91%	âššâšš A DIFFERENT PIPELINE'S COUNT â€” not the model's. It is <code>edges.groupby('gene')['miRNA'].nunique()</code> inside <code>analyses/misc/mirna_comovement.py</code>, i.e. AUTHORED example: <code>CDH1</code> â†’ 12
heur_net_repressed_tumor GENE BOOLEAN 91%	Is the gene net-repressed in tumour, per the RETIRED heuristic lane. True for 282 of 1,409 genes. AUTHORED values: False (1,127) âˆˆ True (282) example: <code>CDH1</code> â†’ False
heur_repression_class GENE CATEGORICAL 91%	Cross-state repression class from the RETIRED heuristic lane (never_repressed / lost_repression / gained_repression / constitutive_repressed). 867 of 1,409 genes are <code>never_repressed</code>. AUTHORED values: never_repressed (867) âˆˆ lost_repression (258) âˆˆ gained_repression (167) âˆˆ constitutive_repressed (115) âˆˆ na (2) example: <code>CDH1</code> â†’ never_repressed
heur_rho_pressure_tumor GENE SIGNED $\hat{r}^1_{\rho}$ 91%	Spearman of the gene's HEURISTIC aggregate pressure against its mRNA in tumour. AUTHORED example: <code>CDH1</code> â†’ 0.05168

ACQUIRED_VS_ âˆˆ 2

reference â€” context, provenance and rarely-quoted detail

2 columns âˆˆ rung gene âˆˆ coverage 91%â€”91% (median 91%) 2x real, signed

acquired_vs_gtex GENE REAL, SIGNED 91%	Repression GAINED relative to the GTEx healthy baseline. Range -11.6 â€” 7.56, 91% filled. AUTHORED example: <code>CDH1</code> â†’ 0.04213
acquired_vs_nat GENE REAL, SIGNED 91%	Repression gained relative to matched NAT â€” the same-platform comparison, and the one to prefer. Range -4.24 â€” 2.83, 91% filled. AUTHORED example: <code>CDH1</code> â†’ 0.3754

PRESSURE_ âˆˆ 1

reference â€” context, provenance and rarely-quoted detail

1 columns âˆˆ rung gene âˆˆ coverage 91%â€”91% (median 91%) 1x real $\hat{\rho}$

pressure_tumor GENE REAL $\hat{\rho}$ 91%	Aggregate incoming pressure on the gene in the named state. AUTHORED example: <code>CDH1</code> â†’ 3.784
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TCGA_ âˆˆ 4

reference â€” context, provenance and rarely-quoted detail

4 columns âˆˆ rung gene âˆˆ coverage 89%â€”89% (median 89%) 3x signed $\hat{r}^1_{\rho}$ âˆˆ 1x count
--

tcga_abund_rho_rna GENE SIGNED $\hat{r}^1_{\rho}$ AGG ARM 89%	TCGA mRNA coupling of an UNWEIGHTED abundance sum â€” the reference $\hat{\rho}$ is judged against. Range -0.36 â€” +0.366, 89% filled. AUTHORED example: <code>CDH1</code> â†’ 0.006381
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tcga_agg_rho_rna GENE SIGNED $\hat{\rho}^2$ 1 AGG ARM 89%	Gene-level $\hat{\rho}^2$ -weighted aggregate coupling to mRNA in TCGA (median $\hat{\rho}^2$ 0.05) against <code>tcga_abund_rho_rna</code> 's $\hat{\rho}^2$ 0.02 the abundance-sum reference. AUTHORED example: <code>CDH1</code> $\hat{\rho}^2$ -0.1922
tcga_agg_rho_rna_raw GENE SIGNED $\hat{\rho}^2$ 1 89%	The $\hat{\rho}^2$ -weighted aggregate mRNA coupling, UNADJUSTED. Range -0.545 to +0.571 , 89% filled. AUTHORED example: <code>CDH1</code> $\hat{\rho}^2$ -0.1921
tcga_n_arms GENE COUNT 89%	Arms behind the TCGA aggregate. Range 1 to 62 , 89% filled. AUTHORED example: <code>CDH1</code> $\hat{\rho}^2$ 23

REALIZATION_ $\hat{\rho}^2$ 1

reference $\hat{\rho}^2$ context, provenance and rarely-quoted detail

1 columns $\hat{\rho}^2$ rung gene $\hat{\rho}^2$ coverage 42% to 42% (median 42%) 1x identifier	realization_owner GENE IDENTIFIER 42% Within-patient realization quantities $\hat{\rho}^2$ the score, its identity allocation, and who owns it. AUTHORED example: <code>CDH1</code> $\hat{\rho}^2$ <code>miR-25-3p/32-5p/92-3p/363-3p/367-3p</code> <i>Defined on: MULTI-FAMILY genes only (an owner needs >1 family to choose between)</i>
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DISC_ $\hat{\rho}^2$ 2

reference $\hat{\rho}^2$ context, provenance and rarely-quoted detail

2 columns $\hat{\rho}^2$ rung gene $\hat{\rho}^2$ coverage 3% to 100% (median 51%) 1x categorical $\hat{\rho}^2$ 1x count	disc_gold_families GENE CATEGORICAL 3% Which seed families this row's discovery-queue edges belong to, semicolon-separated. Usually ONE at the arm rung (an arm has one seed). AUTHORED example: <code>KLF3</code> $\hat{\rho}^2$ <code>miR-130-3p/301-3p/454-3p;miR-17-5p/20-5p/93-5p/106-5p/519-3p</code>
disc_n_gold_edges GENE COUNT 100%	How many of the 157 discovery-queue edges name this gene (max 4). AUTHORED values: 0 (1,510) $\hat{\rho}^2$ 1 (26) $\hat{\rho}^2$ 3 (5) $\hat{\rho}^2$ 2 (5) $\hat{\rho}^2$ 4 (3) example: <code>CDH1</code> $\hat{\rho}^2$ 0

MAX_ $\hat{\rho}^2$ 1

reference $\hat{\rho}^2$ context, provenance and rarely-quoted detail

1 columns $\hat{\rho}^2$ rung gene $\hat{\rho}^2$ coverage 100% to 100% (median 100%) 1x fraction 0 to 1	max_beta_frac_sd GENE FRACTION 0 to 1 100% Maximum over the row's constituent units. AUTHORED example: <code>CDH1</code> $\hat{\rho}^2$ 0.03738
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REGULATORY_ $\hat{\rho}^2$ 1

reference $\hat{\rho}^2$ context, provenance and rarely-quoted detail

1 columns $\hat{\rho}^2$ rung gene $\hat{\rho}^2$ coverage 100% to 100% (median 100%) 1x boolean	
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regulatory_handoff

GENEBOOLEAN100%

Does the gene's globally-dominant regulator differ between healthy and tumour OR between NAT and tumour. AUTHORED

values: False (1,221) True (328)

example: CDH1 True

MEAN_ 1

reference context, provenance and rarely-quoted detail

1 columns rung gene coverage 82%82% (median 82%)

1x real, signed

mean_dTarget

GENEREAL, SIGNED82%

A per-row mean of a lower-rung quantity; check agg_of for which rung it summarises. AUTHORED

example: CDH1 1.18

CONCENTRATION_ 1

reference context, provenance and rarely-quoted detail

1 columns rung gene coverage 53%53% (median 53%)

1x fraction 01

concentration_adj

GENEFRACTION 0153%

(concentration 1/n_fam) / (1 1/n_fam) [0,1] how concentrated the gene is RELATIVE to the most diffuse its own design allows. AUTHORED

example: CDH1 0.1573

FAM_ 1

reference context, provenance and rarely-quoted detail

1 columns rung gene coverage 81%81% (median 81%)

1x signed 11

fam_pooled_rho_adj

GENESIGNED 1181%

Seed-family structure the arm sits in membership, affinity and dominance within it. AUTHORED

example: CDH1 -0.096

gene_family_card.tsv

gene_family

One row per one (seed family, gene) pair. Keyed [gene, family] it cannot express a property of the family itself. That is the seed_family card's job.

61 columns 5,117 rows 12 blocks median coverage 97%

Contents 12 blocks, most important first

1. (bare) 8 CORE

2. n_3 SUPPORTING

3. ctx_16 SUPPORTING

4. beta_7 SUPPORTING

5. comp_16 REFERENCE

6. identity_4 REFERENCE
7. pip_2 REFERENCE

8. w_1 REFERENCE

9. m_1 REFERENCE

10. composition_1 REFERENCE

11. prior_1 REFERENCE

12. retention_1 REFERENCE

(BARE) 8

core the columns findings rest on

8 columns spans family, key this prefix groups columns living on DIFFERENT units coverage 95%100% (median 100%)
3x identifier 3x real 0 1x boolean 1x real, signed

arms FAMILY IDENTIFIER 100%	The arms collapsed into this (gene, family) cell, semicolon-separated (hsa-miR-17-5p;hsa-miR-20a-5p). AUTHORED example: CDH1 / miR-132-3p/212-3p â†’ hsa-miR-132-3p
beta FAMILY REAL 0 100%	FAMILY-rung coupling coefficient readouts.run(level='family') , the same-seed collapse APPLIED. AUTHORED example: CDH1 / miR-132-3p/212-3p â†’ 0.07163
family KEY IDENTIFIER 100%	KEY seed family of the regulator, as used by the family-rung fit. AUTHORED example: CDH1 â†’ miR-132-3p/212-3p
gene KEY IDENTIFIER 100%	KEY HGNC gene symbol. AUTHORED example: miR-9-3p â†’ ITGB1
identified FAMILY BOOLEAN 100%	z > 2 on the UNCALIBRATED SD 27.1% of cells. cal_identified is the honest version. AUTHORED values: False (3,730) True (1,387) example: CDH1 / miR-132-3p/212-3p â†’ False
identity FAMILY REAL, SIGNED 95%	Shapley/LMG identity at the FAMILY rung same estimator as the edge card's identity, different unit. AUTHORED example: CDH1 / miR-132-3p/212-3p â†’ 0.0641
retention FAMILY REAL 0 100%	beta_deconv / beta_core at the family grain the ADJUSTMENT sense of retention (what fraction of the coupling survives the composition block), never the patient-baseline sense. AUTHORED example: CDH1 / miR-132-3p/212-3p â†’ 0.8516

z	beta / beta_sd for the family cell, on the UNCALIBRATED posterior SD. AUTHORED	
	FAMILY REAL $\hat{\mu}_0$ 100%	example: CDH1 / miR-132-3p/212-3p $\hat{\mu}$ 1.844
N_ 3 supporting $\hat{\mu}$ read these when the core raises a question		
3 columns $\hat{\mu}$ spans family, gene $\hat{\mu}$ this prefix groups columns living on DIFFERENT units $\hat{\mu}$ coverage 100%$\hat{\mu}$100% (median 100%) 2x count $\hat{\mu}$ 1x constant		
n_arms	How many arms of THIS family sit in THIS gene's design. $\hat{\mu}$... FAMILY rung, VERIFIED (2026-08-19): it varies within gene for 241 of 1,549 genes, and the per-gene SUM of these equals the gene card's n_arms exactly $\hat{\mu}$ a clean cross-rung consistency check. AUTHORED	
	FAMILY COUNT 100%	values: 1 (4,646) $\hat{\mu}$ 2 (338) $\hat{\mu}$ 3 (87) $\hat{\mu}$ 4 (25) $\hat{\mu}$ 5 (11) $\hat{\mu}$ 6 (6) $\hat{\mu}$ 7 (4) example: CDH1 / miR-132-3p/212-3p $\hat{\mu}$ 1
n_fam	How many families compete at this gene. beta is mathematically INERT at 1. AUTHORED	
	GENE COUNT 100%	example: CDH1 / miR-132-3p/212-3p $\hat{\mu}$ 25
n_samples	Samples the family cell was fit on $\hat{\mu}$ CONSTANT 1,040 across the card. AUTHORED	
	GENE CONSTANT 100%	values: 1040 (5,117) example: CDH1 / miR-132-3p/212-3p $\hat{\mu}$ 1040
CTX_ 16 supporting $\hat{\mu}$ read these when the core raises a question		
16 columns $\hat{\mu}$ rung gene $\hat{\mu}$ SUBSET of the edge block, identical rungs $\hat{\mu}$ coverage 81%$\hat{\mu}$100% (median 96%) 4x signed $\hat{\mu}$1$\hat{\mu}$1 $\hat{\mu}$ 3x real $\hat{\mu}$$\hat{\mu}_0$ $\hat{\mu}$ 3x real, signed $\hat{\mu}$ 3x count $\hat{\mu}$ 1x categorical $\hat{\mu}$ 1x fraction 0$\hat{\mu}$1 $\hat{\mu}$ 1x boolean		
ctx_apriori_class	The a-priori measurability class assigned to the gene's design before any fit (gene_atlas via card_context). Range 4 levels , 100% filled. AUTHORED	
	GENE CATEGORICAL 100%	values: A_COMPETENT (3,402) $\hat{\mu}$ D_UNDEFINED (730) $\hat{\mu}$ B_PARTIAL (590) $\hat{\mu}$ C_WEAK (395) example: CDH1 / miR-132-3p/212-3p $\hat{\mu}$ A_COMPETENT
ctx_ceiling	The UPPER BOUND on what ANY estimator can achieve for this gene $\hat{\mu}$ the 5-fold out-of-fold R^2 of an UNRESTRICTED OLS on the gene's real design (card_context.py:15). AUTHORED	
	GENE SIGNED $\hat{\mu}$1$\hat{\mu}$1 99%	example: CDH1 / miR-132-3p/212-3p $\hat{\mu}$ 0.1535
ctx_collapse	n_arm / n_fam for the gene $\hat{\mu}$ how much curation BUNDLES same-seed mates onto it (1 $\hat{\mu}$ 5). AUTHORED	
	GENE REAL $\hat{\mu}_0$ 96%	example: CDH1 / miR-132-3p/212-3p $\hat{\mu}$ 1.091
ctx_d_collin	Mean pairwise COLLINEARITY among the gene's regulators $\hat{\mu}$ how much they move together (gene_atlas via card_context). Range -0.699 $\hat{\mu}$ +0.495 , 81% filled. AUTHORED	
	GENE SIGNED $\hat{\mu}$1$\hat{\mu}$1 81%	example: CDH1 / miR-132-3p/212-3p $\hat{\mu}$ -0.02497
ctx_d_fam	Design width at FAMILY grain (gene_atlas via card_context). Range -2 $\hat{\mu}$ 25 , 95% filled. AUTHORED	
	GENE REAL, SIGNED 95%	example: CDH1 / miR-132-3p/212-3p $\hat{\mu}$ 0

ctx_dose_max GENE REAL $\hat{A} \geq 0$ 96%	The LARGEST regulator dose in the gene's design (<code>gene_atlas</code> via <code>card_context</code>). Range 0.215 $\hat{A} \leq$ 17.8, 96% filled. AUTHORED example: <code>CDH1 / miR-132-3p/212-3p</code> $\hat{A} \uparrow$ <code>15.38</code>
ctx_dose_med GENE REAL $\hat{A} \geq 0$ 96%	The MEDIAN regulator dose in the gene's design (<code>gene_atlas</code> via <code>card_context</code>). Range 0.215 $\hat{A} \leq$ 17.8, 96% filled. AUTHORED example: <code>CDH1 / miR-132-3p/212-3p</code> $\hat{A} \uparrow$ <code>6.318</code>
ctx_frac_abund GENE FRACTION $0 \hat{A} \leq 1$ 96%	FRACTION of the gene's regulators clearing the abundance gate (<code>gene_atlas</code> via <code>card_context</code>). Range 0 $\hat{A} \leq$ 1, 96% filled. AUTHORED example: <code>CDH1 / miR-132-3p/212-3p</code> $\hat{A} \uparrow$ <code>0.5833</code>
ctx_gap_core GENE SIGNED $\hat{A} \wedge 1 \hat{A} \leq 1$ 95%	The $\hat{I}^2$-vs-decoy MARGIN under the CORE confounder block (<code>gene_atlas</code> via <code>card_context</code>). Range -0.476 $\hat{A} \leq$ +0.541, 95% filled. AUTHORED example: <code>CDH1 / miR-132-3p/212-3p</code> $\hat{A} \uparrow$ <code>-0.2714</code>
ctx_gap_d_dose GENE REAL, SIGNED 95%	Difference in dose between the real design and its decoy. Range -5.41 $\hat{A} \leq$ 1.74, 95% filled. AUTHORED example: <code>CDH1 / miR-132-3p/212-3p</code> $\hat{A} \uparrow$ <code>0.1238</code>
ctx_gap_deconv GENE SIGNED $\hat{A} \wedge 1 \hat{A} \leq 1$ 95%	The same margin under the DECONVOLUTION block (<code>gene_atlas</code> via <code>card_context</code>). Range -0.33 $\hat{A} \leq$ +0.374, 95% filled. AUTHORED example: <code>CDH1 / miR-132-3p/212-3p</code> $\hat{A} \uparrow$ <code>-0.1864</code>
ctx_gap_retention GENE REAL, SIGNED 92%	<code>gap_deconv / gap_core</code> $\hat{A} \leq$ how much of the decoy margin survives adjustment (<code>gene_atlas</code> via <code>card_context</code>). Range -16.5 $\hat{A} \leq$ 17, 92% filled. AUTHORED example: <code>CDH1 / miR-132-3p/212-3p</code> $\hat{A} \uparrow$ <code>0.6867</code>
ctx_measurable GENE BOOLEAN 99%	Is the gene's <code>ctx_ceiling</code> above <code>CEILING_ROBUST</code> (0.02, not 0 $\hat{A} \leq$ MH-144)? AUTHORED values: <code>True</code> (3,652) $\hat{A}$ <code>False</code> (1,432) example: <code>CDH1 / miR-132-3p/212-3p</code> $\hat{A} \uparrow$ <code>True</code>
ctx_n_abund GENE COUNT 96%	How many of the gene's regulators clear the abundance gate (<code>gene_atlas</code> via <code>card_context</code>). Range 0 $\hat{A} \leq$ 53, 96% filled. AUTHORED example: <code>CDH1 / miR-132-3p/212-3p</code> $\hat{A} \uparrow$ <code>7</code>
ctx_n_fam_abund GENE COUNT 96%	How many families clear the abundance gate. Range 0 $\hat{A} \leq$ 34, 96% filled. AUTHORED example: <code>CDH1 / miR-132-3p/212-3p</code> $\hat{A} \uparrow$ <code>6</code>
ctx_n_fam_multi GENE COUNT 96%	How many of the gene's families hold MORE THAN ONE arm (<code>gene_atlas</code> via <code>card_context</code>). Range 0 $\hat{A} \leq$ 14, 96% filled. AUTHORED example: <code>CDH1 / miR-132-3p/212-3p</code> $\hat{A} \uparrow$ <code>1</code>

BETA_ $\hat{A}$ 7

supporting $\hat{A} \leq$ read these when the core raises a question

7 columns $\hat{A}$ <code>rung</code> family $\hat{A}$ <code>coverage</code> 100$\hat{A} \leq$ 100% (median 100%) 4x <code>fraction</code> $0 \hat{A} \leq 1$ $\hat{A}$ 3x <code>real</code> $\hat{A} \geq 0$	
beta_deconv FAMILY REAL $\hat{A} \geq 0$ 100%	The same $\hat{I}^2$ refit under the DECONVOLUTION confounder block. AUTHORED example: <code>CDH1 / miR-132-3p/212-3p</code> $\hat{A} \uparrow$ <code>0.061</code>
beta_frac FAMILY FRACTION $0 \hat{A} \leq 1$ 100%	$\hat{I}^2_f / \hat{I}^2$ averaged over Gibbs DRAWS $\hat{A} \leq$ E[ratio]. AUTHORED example: <code>CDH1 / miR-132-3p/212-3p</code> $\hat{A} \uparrow$ <code>0.05285</code>

beta_frac_abs FAMILY FRACTION 0Å€"1 100%	E[Î²_f] / Î£ E[Î²] Å€" the share from POINT ESTIMATES. AUTHORED example: CDH1 / miR-132-3p/212-3p â†' 0.05296
beta_frac_deconv FAMILY FRACTION 0Å€"1 100%	The share recomputed under the deconvolution block. AUTHORED example: CDH1 / miR-132-3p/212-3p â†' 0.05107
beta_frac_sd FAMILY FRACTION 0Å€"1 100%	Posterior SD of the per-draw share. AUTHORED example: CDH1 / miR-132-3p/212-3p â†' 0.02825
beta_sd FAMILY REAL Å¥ 0 100%	Posterior SD of Î² on the UNCALIBRATED scale. AUTHORED example: CDH1 / miR-132-3p/212-3p â†' 0.03885
beta_sd_deconv FAMILY REAL Å¥ 0 100%	Posterior SD of the deconv-block Î². AUTHORED example: CDH1 / miR-132-3p/212-3p â†' 0.03579

COMP_ Å· 16

reference Å€" context, provenance and rarely-quoted detail

16 columns Å· rung gene Å· coverage 66%Å€"97% (median 81%) 6× categorical Å· 6× signed Å'^1Å€ 1 Å· 3× real, signed Å· 1× real Å¥ 0	
comp_cptac_prot_driver GENE CATEGORICAL 66%	WHICH deconvolved compartment most explains this gene's PROTEIN coupling Å€" prolif 219 Å· CAFs 173 Å· purity 140 Å· PVL 77 Å· Endothelial 67. AUTHORED example: CDH1 / miR-132-3p/212-3p â†' T-cells
comp_cptac_prot_driver_ret GENE REAL, SIGNED 66%	1 Å'^ comp_cptac_prot_driver_share, exactly. Median +0.398 raw, +0.469 gated to Î_r Å‰ 0.10 Å€" quote the gated value. See the mRNA twin. AUTHORED example: CDH1 / miR-132-3p/212-3p â†' 0.5085 Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)
comp_cptac_prot_driver_share GENE REAL, SIGNED 66%	Protein-layer twin of comp_tcga_mrna_driver_share . Same exact complementarity with _driver_ret (8.9e-16 on 918 rows). AUTHORED example: CDH1 / miR-132-3p/212-3p â†' 0.4915
comp_cptac_prot_rho_raw GENE SIGNED Å'^1Å€ 1 81%	Unadjusted aggregate PROTEIN coupling Å€" denominator of comp_cptac_prot_driver_ret . AUTHORED example: CDH1 / miR-132-3p/212-3p â†' -0.1416
comp_cptac_prot_top_budget GENE CATEGORICAL 81%	The compartment the budget loads on most (protein). Range 10 levels, 81% filled. AUTHORED example: CDH1 / miR-132-3p/212-3p â†' Normal Epithelial Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)
comp_cptac_prot_top_budget_r GENE SIGNED Å'^1Å€ 1 81%	As the TCGA mRNA twin, on CPTAC protein Å€" the budget-vs-compartment correlation. Median Å'^0.118, i.e. it runs the OPPOSITE way to mRNA (+0.121). AUTHORED example: CDH1 / miR-132-3p/212-3p â†' 0.3126 Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)
comp_cptac_prot_top_target GENE CATEGORICAL 81%	The compartment the target loads on most (protein). Range 10 levels, 81% filled. AUTHORED example: CDH1 / miR-132-3p/212-3p â†' T-cells Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)

comp_cptac_prot_top_target_r GENE SIGNED $\hat{A}^{\sim 1}$ 81%	As the TCGA mRNA twin, on CPTAC protein $\hat{A}^{\sim}$ the target-vs-compartment correlation. Median +0.203. AUTHORED example: CDH1 / miR-132-3p/212-3p $\hat{A}^{\dagger}$ -0.2385 <i>Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)</i>
comp_tcga_mrna_driver GENE CATEGORICAL 81%	WHICH compartment drives the gene's mRNA coupling $\hat{A}^{\sim}$ the one whose removal moves it most (CAFs 279 $\hat{A}^{\sim}$ purity 174 $\hat{A}^{\sim}$ prolif 148 $\hat{A}^{\sim}$ Endothelial 122 $\hat{A}^{\sim}$ Myeloid 109). AUTHORED example: CDH1 / miR-132-3p/212-3p $\hat{A}^{\dagger}$ T-cells
comp_tcga_mrna_driver_ret GENE REAL, SIGNED 81%	1 $\hat{A}^{\sim}$ comp_tcga_mrna_driver_share , exactly (see there). The RETENTION orientation: what fraction of the coupling survives removing the driver compartment. AUTHORED example: CDH1 / miR-132-3p/212-3p $\hat{A}^{\dagger}$ 0.8681
comp_tcga_mrna_driver_share GENE REAL $\hat{A}^{\sim} \neq 0$ 81%	Share of the gene's mRNA coupling attributable to the single compartment that most changes it $\hat{A}^{\sim}$ axiom 8's composition GATE, the axis that separates the harm tails at p=0.00999 where the ensemble axis gives p=0.10. AUTHORED example: CDH1 / miR-132-3p/212-3p $\hat{A}^{\dagger}$ 0.1319
comp_tcga_mrna_rho_raw GENE SIGNED $\hat{A}^{\sim 1}$ 97%	The gene's UNADJUSTED aggregate mRNA coupling $\hat{A}^{\sim}$ the DENOMINATOR of comp_tcga_mrna_driver_ret . AUTHORED example: CDH1 / miR-132-3p/212-3p $\hat{A}^{\dagger}$ -0.1921
comp_tcga_mrna_top_budget GENE CATEGORICAL 97%	The compartment the gene's miRNA BUDGET loads on most $\hat{A}^{\sim}$ a property of the REGULATORS. AUTHORED example: CDH1 / miR-132-3p/212-3p $\hat{A}^{\dagger}$ Normal Epithelial
comp_tcga_mrna_top_budget_r GENE SIGNED $\hat{A}^{\sim 1}$ 97%	Correlation between the gene's miRNA PRESSURE BUDGET and the fraction of the compartment that budget loads on most (comp_tcga_mrna_top_budget), in TCGA mRNA. AUTHORED example: CDH1 / miR-132-3p/212-3p $\hat{A}^{\dagger}$ 0.2545
comp_tcga_mrna_top_target GENE CATEGORICAL 97%	The compartment the TARGET GENE's expression loads on most. AUTHORED example: CDH1 / miR-132-3p/212-3p $\hat{A}^{\dagger}$ prolif
comp_tcga_mrna_top_target_r GENE SIGNED $\hat{A}^{\sim 1}$ 97%	Correlation between the TARGET GENE's own expression and the fraction of the compartment it loads on most. AUTHORED example: CDH1 / miR-132-3p/212-3p $\hat{A}^{\dagger}$ 0.3289

IDENTITY_ $\hat{A}^{\sim}$ 4

reference $\hat{A}^{\sim}$ context, provenance and rarely-quoted detail

4 columns $\hat{A}^{\sim}$ rung family $\hat{A}^{\sim}$ coverage 95%$\hat{A}^{\sim}$97% (median 95%) 2x fraction $\hat{A}^{\sim}$1 $\hat{A}^{\sim}$ 1x real, signed $\hat{A}^{\sim}$ 1x boolean	
identity_abs FAMILY FRACTION $\hat{A}^{\sim}$1 95%	identity / $\hat{A}^{\sim}$ identity per gene $\hat{A}^{\sim}$ the BOUNDED companion to identity , always in [0,1] (verified: 0 non-NaN values outside it on either card). AUTHORED example: CDH1 / miR-132-3p/212-3p $\hat{A}^{\dagger}$ 0.063
identity_coherence FAMILY FRACTION $\hat{A}^{\sim}$1 95%	1 / $\hat{A}^{\sim}$ identity per GENE, in (0,1] $\hat{A}^{\sim}$ identity's own sign coherence. 1.0 when nothing cancels, $\hat{A}^{\dagger}$ 0 as suppressor cancellation grows. AUTHORED example: CDH1 / miR-132-3p/212-3p $\hat{A}^{\dagger}$ 0.983

identity_deconv FAMILY REAL, SIGNED 97%	Shapley/LMG attribution on R^2 with bagged-NNLS weights " WHO regulates the gene, collinearity-fair. AUTHORED example: CDH1 / miR-132-3p/212-3p " 0.04796
identity_reliable FAMILY BOOLEAN 95%	identity_coherence " 0.5 AND identity " 1. Admits 93.3% of edge rows and 92.6% of gene_family rows, and excludes every row whose identity exceeds 1 (119/119 and 118/118). AUTHORED values: True (4,740) " False (146) example: CDH1 / miR-132-3p/212-3p " True

PIP_ " 2
reference " context, provenance and rarely-quoted detail

2 columns " rung family " coverage 100%"100% (median 100%) 1x constant " 1x fraction 0"1	
pip_dense FAMILY CONSTANT 100%	" CONSTANT by construction ("1 dense readout). AUTHORED values: 1.0 (5,117) example: CDH1 / miR-132-3p/212-3p " 1
pip_discovery FAMILY FRACTION 0"1 100%	Posterior inclusion probability. pip_dense comes from the pi=1 coupling readout; pip_discovery from the evidence-pi readout. AUTHORED example: CDH1 / miR-132-3p/212-3p " 0.3261

W_ " 1
reference " context, provenance and rarely-quoted detail

1 columns " rung gene " coverage 100%"100% (median 100%) 1x real " 0	
w_max GENE REAL " 0 100%	Maximum curated EVIDENCE weight for this (gene, family) cell. Family-rung here; gene-rung on the gene card. AUTHORED example: CDH1 / miR-132-3p/212-3p " 13.24

M_ " 1
reference " context, provenance and rarely-quoted detail

1 columns " rung family " coverage 100%"100% (median 100%) 1x real " 0	
m_nnl FAMILY REAL " 0 100%	Bagged-NNLS weight for this (gene, family) cell. AUTHORED example: CDH1 / miR-132-3p/212-3p " 0.07211

COMPOSITION_ " 1
reference " context, provenance and rarely-quoted detail

1 columns " rung family " coverage 100%"100% (median 100%) 1x categorical	
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composition_class

FAMILY CATEGORICAL 100%

Composition-adjustment verdict for this row (e.g. composition_explained when the effect does not survive the deconvolution block). AUTHORED

values: cell_intrinsic (3,378) partial (1,095) composition_explained (644)

example: CDH1 / miR-132-3p/212-3p cell_intrinsic

PRIOR_ 1

reference context, provenance and rarely-quoted detail

1 columns rung family coverage 100%100% (median 100%)

1x fraction 1

prior_pi

FAMILY FRACTION 1 100%

The evidence prior fed to the discovery readout. AUTHORED

example: CDH1 / miR-132-3p/212-3p 0.2962

RETENTION_ 1

reference context, provenance and rarely-quoted detail

1 columns rung family coverage 100%100% (median 100%)

1x boolean

retention_reliable

FAMILY BOOLEAN 100%

Fraction of an effect surviving covariate adjustment (adjusted / raw). AUTHORED

values: False (3,730) True (1,387)

example: CDH1 / miR-132-3p/212-3p False

arm

296 columns **2,450** rows **44** blocks median coverage **30%**

1. adm_ 4 Â· CORE	23. sub_ 6 Â· REFERENCE
2. arm_ 7 Â· CORE	24. cov_ 22 Â· REFERENCE
3. seed_ 1 Â· CORE	25. seq_ 4 Â· REFERENCE
4. arb_ 16 Â· SUPPORTING	26. iso_ 5 Â· REFERENCE
5. healthy_ 3 Â· SUPPORTING	27. cur_he_ 2 Â· REFERENCE
6. site_ 12 Â· SUPPORTING	28. field_ 5 Â· REFERENCE
7. fame_ 28 Â· SUPPORTING	29. bc_ 11 Â· REFERENCE
8. (bare) 1 Â· SUPPORTING	30. famrole_ 6 Â· REFERENCE
9. hly_ 7 Â· SUPPORTING	31. gctx_ 16 Â· REFERENCE
10. dose_ 3 Â· SUPPORTING	32. surv_ 8 Â· REFERENCE
11. ago_ 4 Â· SUPPORTING	33. aid_ 4 Â· REFERENCE
12. abund_ 5 Â· SUPPORTING	34. wshift_ 6 Â· REFERENCE
13. attr_ 14 Â· SUPPORTING	35. cnv_ 9 Â· REFERENCE
14. ctx_arm_ 2 Â· REFERENCE	36. tier_ 5 Â· REFERENCE
15. real_ 11 Â· REFERENCE	37. foc_ 7 Â· REFERENCE
16. grank_ 3 Â· REFERENCE	38. surrogate_ 2 Â· REFERENCE
17. ts_ 10 Â· REFERENCE	39. cnvc_ 5 Â· REFERENCE
18. isoc_ 5 Â· REFERENCE	40. disc_ 2 Â· REFERENCE
19. dGlobal_ 3 Â· REFERENCE	41. cload_ 3 Â· REFERENCE
20. chim_ 5 Â· REFERENCE	42. model_ 1 Â· REFERENCE
21. fst_ 16 Â· REFERENCE	43. comv_ 2 Â· REFERENCE
22. sdsz_ 4 Â· REFERENCE	44. hx_ 1 Â· REFERENCE

4 columns $\hat{\cdot}$ `rung arm` $\hat{\cdot}$ AGGREGATE of the edge block $\hat{\cdot}$ verified: `n_with_site` and `n_admissible` reproduce a direct rollup on 100% of arms (`n_edges` only 77.5%, the MH-252 universe gap) $\hat{\cdot}$ coverage **14% $\hat{\cdot}$ 29%** (median 29%) $\hat{\cdot}$ `3x count` $\hat{\cdot}$ `1x fraction` $\hat{\cdot}$ 1

example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 33

adm_n_edges ARM COUNT 29%	ADMISSIBILITY “ could this edge repress in breast AT ALL? has_site (seed match in the 3'UTR) AND expressed (arm present in TCGA tumour OR GTEx healthy breast). AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p “ 35
adm_n_with_site ARM COUNT 29%	Edges of this arm whose target carries a seed site “ 89.9% of edges (4,436/4,937) , so the site requirement is NOT the binding constraint; the evidence bar is (admissible drops it to 60.4%). 19.7% of arms have none. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p “ 33 Defined on: edges/arms with an admissibility tag (4,937 edges; has_site 89.9% ^ expressed 63.9% ^ admissible 60.4%)

ARM_ 7
core “ the columns findings rest on

7 columns ^ rung arm ^ coverage 15%“20% (median 20%) 4x real, signed ^ 2x real “ 0 ^ 1x percent	
arm_iqr ARM REAL “ 0 20%	Interquartile range of the arm's tumour RPM “ its DYNAMIC RANGE. AUTHORED example: hsa-miR-200c-3p / miR-200bc-3p/429 “ 1.1
arm_lfc_HLY_NAT_raw ARM REAL, SIGNED 15%	log-FC GTEx-healthy “ TCGA-NAT, raw. AUTHORED example: hsa-miR-200c-3p / miR-200bc-3p/429 “ 7.94
arm_lfc_HLY_TUM_QN ARM REAL, SIGNED 16%	log-FC GTEx-healthy “ TCGA-tumour, on quantile-normalised values. AUTHORED example: hsa-miR-200c-3p / miR-200bc-3p/429 “ 5.11
arm_lfc_HLY_TUM_raw ARM REAL, SIGNED 15%	As arm_lfc_HLY_TUM_QN but against the RAW GTEx median. AUTHORED example: hsa-miR-200c-3p / miR-200bc-3p/429 “ 10
arm_lfc_NAT_TUM ARM REAL, SIGNED 20%	log-FC of the arm's median abundance, matched NAT “ tumour. AUTHORED example: hsa-miR-200c-3p / miR-200bc-3p/429 “ 2.06
arm_med_rpm ARM REAL “ 0 20%	The arm's MEDIAN RPM across tumour samples “ its abundance level, gene-free. Defined on the 19.8% of arms with tumour expression data. AUTHORED example: hsa-miR-200c-3p / miR-200bc-3p/429 “ 13.4 Defined on: edges present in the progression panel (arm expressed + state measured)
arm_pct_above_floor ARM PERCENT 20%	“ THE NAME READS BACKWARDS . Computed as $100 \cdot \text{mean}(x > \text{FLOOR})$ “ the percent of samples ABOVE the detection floor, i.e. AUTHORED example: hsa-miR-200c-3p / miR-200bc-3p/429 “ 100
SEED_ 1 core “ the columns findings rest on 1 columns ^ rung arm ^ coverage 100%“100% (median 100%) 1x identifier	
seed_family ARM IDENTIFIER 100%	KEY “ the seed family itself, one row per family. AUTHORED example: hsa-let-7a-5p “ let-7-5p/98-5p

supporting “ read these when the core raises a question

```
16 columns  Â·  rung arm  Â·  coverage 28%â€³30% (median 30%)
12x count  Â·  2x real  â‰¥ 0  Â·  1x fraction 0â€³1  Â·  1x signed  âˆ'1â€³1
```

example: hsa-let-7a-5p / let-7-5p/98-5p $\hat{+}$ 0.1316

example: hsa-let-7a-5p / let-7-5p/98-5p $\hat{+}$ 0.2173

example: hsa-let-7a-5p / let-7-5p/98-5p $\hat{+}$ 1

example: hsa-let-7a-5p / let-7-5p/98-5p $\hat{\alpha}'$ 0.02837

example: hsa-let-7a-5p / let-7-5p/98-5p $\hat{+}$ ' 18

example: hsa-let-7a-5p / let-7-5p/98-5p $\hat{+}$ 5

example: hsa-let-7a-5p / let-7-5p/98-5p $\hat{+}$ 6

example: hsa-let-7a-5p / let-7-5p/98-5p $\hat{+}$ 5

example: hsa-let-7a-5p / let-7-5p/98-5p $\hat{+}$ 4

example: hsa-let-7a-5p / let-7-5p/98-5p $\hat{+}$ 4

example: hsa-let-7a-5p / let-7-5p/98-5p $\hat{+}$ 0

example: hsa-let-7a-5p / let-7-5p/98-5p $\hat{+}$ 0

example: hsa-let-7a-5p / let-7-5p/98-5p $\hat{+}$ 0

example: `hsa-let-7a-5p / let-7-5p/98-5p` $\hat{+}$ 38

example: hsa-let-7a-5p / let-7-5p/98-5p $\hat{+}$ 5

arb_n_identity_reliable ARM COUNT AGG EDGE 30%	Edges of this arm whose <code>identity</code> share is usable (<code>identity_reliable</code>). AUTHORED example: <code>hsa-let-7a-5p / let-7-5p/98-5p</code> $\hat{+}$ 33
---	---

HEALTHY_ 3

supporting $\hat{+}$ read these when the core raises a question

3 columns $\hat{+}$ <code>rung arm</code> $\hat{+}$ identical column set to the edge block $\hat{+}$ coverage 16%$\hat{+}$20% (median 16%) 1 $\times$ categorical $\hat{+}$ 1 $\times$ real $\hat{+}$ 0 $\hat{+}$ 1 $\times$ boolean

healthy_leg ARM CATEGORICAL 20%	Provenance of this arm's GTEx healthy leg $\hat{+}$ <code>measured</code> , <code>collapse_blind</code> or <code>true_absent</code> . AUTHORED values: <code>measured</code> (230) $\hat{+}$ <code>true_absent</code> (168) $\hat{+}$ <code>collapse_blind</code> (100) example: <code>hsa-miR-200c-3p / miR-200bc-3p/429</code> $\hat{+}$ <code>measured</code>
---	---

healthy_potential ARM REAL $\hat{+}$ 0 16%	The arm's IMPUTED healthy abundance $\hat{+}$ a repression-POTENTIAL proxy, collapse-repaired (0.27 $\hat{+}$ 17.9). AUTHORED example: <code>hsa-miR-200c-3p / miR-200bc-3p/429</code> $\hat{+}$ 8.24
---	--

healthy_uninformative ARM BOOLEAN 16%	Is the healthy leg both <code>collapse_blind</code> AND high-potential $\hat{+}$ i.e. the one combination where an <i>*acquired*</i> call is genuinely unsafe? True on 143 of 3,490 edges (4%). AUTHORED values: <code>False</code> (392) $\hat{+}$ <code>True</code> (7) example: <code>hsa-miR-200c-3p / miR-200bc-3p/429</code> $\hat{+}$ <code>False</code>
---	---

SITE_ 12

supporting $\hat{+}$ read these when the core raises a question

12 columns $\hat{+}$ <code>rung arm</code> $\hat{+}$ coverage 29%$\hat{+}$29% (median 29%) 7 $\times$ count $\hat{+}$ 2 $\times$ fraction 0 $\hat{+}$ 1 $\hat{+}$ 2 $\times$ real, signed $\hat{+}$ 1 $\times$ real $\hat{+}$ 0
--

site_frac_8mer ARM FRACTION 0$\hat{+}$1 29%	8mer share of the arm's CANONICAL sites (<code>site_n_8mer / site_n_canonical</code> ; median 0.150). AUTHORED example: <code>hsa-let-7a-5p / let-7-5p/98-5p</code> $\hat{+}$ 0.2169
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site_frac_canonical ARM FRACTION 0$\hat{+}$1 29%	Canonical share of the arm's TOTAL sites (<code>site_n_canonical / site_n_total</code> ; median 0.054) $\hat{+}$ a different denominator from <code>site_frac_8mer</code> . AUTHORED example: <code>hsa-let-7a-5p / let-7-5p/98-5p</code> $\hat{+}$ 0.03913
---	--

site_n_6mer ARM COUNT AGG EDGE 29%	Count of 6mer sites for this arm in the MANE 3'UTR seed scan , scoped to Hallmark genes. AUTHORED example: <code>hsa-let-7a-5p / let-7-5p/98-5p</code> $\hat{+}$ 1459
---	---

site_n_7mer ARM COUNT AGG EDGE 29%	Count of 7mer sites for this arm in the MANE 3'UTR seed scan , scoped to Hallmark genes. AUTHORED example: <code>hsa-let-7a-5p / let-7-5p/98-5p</code> $\hat{+}$ 278
---	--

site_n_8mer ARM COUNT AGG EDGE 29%	Count of 8mer sites for this arm in the MANE 3'UTR seed scan , scoped to Hallmark genes. AUTHORED example: <code>hsa-let-7a-5p / let-7-5p/98-5p</code> $\hat{+}$ 77
---	---

site_n_canonical ARM COUNT AGG EDGE 29%	Count of CANONICAL sites (6/7/8mer) for this arm in the MANE 3'UTR seed scan , scoped to Hallmark genes. AUTHORED example: <code>hsa-let-7a-5p / let-7-5p/98-5p</code> $\hat{+}$ 355
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fame_assay_nonfunctional_mti_studies ARM COUNT 96%	Curated studies for this arm using nonfunctional_mti assays â€” a MARGINAL of the assay Ã— functional-class cross-tab, so it is the sum of that row's cells. Range 0 â€” 35, 96% of arms. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 13
fame_assay_other_functional_mti_studies ARM COUNT 96%	Curated studies for this arm that are BOTH assays outside the four named classes AND carry strong functional MTI evidence â€” a CELL of the assay Ã— functional-class cross-tab. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 4
fame_assay_other_functional_mti_weak_studies ARM COUNT 96%	Curated studies for this arm that are BOTH assays outside the four named classes AND carry weak functional MTI evidence â€” a CELL of the assay Ã— functional-class cross-tab. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 2
fame_assay_other_nonfunctional_mti_studies ARM COUNT 96%	Curated studies for this arm that are BOTH assays outside the four named classes AND carry explicitly NON-functional MTI evidence â€” a CELL of the assay Ã— functional-class cross-tab. AUTHORED values: 0.0 (2,225) âˆˆ 1.0 (77) âˆˆ 2.0 (17) âˆˆ 3.0 (10) âˆˆ 5.0 (6) âˆˆ 4.0 (4) âˆˆ 6.0 (3) âˆˆ 7.0 (1) example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 1
fame_assay_other_studies ARM COUNT 96%	Curated studies for this arm using assays outside the four named classes â€” a MARGINAL of the assay Ã— functional-class cross-tab, so it is the sum of that row's cells. Range 0 â€” 259, 96% of arms. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 6
fame_assay_protein_functional_mti_studies ARM COUNT 96%	Curated studies for this arm that are BOTH protein-level assays (western, ELISA) AND carry strong functional MTI evidence â€” a CELL of the assay Ã— functional-class cross-tab. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 28
fame_assay_protein_functional_mti_weak_studies ARM COUNT 96%	Curated studies for this arm that are BOTH protein-level assays (western, ELISA) AND carry weak functional MTI evidence â€” a CELL of the assay Ã— functional-class cross-tab. AUTHORED values: 0.0 (2,225) âˆˆ 1.0 (75) âˆˆ 2.0 (22) âˆˆ 3.0 (10) âˆˆ 4.0 (6) âˆˆ 5.0 (3) âˆˆ 10.0 (1) âˆˆ 25.0 (1) example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 0
fame_assay_protein_nonfunctional_mti_studies ARM COUNT 96%	Curated studies for this arm that are BOTH protein-level assays (western, ELISA) AND carry explicitly NON-functional MTI evidence â€” a CELL of the assay Ã— functional-class cross-tab. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 10
fame_assay_protein_studies ARM COUNT 96%	Curated studies for this arm using protein-level assays (western, ELISA) â€” a MARGINAL of the assay Ã— functional-class cross-tab, so it is the sum of that row's cells. Range 0 â€” 61, 96% of arms. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 31
fame_assay_reporter_functional_mti_studies ARM COUNT 96%	Curated studies for this arm that are BOTH reporter assays (luciferase) AND carry strong functional MTI evidence â€” a CELL of the assay Ã— functional-class cross-tab. Range 0 â€” 65, 96% of arms. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 31

fame_assay_reporter__nonfunctional_mti_studies ARM COUNT 96%	Curated studies for this arm that are BOTH reporter assays (luciferase) AND carry explicitly NON-functional MTI evidence â€” a CELL of the assay ðŸ“š functional-class cross-tab. AUTHORED example: <code>hsa-let-7a-5p / let-7-5p/98-5p</code> â†’ 10
fame_assay_reporter_studies ARM COUNT 96%	Curated studies for this arm using reporter assays (luciferase) â€” a MARGINAL of the assay ðŸ“š functional-class cross-tab, so it is the sum of that row's cells. Range 0 â€” 73, 96% of arms. AUTHORED example: <code>hsa-let-7a-5p / let-7-5p/98-5p</code> â†’ 34
fame_assay_rna__functional_mti_studies ARM COUNT 96%	Curated studies for this arm that are BOTH RNA-level assays (qPCR, microarray) AND carry strong functional MTI evidence â€” a CELL of the assay ðŸ“š functional-class cross-tab. AUTHORED example: <code>hsa-let-7a-5p / let-7-5p/98-5p</code> â†’ 22
fame_assay_rna__functional_mti_weak_studies ARM COUNT 96%	Curated studies for this arm that are BOTH RNA-level assays (qPCR, microarray) AND carry weak functional MTI evidence â€” a CELL of the assay ðŸ“š functional-class cross-tab. AUTHORED example: <code>hsa-let-7a-5p / let-7-5p/98-5p</code> â†’ 9
fame_assay_rna__nonfunctional_mti_studies ARM COUNT 96%	Curated studies for this arm that are BOTH RNA-level assays (qPCR, microarray) AND carry explicitly NON-functional MTI evidence â€” a CELL of the assay ðŸ“š functional-class cross-tab. AUTHORED example: <code>hsa-let-7a-5p / let-7-5p/98-5p</code> â†’ 5
fame_assay_rna_studies ARM COUNT 96%	Curated studies for this arm using RNA-level assays (qPCR, microarray) â€” a MARGINAL of the assay ðŸ“š functional-class cross-tab, so it is the sum of that row's cells. Range 0 â€” 638, 96% of arms. AUTHORED example: <code>hsa-let-7a-5p / let-7-5p/98-5p</code> â†’ 30
fame_assay_total_studies ARM COUNT 96%	Distinct STUDIES (<code>study_id.nunique()</code>) behind this arm's curated edges â€” median 57, max 930. See <code>fame_assay_unique_targets</code>: different definitions, one axis (spearman 1.0000). AUTHORED example: <code>hsa-let-7a-5p / let-7-5p/98-5p</code> â†’ 189
fame_assay_unique_targets ARM COUNT 96%	Distinct TARGET GENES this arm has in the curated literature (<code>gene.nunique()</code>). AUTHORED example: <code>hsa-let-7a-5p / let-7-5p/98-5p</code> â†’ 189
fame_led_frac_ltfunc ARM FRACTION 0â€”1 AGG EDGE 96%	Their share of the arm's publications â€” median 0.00. Range 0 â€” 1, 96% filled. AUTHORED example: <code>hsa-let-7a-5p / let-7-5p/98-5p</code> â†’ 0.4565 <i>Defined on: arms with miRTarBase curation / ledger PMIDs â€” â†’ FAME axes, not targetome (MH-208)</i>
fame_led_n_classes ARM COUNT AGG EDGE 96%	How many distinct evidence CLASSES this arm's curated edges span (1â€”7). AUTHORED values: 1.0 (1,466) â†” 5.0 (246) â†” 6.0 (241) â†” 4.0 (166) â†” 2.0 (137) â†” 3.0 (76) â†” 7.0 (18) example: <code>hsa-let-7a-5p / let-7-5p/98-5p</code> â†’ 6
fame_led_n_ltfunc_pmid ARM COUNT AGG EDGE 96%	Low-throughput FUNCTIONAL publications â€” the strongest rung. Range 0 â€” 197, 96% filled. AUTHORED example: <code>hsa-let-7a-5p / let-7-5p/98-5p</code> â†’ 42 <i>Defined on: arms with miRTarBase curation / ledger PMIDs â€” â†’ FAME axes, not targetome (MH-208)</i>
fame_led_n_weak ARM COUNT AGG EDGE 94%	Weak-evidence publications for this arm. Range 1 â€” 754, 94% filled. AUTHORED example: <code>hsa-let-7a-5p / let-7-5p/98-5p</code> â†’ 206 <i>Defined on: arms with miRTarBase curation / ledger PMIDs â€” â†’ FAME axes, not targetome (MH-208)</i>

fame_n_genes_curated

ARM

COUNT

AGG EDGE

96%

Distinct genes this arm has in the curated literature. Range **1** – **3615**, 96% filled.

AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p – 3216

Defined on: arms with miRTarBase curation / ledger PMIDs – FAME axes, not targetome (MH-208)

fame_npmid

ARM

COUNT

AGG EDGE

96%

Distinct PMIDs citing this arm.

AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p – 92

(BARE) – 1

supporting – read these when the core raises a question

1 columns – rung key – coverage 100 – 100% (median 100%)

1x identifier

arm

KEY

IDENTIFIER

100%

KEY – mature miRNA arm (e.g. hsa-miR-21-5p). Normalised through `_arm_key()`, which gates against the canonical universe and LOGS drops rather than silently dropping.

AUTHORED

example: miR-588 – hsa-miR-4701-5p

HLY_ – 7

supporting – read these when the core raises a question

7 columns – rung arm – coverage 17 – 91% (median 91%)

2x real – 1x categorical – 1x boolean – 1x percent – 1x count – 1x fraction – 1

hly_baseline_src

ARM

CATEGORICAL

91%

WHERE the healthy baseline came from – `floor0` (1,137) / `mited_anchor` (634) / `gtex` (417) / `gtex_family` (49). Range **4 levels**, 91% filled.

AUTHORED

values: floor0 (1,137) – mited_anchor (634) – gtex (417) – gtex_family (49)

example: hsa-let-7a-5p / let-7-5p/98-5p – mited_anchor

Defined on: arms with a GTEx healthy anchor – MEASURED leg only; floor0 is NEVER emitted as a value

hly_from_seedmate

ARM

BOOLEAN

91%

Was this arm's healthy baseline borrowed from a SEED-MATE rather than measured directly (49 arms)?

AUTHORED

values: False (2,188) – True (49)

example: hsa-let-7a-5p / let-7-5p/98-5p – False

hly_gtex_median

ARM

REAL – 0

17%

The arm's median GTEx healthy abundance. Range **0.0803** – **18.1**, 17% filled.

AUTHORED

example: hsa-miR-200c-3p / miR-200bc-3p/429 – 11.62

Defined on: arms with a GTEx healthy anchor – MEASURED leg only; floor0 is NEVER emitted as a value

hly_gtex_pct

ARM

PERCENT

17%

Its GTEx percentile. Range **0.0514** – **1**, 17% filled.

AUTHORED

example: hsa-miR-200c-3p / miR-200bc-3p/429 – 0.9636

Defined on: arms with a GTEx healthy anchor – MEASURED leg only; floor0 is NEVER emitted as a value

hly_nat_detect_n

ARM

COUNT

91%

In how many of the ~104 matched NAT samples this arm is detected above the floor.

AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p – 104

hly_nat_median

ARM

REAL – 0

74%

Median NAT abundance (fill 74.0%).

AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p – 15.09

hly_tumor_prev

ARM

FRACTION – 1

91%

Prevalence of the arm above the detection floor in tumour (median 0.01).

AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p – 1

supporting “ read these when the core raises a question

dose_comp_retention

Arm-rung version of the composition dose retention. Range **0.51** – **1.66**, 12% filled. AUTHORED

Defined on: edges with a matched NAT leg (n~104 paired participants)

Is this arm's NAT47 tumour dose a composition or proliferation artifact (either gated retention < 0.4)?

```
example: hsa-miR-200c-3p / miR-200bc-3p/429  +  False
```

Arm-rung version of the proliferation dose retention. Range **0.69** $\hat{=}$ **1.25**, 12% filled. AUTHORED

Defined on: edges with a matched NAT leg (n~104 paired participants)

supporting “ read these when the core raises a question

Fraction of the arm's Manakov AGO-CLIP reads assigned to it rather than its sibling arm – the loading dominance. Median 0.658 on the 27.7% of arms with CLIP coverage. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p` $\hat{+}$ 0.9937

Provenance of `ago_dom` â€” which CLIP dataset the AGO-loading dominance came from. **AUTHORED**

example: hsa-let-7a-5p / let-7-5p/98-5p â†’ manakov

AGO/RISC loading gate for the arm. AUTHORED

```
values: guide (288) Â· co_loaded (204) Â· passenger (186)
```

example: hsa-let-7a-5p / let-7-5p/98-5p â†’ guide

Raw AGO-CLIP read count (median 6, max 210,722). AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 9900

supporting " read these when the core raises a question

Fraction of samples where the arm clears the floor. Range 0 – 1, 91% filled. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p $\hat{+}$ 1

Defined on: arms detected in the TCGA miRNA matrix (2,236 of 2,450)

abund_iqr ARM REAL 0 91%	Its interquartile range. Range 0 to 4.59 , 91% filled. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p at 0.9309 Defined on: arms detected in the TCGA miRNA matrix (2,236 of 2,450)
abund_med ARM REAL 0 91%	The arm's median abundance across tumour samples. Range 0.105 to 17.9 , 91% filled. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p at 14.8 Defined on: arms detected in the TCGA miRNA matrix (2,236 of 2,450)
abund_sd ARM REAL 0 88%	Its dispersion AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p at 0.7103
abund_sd_tertile ARM CATEGORICAL 88%	Dispersion tertile 718 / 716 / 718 by construction , so it carries no information beyond abund_sd 's rank. AUTHORED values: high (718) · low (718) · mid (716) example: hsa-let-7a-5p / let-7-5p/98-5p at high

ATTR_ 14
supporting read these when the core raises a question

14 columns · rung arm · coverage **6% to 29%** (median 15%)
11x fraction 0 to 1 · 2x count · 1x signed -1 to 1

attr_max_budget_share ARM FRACTION 0 to 1 15%	Largest value of the arm's share of its gene's DOSE budget over this arm's cells. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p at 0.2604
attr_max_credit_share ARM FRACTION 0 to 1 6%	Largest value of the arm's share of attribution CREDIT over this arm's cells. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p at 0.8519
attr_max_joint_share ARM FRACTION 0 to 1 15%	Largest JOINT attribution share among this arm's edges. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p at 0.3167
attr_max_nested_share ARM FRACTION 0 to 1 15%	Largest NESTED attribution share. See attr_max_joint_share : different values (agree on 35.4%), one axis (spearman +0.962). AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p at 0.3167 Defined on: arms with a Shapley/attribution decomposition over their edges (728)
attr_max_within_share ARM FRACTION 0 to 1 15%	Largest WITHIN-cell attribution share for this arm. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p at 1
attr_med_budget_share ARM FRACTION 0 to 1 15%	Median value of the arm's share of its gene's DOSE budget over this arm's cells. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p at 0.1036
attr_med_credit_share ARM FRACTION 0 to 1 6%	Median value of the arm's share of attribution CREDIT over this arm's cells. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p at 0.2493
attr_med_dose_share ARM FRACTION 0 to 1 29%	Median dose share of this arm within its family cells. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p at 0.1582
attr_med_joint_share ARM FRACTION 0 to 1 15%	Median value of the JOINT (Shapley) share over this arm's cells. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p at 0.07153
attr_med_nested_share ARM FRACTION 0 to 1 15%	Median value of the NESTED share over this arm's cells. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p at 0.1393

attr_med_phi_joint ARM SIGNED $\hat{A}^{-1}\hat{A}\epsilon 1$ 15%	Median value of the joint Shapley $\hat{I}^{\dagger}$ over this arm's cells. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p $\hat{A}^{\dagger}$ 0.008336
attr_med_within_share ARM FRACTION $0\hat{A}\epsilon^{-1}$ 15%	Median value of the WITHIN-cell share over this arm's cells. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p $\hat{A}^{\dagger}$ 0.33
attr_n_edges ARM COUNT 22%	Attribution rollup over the arm's edges $\hat{\epsilon}$ how much of the gene's coupling this arm is credited with, and over how many reliable edges. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p $\hat{A}^{\dagger}$ 23 <i>Defined on: arms with a Shapley/attribution decomposition over their edges (728)</i>
attr_n_reliable ARM COUNT 15%	Reliable edges behind this arm's attr_* shares (median 2). AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p $\hat{A}^{\dagger}$ 4

CTX_ARM_ $\hat{A}$ 2
reference $\hat{\epsilon}$ context, provenance and rarely-quoted detail

2 columns $\hat{A}$ rung **arm** $\hat{A}$ the two ARM-RUNG members of ctx_, which the other cards do not carry $\hat{A}$ coverage **29% $\hat{\epsilon}$ 29%** (median 29%)
1x boolean $\hat{A}$ 1x real $\hat{A}\%¥0$

ctx_arm_abundant ARM BOOLEAN 29%	Is <code>ctx_arm_dose >= log2(11)</code> , i.e. is this arm abundant enough to plausibly repress? Median 0.00 $\hat{\epsilon}$ most arms are NOT. AUTHORED values: False (499) $\hat{A}$ True (213) example: hsa-let-7a-5p / let-7-5p/98-5p $\hat{A}^{\dagger}$ True
ctx_arm_dose ARM REAL $\hat{A}\%¥0$ 29%	This arm's median log2(RPM+1) across tumour samples (0.13 $\hat{\epsilon}$ 17.92, median 1.08). AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p $\hat{A}^{\dagger}$ 14.8

REAL_ $\hat{A}$ 11
reference $\hat{\epsilon}$ context, provenance and rarely-quoted detail

11 columns $\hat{A}$ rung **arm** $\hat{A}$ coverage **9% $\hat{\epsilon}$ 24%** (median 24%)
7x count $\hat{A}$ 2x signed $\hat{A}^{-1}\hat{A}\epsilon|1$ $\hat{A}$ 2x fraction $0\hat{A}\epsilon^{-1}$

real_best_coupling ARM SIGNED $\hat{A}^{-1}\hat{A}\epsilon 1$ 24%	Most negative realized coupling among this arm's scored genes (median $\hat{A}^{-0.08}$, min $\hat{A}^{-0.56}$). AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p $\hat{A}^{\dagger}$ -0.306
real_frac_c10 ARM FRACTION $0\hat{A}\epsilon^{-1}$ 9%	Fraction of this arm's scored genes coupling below $\hat{A}^{-0.10}$. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p $\hat{A}^{\dagger}$ 0.2609
real_frac_sig ARM FRACTION $0\hat{A}\epsilon^{-1}$ 9%	Fraction of this arm's scored genes reaching significance. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p $\hat{A}^{\dagger}$ 0.2609
real_med_coupling ARM SIGNED $\hat{A}^{-1}\hat{A}\epsilon 1$ 24%	The MEDIAN realized coupling over this arm's scored genes. Range -0.418 $\hat{\epsilon}$ +0.222 , 24% filled. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p $\hat{A}^{\dagger}$ 0.046 <i>Defined on: arms with $\hat{A}\%¥1$ calibrated-coupling-scored edge (593) $\hat{\epsilon}$ the realization LADDER; rates are gated at $n\hat{A}\%¥5$</i>
real_n_c05 ARM COUNT 24%	How many of this arm's scored genes couple below $\hat{A}^{-0.05}$. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p $\hat{A}^{\dagger}$ 6

real_n_c10 ARM COUNT 24%	How many of this arm's scored genes couple below $\hat{a}^0$ 0.10 . AUTHORED example: <code>hsa-let-7a-5p / let-7-5p/98-5p</code> $\hat{a}^\dagger$ 6
real_n_c15 ARM COUNT 24%	How many of this arm's scored genes couple below $\hat{a}^0$ 0.15 . AUTHORED example: <code>hsa-let-7a-5p / let-7-5p/98-5p</code> $\hat{a}^\dagger$ 6
real_n_c20 ARM COUNT 24%	How many of this arm's scored genes couple below $\hat{a}^0$ 0.20 . AUTHORED example: <code>hsa-let-7a-5p / let-7-5p/98-5p</code> $\hat{a}^\dagger$ 6
real_n_c30 ARM COUNT 24%	How many of this arm's scored genes couple below $\hat{a}^0$ 0.30 . AUTHORED example: <code>hsa-let-7a-5p / let-7-5p/98-5p</code> $\hat{a}^\dagger$ 1
real_n_scored ARM COUNT 24%	Genes scored for this arm's realized coupling (median 3). $\hat{a}^\infty$... Its <code>n_c05</code> $\hat{a}^\infty$ <code>n_c30</code> ladder is properly nested $\hat{a}^\infty$ 0 violations (verified). AUTHORED example: <code>hsa-let-7a-5p / let-7-5p/98-5p</code> $\hat{a}^\dagger$ 23 <i>Defined on: arms with $\hat{a}^\infty \geq 1$ calibrated-coupling-scored edge (593) $\hat{a}^\infty$ the realization LADDER; rates are gated at $\hat{n} \geq 5$</i>
real_n_sig ARM COUNT 24%	How many of this arm's scored genes reach significance. AUTHORED example: <code>hsa-let-7a-5p / let-7-5p/98-5p</code> $\hat{a}^\dagger$ 6

GRANK_ $\hat{A}$ 3
reference $\hat{a}^\infty$ context, provenance and rarely-quoted detail

3 columns $\hat{A}$ <code>rung arm</code> $\hat{A}$ coverage 15%$\hat{a}^\infty$20% (median 20%) 3x count	
grank_HLY ARM COUNT 15%	The arm's GLOBAL abundance percentile across the whole cohort in the GTEx-healthy state. AUTHORED example: <code>hsa-miR-200c-3p / miR-200bc-3p/429</code> $\hat{a}^\dagger$ 96
grank_NAT ARM COUNT 20%	The arm's GLOBAL abundance percentile across the whole cohort in matched normal (NAT). AUTHORED example: <code>hsa-miR-200c-3p / miR-200bc-3p/429</code> $\hat{a}^\dagger$ 99
grank_TUM ARM COUNT 20%	The arm's GLOBAL abundance percentile across the whole cohort in tumour. AUTHORED example: <code>hsa-miR-200c-3p / miR-200bc-3p/429</code> $\hat{a}^\dagger$ 99

TS_ $\hat{A}$ 10
reference $\hat{a}^\infty$ context, provenance and rarely-quoted detail

10 columns $\hat{A}$ <code>rung arm</code> $\hat{A}$ coverage 13%$\hat{a}^\infty$13% (median 13%) 5x count $\hat{A}$ 2x real, signed $\hat{A}$ 1x signed $\hat{a}^1$$\hat{a}^\infty$1 $\hat{A}$ 1x fraction 0$\hat{a}^\infty$1 $\hat{A}$ 1x real $\hat{a}^\infty$0	
ts_ctx_best ARM REAL, SIGNED 13%	Best (most negative) TargetScan context++ score for this arm $\hat{a}^\infty$ range [$\hat{a}^2$ 0.02, $\hat{a}^0$ 0.34], all negative by construction (context++ scores repression as a negative number). AUTHORED example: <code>hsa-let-7a-5p / let-7-5p/98-5p</code> $\hat{a}^\dagger$ -0.809
ts_ctx_mean ARM SIGNED $\hat{a}^1$$\hat{a}^\infty$1 13%	TargetScan context++ scan: mean context++ score. AUTHORED example: <code>hsa-let-7a-5p / let-7-5p/98-5p</code> $\hat{a}^\dagger$ -0.283
ts_frac_8mer ARM FRACTION 0$\hat{a}^\infty$1 13%	TargetScan context++ scan: the 8mer share of this arm's sites. AUTHORED example: <code>hsa-let-7a-5p / let-7-5p/98-5p</code> $\hat{a}^\dagger$ 0.3189

ts_kd_med ARM REAL, SIGNED 13%	TargetScan context++ scan: median predicted K_D. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p â†’ -5.181
ts_n_7mer_A1 ARM COUNT 13%	TargetScan context++ scan: 7mer-A1 sites. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 463
ts_n_7mer_m8 ARM COUNT 13%	TargetScan context++ scan: 7mer-m8 sites. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 481
ts_n_8mer ARM COUNT 13%	TargetScan context++ scan: 8mer sites. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 442
ts_n_genes ARM COUNT AGG EDGE 13%	Genes with a TargetScan context++ site â€” median 497, fill 12.9%, the SPARSEST universe on the card. Correlates +0.726 with the MANE seed scan and only +0.231 with scanMiR-any. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 1207
ts_n_sites ARM COUNT AGG EDGE 13%	TargetScan context++ scan: all sites. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 1386
ts_sites_per_gene ARM REAL ðŸšŸ 0 13%	TargetScan context++ scan: sites per targeted gene. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 1.148

ISOC_ ðŸšŸ 5

reference â€” context, provenance and rarely-quoted detail

5 columns ðŸšŸ rung arm ðŸšŸ coverage 91%â€”91% (median 91%) 3x fraction ðŸšŸ1 ðŸšŸ 1x real ðŸšŸ 0 ðŸšŸ 1x count	
isoc_canonical_frac ARM FRACTION ðŸšŸ1 91%	Fraction of the arm's isomiR reads matching the canonical sequence. Range 0.151 â€” 1, 91% filled. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.9956 Defined on: arms in the isomiR seed-COMPOSITION table â€” where the reads go; â isoc_orphan_frac flags MH-153 mass loss
isoc_donated_frac ARM FRACTION ðŸšŸ1 91%	Fraction reassigned to a different family. Range 0 â€” 0.5, 91% filled. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.0029 Defined on: arms in the isomiR seed-COMPOSITION table â€” where the reads go; â isoc_orphan_frac flags MH-153 mass loss
isoc_med_reads ARM REAL ðŸšŸ 0 91%	Median isomiR reads for this arm â€” median 1.00, max 528,927. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 2.229e+04
isoc_n_dest_families ARM COUNT 91%	How many families received reads from this arm. Range 0 â€” 6, 91% filled. AUTHORED values: 0.0 (1,283) ðŸšŸ 1.0 (623) ðŸšŸ 2.0 (238) ðŸšŸ 3.0 (59) ðŸšŸ 4.0 (13) ðŸšŸ 5.0 (3) ðŸšŸ 6.0 (3) example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 1 Defined on: arms in the isomiR seed-COMPOSITION table â€” where the reads go; â isoc_orphan_frac flags MH-153 mass loss
isoc_orphan_frac ARM FRACTION ðŸšŸ1 91%	Fraction belonging to no known family. Range 0 â€” 0.823, 91% filled. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.0034 Defined on: arms in the isomiR seed-COMPOSITION table â€” where the reads go; â isoc_orphan_frac flags MH-153 mass loss

DGLOBAL_ 3

reference “ context, provenance and rarely-quoted detail

3 columns rung arm coverage 15%20% (median 15%) 3× real, signed	
dGlobal_HLY_NAT ARMREAL, SIGNED15%	Change in the arm's GLOBAL (cohort-wide) abundance percentile from the GTEx-healthy state to matched normal (NAT). AUTHORED example: hsa-miR-200c-3p / miR-200bc-3p/429 2
dGlobal_HLY_TUM ARMREAL, SIGNED15%	Change in the arm's GLOBAL (cohort-wide) abundance percentile from the GTEx-healthy state to tumour. AUTHORED example: hsa-miR-200c-3p / miR-200bc-3p/429 3
dGlobal_NAT_TUM ARMREAL, SIGNED20%	Change in the arm's GLOBAL (cohort-wide) abundance percentile from matched normal (NAT) to tumour. AUTHORED example: hsa-miR-200c-3p / miR-200bc-3p/429 1

CHIM_ 5

reference “ context, provenance and rarely-quoted detail

5 columns rung arm coverage 21%63% (median 60%) 4× count 1× real 0	
chim_manakov_n_genes ARMCOUNTAGG EDGE60%	Genes with a Manakov chimeric duplex for this arm. Range 0 “ 13825, 60% filled. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p 7297 Defined on: arms with chimeric ligation evidence (1,538) “ ~90% Manakov/HEK293T, per-source, NEVER pooled
chim_manakov_weight ARMREAL 0AGG EDGE60%	Manakov chimeric support summed over this arm's edges (median 6, max 210,722). AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p 9900
chim_n_sources ARMCOUNTAGG EDGE63%	How many chimeric sources support this arm. Range 1 “ 4, 63% filled. AUTHORED values: 1.0 (1,079) 2.0 (229) 4.0 (128) 3.0 (102) example: hsa-let-7a-5p / let-7-5p/98-5p 4 Defined on: arms with chimeric ligation evidence (1,538) “ ~90% Manakov/HEK293T, per-source, NEVER pooled
chim_tarbase_n_genes ARMCOUNTAGG EDGE21%	Genes with a TarBase chimeric duplex. Range 1 “ 2701, 21% filled. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p 2701 Defined on: arms with chimeric ligation evidence (1,538) “ ~90% Manakov/HEK293T, per-source, NEVER pooled
chim_tarbase_weight ARMCOUNTAGG EDGE21%	TarBase chimeric support for the arm (median 8, max 10,255), defined on 20.9% of arms “ a third of the Manakov coverage. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p 1.026e+04 Defined on: arms with chimeric ligation evidence (1,538) “ ~90% Manakov/HEK293T, per-source, NEVER pooled

FST_ 16

reference “ context, provenance and rarely-quoted detail

16 columns 16 columns 16 columns 3%100% (median 20%) 6x count 5x boolean 3x fraction 1 2x signed 11		
fst_d_share_HLY_TUM ARM SIGNED 11 3%	As fst_d_share_NAT_TUM but GTEx-healthy 1' tumour, same common-support correction. n=68 (HLY is the sparsest leg at 17.0%). Post-fix mean 0.0000, 47.1% positive, p=0.93. AUTHORED example: hsa-miR-26b-5p / miR-26-5p 1' 0.2402	
fst_d_share_NAT_TUM ARM SIGNED 11 6%	Change in the arm's share of its seed family's linear dose, matched NAT 1' tumour, re-normalised over the members measured in BOTH states. AUTHORED example: hsa-miR-200c-3p / miR-200bc-3p/429 1' -0.0072	
fst_dominance_switch ARM BOOLEAN 7%	Did the family's DOMINANT arm change between healthy and tumour (103 true / 74 false)? AUTHORED values: 1.0 (103) 0.0 (74) example: hsa-miR-200c-3p / miR-200bc-3p/429 1' 0	
fst_hly_family_measured ARM BOOLEAN 100%	Do ALL members of this arm's family have a measured GTEx level. AUTHORED values: False (2,142) True (308) example: hsa-let-7a-5p / let-7-5p/98-5p 1' False	
fst_hly_measured ARM BOOLEAN 100%	Does this arm have a MEASURED healthy leg for the family-share comparison. AUTHORED values: False (2,033) True (417) example: hsa-let-7a-5p / let-7-5p/98-5p 1' False	
fst_is_dominant_HLY ARM BOOLEAN 17%	Is the arm its family's dominant member in the healthy state. AUTHORED values: True (345) False (72) example: hsa-miR-200c-3p / miR-200bc-3p/429 1' True	
fst_is_dominant_TUM ARM BOOLEAN 20%	Is it dominant in tumour. AUTHORED values: True (394) False (91) example: hsa-miR-200c-3p / miR-200bc-3p/429 1' True	
fst_n_common_HLY_TUM ARM COUNT 9%	Members measured in BOTH GTEx-healthy and tumour “ the denominator behind fst_d_share_HLY_TUM. See fst_n_common_NAT_TUM. AUTHORED values: 1.0 (162) 2.0 (48) 4.0 (8) 6.0 (6) 3.0 (6) example: hsa-miR-200c-3p / miR-200bc-3p/429 1' 1	
fst_n_common_NAT_TUM ARM COUNT 20%	How many of the family's arms are measured in BOTH NAT and tumour “ the DENOMINATOR behind fst_d_share_NAT_TUM , shipped because the column was wrong for want of it. AUTHORED values: 1.0 (343) 2.0 (86) 4.0 (16) 3.0 (12) 8.0 (8) 7.0 (7) 6.0 (6) example: hsa-miR-200c-3p / miR-200bc-3p/429 1' 2	
fst_n_members ARM COUNT 100%	How many arms are in this arm's seed family. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p 1' 10	
fst_rank_HLY ARM COUNT 17%	The arm's rank within its family by healthy dose. AUTHORED values: 1.0 (348) 2.0 (47) 3.0 (13) 4.0 (6) 5.0 (2) 6.0 (1) example: hsa-miR-200c-3p / miR-200bc-3p/429 1' 1	
fst_rank_NAT ARM COUNT 74%	Its rank by NAT dose. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p 1' 1	

fst_rank_TUM ARM COUNT 20%	Its rank by tumour dose. AUTHORED values: 1.0 (412) · 2.0 (46) · 3.0 (11) · 4.0 (9) · 7.0 (2) · 5.0 (2) · 6.0 (2) · 8.0 (1) example: hsa-miR-200c-3p / miR-200bc-3p/429 â†’ 1
fst_share_HLY ARM FRACTION 0â€”1 17%	Share of family dose in GTEx-healthy, from hly_gtex_median (fill 17.0% â€” the sparsest leg). Same singleton degeneracy; and cross-platform, so contrast only. AUTHORED example: hsa-miR-200c-3p / miR-200bc-3p/429 â†’ 0.9533
fst_share_NAT ARM FRACTION 0â€”1 74%	Share of family dose in matched NAT, from hly_nat_median (fill 74.0% â€” the best-covered leg, 3.7Å— TUM). Same singleton degeneracy: median 1.000, mask on fst_n_members > 1. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.4124
fst_share_TUM ARM FRACTION 0â€”1 20%	The arm's fraction of its seed family's total linear dose in tumour, from arm_med_rpm (fill 19.8%). AUTHORED example: hsa-miR-200c-3p / miR-200bc-3p/429 â†’ 0.945

SDSZ_ · 4

reference â€” context, provenance and rarely-quoted detail

4 columns · rung arm · coverage 97%â€”97% (median 97%) 2× count · 1× real â‰¥ 0 · 1× fraction 0â€”1
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sdsz_N_7mer_plus ARM COUNT AGG EDGE 97%	7mer-and-better sites for this seed. Range 102 â€” 8814, 97% filled. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 3013 Defined on: arms in the MANE 3'UTR seed scan â€” â‰” a FOURTH targetome universe, never blend (2,370)
sdsz_N_8mer ARM COUNT AGG EDGE 97%	8mer sites for this seed across the scanned universe. Range 8 â€” 3030, 97% filled. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 713 Defined on: arms in the MANE 3'UTR seed scan â€” â‰” a FOURTH targetome universe, never blend (2,370)
sdsz_N_eff ARM REAL â‰¥ 0 AGG EDGE 97%	Effective seed-site count over the scanned universe (median 2,094, fill 96.7% â€” the best-covered targetome axis). AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 1863
sdsz_rarity ARM FRACTION 0â€”1 97%	The arm's seed RARITY, min-max normalised to 0â€”1 (median 0.11). AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.1253

SUB_ · 6

reference â€” context, provenance and rarely-quoted detail

6 columns · rung arm · coverage 30%â€”30% (median 30%) 2× count · 2× signed âˆ’1â€”1 · 1× categorical · 1× fraction 0â€”1
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sub_best_frac ARM FRACTION 0â€”1 30%	Subtype rollup: fraction of the arm's edges whose best subtype is the modal one. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.3684
sub_best_mode ARM CATEGORICAL 30%	Subtype rollup: the arm's most common best-subtype. AUTHORED values: Basal (278) · Her2 (240) · LumB (105) · LumA (105) example: hsa-let-7a-5p / let-7-5p/98-5p â†’ Her2

sub_med_rho_best ARM SIGNED $\hat{\rho}^1$ 30%	Median coupling in the arm's BEST subtype. AUTHORED example: <code>hsa-let-7a-5p / let-7-5p/98-5p</code> $\hat{\rho}^1$ -0.03409
sub_med_rho_pool ARM SIGNED $\hat{\rho}^1$ 30%	Median pooled-subtype coupling (median $\hat{\rho}^1$ 0.0064) $\hat{\rho}^1$ the unselected reference for <code>sub_med_rho_best</code>. AUTHORED example: <code>hsa-let-7a-5p / let-7-5p/98-5p</code> $\hat{\rho}^1$ 0.06516 Defined on: arms with a per-PAM50 edge-heterogeneity fit (728) $\hat{\rho}^1$ $\hat{\rho}^1$ DISTRIBUTIONAL (MH-165), not a validated label
sub_n_edges ARM COUNT 30%	Subtype rollup: edges behind the arm's subtype summary. AUTHORED example: <code>hsa-let-7a-5p / let-7-5p/98-5p</code> $\hat{\rho}^1$ 38
sub_n_q05 ARM COUNT 30%	Subtype rollup: how many of the arm's edges reach $q < 0.05$ in ANY subtype. AUTHORED values: 0.0 (659) $\hat{\rho}^1$ 1.0 (54) $\hat{\rho}^1$ 2.0 (10) $\hat{\rho}^1$ 3.0 (3) $\hat{\rho}^1$ 5.0 (2) example: <code>hsa-let-7a-5p / let-7-5p/98-5p</code> $\hat{\rho}^1$ 0

COV_ $\hat{\rho}^1$ 22

reference $\hat{\rho}^1$ context, provenance and rarely-quoted detail

22 columns $\hat{\rho}^1$ <code>rung arm</code> $\hat{\rho}^1$ coverage 100%100% (median 100%) 22× boolean	
cov_ago_dom ARM BOOLEAN 100%	Was this arm ever SCANNED for AGO-loading dominance from Manakov CLIP? True on 678 of 2,450 arms. AUTHORED values: False (1,772) $\hat{\rho}^1$ True (678) example: <code>hsa-let-7a-5p / let-7-5p/98-5p</code> $\hat{\rho}^1$ True
cov_attr ARM BOOLEAN 100%	Was this arm ever SCANNED for the Shapley/NNLS attribution block? True on 546 of 2,450 arms. AUTHORED values: False (1,904) $\hat{\rho}^1$ True (546) example: <code>hsa-let-7a-5p / let-7-5p/98-5p</code> $\hat{\rho}^1$ True
cov_beta ARM BOOLEAN 100%	Was this arm scanned by the learned-$\hat{\rho}^1$ block. Range 728 true / 1,722 false, 100% filled. AUTHORED values: False (1,722) $\hat{\rho}^1$ True (728) example: <code>hsa-let-7a-5p / let-7-5p/98-5p</code> $\hat{\rho}^1$ True
cov_cnv ARM BOOLEAN 100%	Was this arm ever SCANNED for the arm-locus copy-number block? True on 856 of 2,450 arms. AUTHORED values: False (1,594) $\hat{\rho}^1$ True (856) example: <code>hsa-let-7a-5p / let-7-5p/98-5p</code> $\hat{\rho}^1$ True
cov_comp ARM BOOLEAN 100%	Was this arm ever SCANNED for the compartment-loading (<code>cload_</code>) block? True on 205 of 2,450 arms. AUTHORED values: False (2,245) $\hat{\rho}^1$ True (205) example: <code>hsa-let-7a-5p / let-7-5p/98-5p</code> $\hat{\rho}^1$ True
cov_cptac ARM BOOLEAN 100%	Was it scanned by the CPTAC protein block. Range 2,071 true / 379 false, 100% filled. AUTHORED values: True (2,071) $\hat{\rho}^1$ False (379) example: <code>hsa-let-7a-5p / let-7-5p/98-5p</code> $\hat{\rho}^1$ True Defined on: arms measurable in CPTAC (2,095)

cov_curated ARM BOOLEAN 100%	Was this arm ever SCANNED for the curated-literature blocks (fame_ , cur_)? True on 875 of 2,450 arms. AUTHORED values: False (1,575) Â· True (875) example: hsa-let-7a-5p / let-7-5p/98-5p â†’ True
cov_expr ARM BOOLEAN 100%	Was this arm ever SCANNED for the expression / abundance blocks? True on 2,231 of 2,450 arms. AUTHORED values: True (2,231) Â· False (219) example: hsa-let-7a-5p / let-7-5p/98-5p â†’ True
cov_famrole ARM BOOLEAN 100%	Was this arm ever SCANNED for the family-role block? True on 2,231 of 2,450 arms. AUTHORED values: True (2,231) Â· False (219) example: hsa-let-7a-5p / let-7-5p/98-5p â†’ True
cov_field ARM BOOLEAN 100%	Was this arm ever SCANNED for the focal-field correlation block? True on 571 of 2,450 arms. AUTHORED values: False (1,879) Â· True (571) example: hsa-let-7a-5p / let-7-5p/98-5p â†’ True
cov_focal ARM BOOLEAN 100%	Was this arm ever SCANNED for the focal-locus CNV block? True on 733 of 2,450 arms. AUTHORED values: False (1,717) Â· True (733) example: hsa-let-7a-5p / let-7-5p/98-5p â†’ True
cov_fst ARM BOOLEAN 100%	Was the family-share block computed for this arm at all â€” i.e. did it have the abundance input. AUTHORED values: False (1,965) Â· True (485) example: hsa-let-7a-5p / let-7-5p/98-5p â†’ False
cov_gctx ARM BOOLEAN 100%	Was this arm ever SCANNED for the genomic-context block? True on 714 of 2,450 arms. AUTHORED values: False (1,736) Â· True (714) example: hsa-let-7a-5p / let-7-5p/98-5p â†’ True
cov_isocomp ARM BOOLEAN 100%	Was this arm ever SCANNED for the isomiR-composition block? True on 2,222 of 2,450 arms. AUTHORED values: True (2,222) Â· False (228) example: hsa-let-7a-5p / let-7-5p/98-5p â†’ True
cov_isomir ARM BOOLEAN 100%	Was this arm ever SCANNED for the isomiR seed-shift block? True on 687 of 2,450 arms. AUTHORED values: False (1,763) Â· True (687) example: hsa-let-7a-5p / let-7-5p/98-5p â†’ True
cov_meth ARM BOOLEAN 100%	Was this arm ever SCANNED for the locus-methylation (bc_meth_) block? True on 343 of 2,450 arms. AUTHORED values: False (2,107) Â· True (343) example: hsa-let-7a-5p / let-7-5p/98-5p â†’ True
cov_real ARM BOOLEAN 100%	Was this arm ever SCANNED for the realized-coupling (real_) block? True on 590 of 2,450 arms. AUTHORED values: False (1,860) Â· True (590) example: hsa-let-7a-5p / let-7-5p/98-5p â†’ True

cov_seed_scan ARM BOOLEAN 100%	Was this arm ever SCANNED for the MANE 3'UTR seed scan (site_)? AUTHORED values: True (2,370) · False (80) example: hsa-let-7a-5p / let-7-5p/98-5p â†’ True
cov_seq_scan ARM BOOLEAN 100%	Was this arm ever SCANNED for the scanMiR K_D scan (seq_)? AUTHORED values: False (1,714) · True (736) example: hsa-let-7a-5p / let-7-5p/98-5p â†’ True
cov_site_types ARM BOOLEAN 100%	Was this arm ever SCANNED for the site-type breakdown? AUTHORED values: False (1,732) · True (718) example: hsa-let-7a-5p / let-7-5p/98-5p â†’ True
cov_subtype ARM BOOLEAN 100%	Was this arm ever SCANNED for the PAM50 subtype block (sub_)? AUTHORED values: False (1,722) · True (728) example: hsa-let-7a-5p / let-7-5p/98-5p â†’ True
cov_targetscan ARM BOOLEAN 100%	Was this arm ever SCANNED for the TargetScan context++ block (ts_)? AUTHORED values: False (2,133) · True (317) example: hsa-let-7a-5p / let-7-5p/98-5p â†’ True

SEQ_ 4

reference “ context, provenance and rarely-quoted detail

4 columns · rung arm · coverage 30%“30% (median 30%) 2× count · 2× real % 0	
seq_n_genes_any ARM COUNT AGG EDGE 30%	Genes with ANY scanMiR K_D site “ AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 1.124e+04
seq_n_genes_strong ARM COUNT AGG EDGE 30%	Genes with a STRONG scanMiR K_D site “ median 3,632, spearman +0.822 with sdsz_N_eff and +0.797 with site_n_genes_canonical , but only +0.566 with ts_n_genes . AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 4550
seq_promisc ARM REAL % 0 30%	log-scale promiscuity over STRONG scanMiR sites (4.45 “ 8.94). AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 8.423
seq_promisc_any ARM REAL % 0 30%	Promiscuity over ANY scanMiR site (6.46 “ 9.45) “ the permissive cut. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 9.327

ISO_ 5

reference “ context, provenance and rarely-quoted detail

5 columns · rung arm · coverage 28%“28% (median 28%) 2× fraction 0“1 · 1× count · 1× real % 0 · 1× real, signed	
iso_n_samples ARM COUNT 28%	Samples behind the isomiR estimate. Range 3 “ 1078, 28% filled. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 1078 Defined on: arms with a 5'-isomiR seed-shift measurement (687) “ š gate on iso_n_samples, the median is n-DEPENDENT

iso_rpm_summed ARM REAL $\hat{\mu}_0$ 28%	âš A SUM ACROSS SAMPLES, not a per-sample RPM â€” the name misleads. Max 270,988,199, which is impossible for a per-million unit; divided by <code>iso_n_samples</code> the scale is sane (median 57, max 251,381). AUTHORED example: <code>hsa-let-7a-5p / let-7-5p/98-5p</code> â† 3.553e+07
iso_seed_shift_frac ARM FRACTION $\hat{0}\hat{1}$ 28%	Fraction of the arm's reads carrying a SHIFTED seed. Range 0 â€“ 0.868, 28% filled. AUTHORED example: <code>hsa-let-7a-5p / let-7-5p/98-5p</code> â† 0.004 Defined on: arms with a 5'-isomiR seed-shift measurement (687) â€” gate on <code>iso_n_samples</code>, the median is n-DEPENDENT
iso_seed_shift_sd ARM FRACTION $\hat{0}\hat{1}$ 28%	Dispersion of that shift across samples. Range 0 â€“ 0.354, 28% filled. AUTHORED example: <code>hsa-let-7a-5p / let-7-5p/98-5p</code> â† 0.004 Defined on: arms with a 5'-isomiR seed-shift measurement (687) â€” gate on <code>iso_n_samples</code>, the median is n-DEPENDENT
iso_top_shift_nt ARM REAL, SIGNED 28%	The most common shift, in nucleotides. Range -6 â€“ 6, 28% filled. AUTHORED example: <code>hsa-let-7a-5p / let-7-5p/98-5p</code> â† 3 Defined on: arms with a 5'-isomiR seed-shift measurement (687) â€” gate on <code>iso_n_samples</code>, the median is n-DEPENDENT

CUR_HE_ $\hat{\cdot}$ 2

reference â€” context, provenance and rarely-quoted detail

2 columns $\hat{\cdot}$ rung arm $\hat{\cdot}$ coverage 36%â€“36% (median 36%) 2x count	
cur_he_degree ARM COUNT AGG EDGE 36%	How many HE genes this arm regulates in the curated design â€” the arm's curation degree (median 3, max 125). AUTHORED example: <code>hsa-let-7a-5p / let-7-5p/98-5p</code> â† 51 Defined on: arms with miRTarBase curation / ledger PMIDs â€” FAME axes, not targetome (MH-208)
cur_he_degree_expr ARM COUNT AGG EDGE 36%	As <code>cur_he_degree</code> but counting only targets above the expression floor. AUTHORED example: <code>hsa-let-7a-5p / let-7-5p/98-5p</code> â† 43

FIELD_ $\hat{\cdot}$ 5

reference â€” context, provenance and rarely-quoted detail

5 columns $\hat{\cdot}$ rung arm $\hat{\cdot}$ coverage 23%â€“23% (median 23%) 4x signed $\hat{\cdot}$1â€ 1 $\hat{\cdot}$ 1x count	
field_excess_over_perm ARM SIGNED $\hat{\cdot}$1â€ 1 23%	â”â” NOT A RETENTION RATIO â€” a FIFTH sense of the word in this codebase. Measured: it equals field_r_own â” field_r_perm (exact within rounding on 100% of 571 arms, max residual 0.001), i.e. AUTHORED example: <code>hsa-let-7a-5p / let-7-5p/98-5p</code> â† 0.363
field_n ARM COUNT 23%	Patients behind the field correlation. Range 40 â€“ 103, 23% filled. AUTHORED example: <code>hsa-let-7a-5p / let-7-5p/98-5p</code> â† 103 Defined on: arms with a within-patient NAT field-effect fit (571)
field_r_adj ARM SIGNED $\hat{\cdot}$1â€ 1 23%	The same, covariate-adjusted. Range -0.344 â€“ +0.665, 23% filled. AUTHORED example: <code>hsa-let-7a-5p / let-7-5p/98-5p</code> â† 0.367 Defined on: arms with a within-patient NAT field-effect fit (571)

field_r_own

ARM

SIGNED -1

23%

The arm's own field correlation (median +0.11 against a permutation null centred at 0.000). Range **-0.304** to **+0.624**, 23% filled.

AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p at 0.343

| Defined on: arms with a within-patient NAT field-effect fit (571)

field_r_perm

ARM

SIGNED -1

23%

The permutation null for field_r_own median **0.000**, range [-0.11, +0.09].

AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p at -0.02

BC_ 11

reference context, provenance and rarely-quoted detail

11 columns

run arm

coverage **14%** to **36%** (median 14%)

4x fraction

4x count

1x boolean

1x signed -1

1x categorical

bc_fam_context_homogeneous

ARM

BOOLEAN

36%

The arm's SEED-FAMILY property fam_context_homogeneous, broadcast down to the arm.

AUTHORED

values: True (720) False (160)

example: hsa-let-7a-5p / let-7-5p/98-5p at False

bc_fam_dose_coding_host_fraction

ARM

FRACTION 1

36%

The arm's SEED-FAMILY property fam_dose_coding_host_frac, broadcast down to the arm.

AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p at 0.121

bc_fam_dose_cotranscribed_fraction

ARM

FRACTION 1

36%

The arm's SEED-FAMILY property fam_dose_cotranscribed_frac, broadcast down to the arm.

AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p at 0.105

bc_fam_n_distinct_coding_hosts

ARM

COUNT

36%

The arm's SEED-FAMILY property fam_n_distinct_coding_hosts, broadcast down to the arm.

AUTHORED

values: 1.0 (424) 0.0 (379) 2.0 (45) 3.0 (32)

example: hsa-let-7a-5p / let-7-5p/98-5p at 3

bc_fam_n_members

ARM

COUNT

36%

The arm's SEED-FAMILY property fam_n_members, broadcast down to the arm.

AUTHORED

values: 1.0 (606) 2.0 (139) 3.0 (69) 5.0 (22) 10.0 (21) 8.0 (8) 4.0 (8) 6.0 (7)

example: hsa-let-7a-5p / let-7-5p/98-5p at 10

bc_meth_dbeta

ARM

SIGNED -1

14%

Tumour-minus-normal methylation beta at this arm's locus (median -0.03, range [-0.32, +0.23].

AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p at 0.1451

bc_meth_direction

ARM

CATEGORICAL

14%

Tumour-vs-normal methylation direction at this arm's locus.

AUTHORED

values: none (287) hypo (46) hyper (10)

example: hsa-let-7a-5p / let-7-5p/98-5p at none

bc_meth_n CGI

ARM

COUNT

14%

Locus methylation: n CGI.

AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p at 0

bc_meth_n probes

ARM

COUNT

14%

Locus methylation: n probes.

AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p at 1

bc_meth_normal_beta

ARM

FRACTION 1

14%

Locus methylation: normal_beta.

AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p at 0.6494

bc_meth_tumour_beta

ARM

FRACTION 0â€“1

14%

Locus methylation: tumour_beta.

AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.7945

FAMROLE_ 6

reference â€” context, provenance and rarely-quoted detail

6 columns â€• rung arm â€• coverage 88%â€“100% (median 91%)
3x count â€• 1x fraction 0â€“1 â€• 1x categorical â€• 1x boolean

famrole_abund_rank

ARM

COUNT

91%

The arm's rank by abundance within its seed family. Range 1 â€“ 14, 91% filled.

AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 1

Defined on: arms with a seed family â€” the arm's ROLE inside it; â” DEGENERATE at famrole_n_members==1

famrole_abund_share

ARM

FRACTION 0â€“1

91%

Its share of the family's abundance â€” median 1.000, because most families are singletons. Range 0 â€“ 1, 91% filled.

AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.465

Defined on: arms with a seed family â€” the arm's ROLE inside it; â” DEGENERATE at famrole_n_members==1

famrole_class

ARM

CATEGORICAL

100%

The arm's role in its seed family â€” sole on 1,639 of 2,450 (66.9%), dominant 282, co and the rest below.

AUTHORED

values: sole (1,639) â€• dominant (282) â€• co (267) â€• minor (262)

example: hsa-let-7a-5p / let-7-5p/98-5p â†’ co

famrole_is_dominant

ARM

BOOLEAN

91%

Does the arm carry >50% of its family's abundance. Range 1,779 true / 452 false, 91% filled.

AUTHORED

values: True (1,779) â€• False (452)

example: hsa-let-7a-5p / let-7-5p/98-5p â†’ False

Defined on: arms with a seed family â€” the arm's ROLE inside it; â” DEGENERATE at famrole_n_members==1

famrole_n_members

ARM

COUNT

100%

How many arms are in the family. Range 1 â€“ 22, 100% filled.

AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 10

Defined on: arms with a seed family â€” the arm's ROLE inside it; â” DEGENERATE at famrole_n_members==1

famrole_var_rank

ARM

COUNT

88%

Its rank by variance within the family. Range 1 â€“ 14, 88% filled.

AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 7

Defined on: arms with a seed family â€” the arm's ROLE inside it; â” DEGENERATE at famrole_n_members==1

GCTX_ 16

reference â€” context, provenance and rarely-quoted detail

16 columns â€• rung arm â€• coverage 24%â€“29% (median 29%)
6x categorical â€• 4x boolean â€• 3x identifier â€• 1x constant â€• 1x count â€• 1x real â‰¥ 0

gctx_clustered

ARM

BOOLEAN

29%

Does this arm's precursor sit in a genomic CLUSTER with other miRNAs (310 true / 404 false)?

AUTHORED

values: False (404) â€• True (310)

example: hsa-let-7a-5p / let-7-5p/98-5p â†’ True

gctx_context_mixed ARM BOOLEAN 29%	Do this arm's multiple loci DISAGREE about their genomic context â€” 20 arms of 714? AUTHORED values: False (694) Â· True (20) example: <code>hsa-let-7a-5p / let-7-5p/98-5p â†’ False</code>
gctx_coord_source ARM CONSTANT 29%	Provenance of the genomic-context coordinates. AUTHORED values: gencode (714) example: <code>hsa-let-7a-5p / let-7-5p/98-5p â†’ gencode</code>
gctx_host ARM IDENTIFIER 27%	The HOST GENE the precursor sits inside (356 distinct; <code>ENSG00000269842</code>, <code>MEG9</code>, <code>MEG3</code> most common). AUTHORED example: <code>hsa-let-7a-5p / let-7-5p/98-5p â†’ MIR100HG</code>
gctx_host_region ARM IDENTIFIER 27%	WHERE in the host the precursor sits (74 distinct; <code>intron 2</code> and <code>intron 1</code> most common, 61 arms each). An intronic miRNA is spliced from its host's pre-mRNA; an exonic one competes with it. AUTHORED example: <code>hsa-let-7a-5p / let-7-5p/98-5p â†’ exon 41</code> <i>Defined on: arms with a classified precursor locus (<code>genomic_context.classify_he_arms</code> â€” 714)</i>
gctx_host_type ARM CATEGORICAL 27%	The host's BIOTYPE â€” <code>protein_coding</code> 364 Â· <code>lncRNA</code> 303, plus 5 rarer levels. AUTHORED values: <code>protein_coding</code> (364) Â· <code>lncRNA</code> (303) Â· <code>transcribed_unprocessed_pseudogene</code> (2) Â· <code>TEC</code> (1) Â· <code>transcribed_unitary_pseudogene</code> (1) Â· <code>processed_pseudogene</code> (1) Â· <code>IG_V_gene</code> (1) example: <code>hsa-let-7a-5p / let-7-5p/98-5p â†’ lncRNA</code>
gctx_in_exon ARM BOOLEAN 29%	Is the precursor inside a host EXON (218 true / 496 false) rather than an intron? AUTHORED values: False (496) Â· True (218) example: <code>hsa-let-7a-5p / let-7-5p/98-5p â†’ True</code>
gctx_mir_class ARM CATEGORICAL 29%	The arm's genomic class â€” <code>sense_coding_host</code> (337) / <code>sense_lncRNA_host</code> / <code>antisense_overlap</code> / intergenic. Range 4 levels, 29% filled. AUTHORED values: <code>sense_coding_host</code> (337) Â· <code>sense_lncRNA_host</code> (296) Â· <code>intergenic</code> (41) Â· <code>antisense_overlap</code> (40) example: <code>hsa-let-7a-5p / let-7-5p/98-5p â†’ sense_lncRNA_host</code> <i>Defined on: arms with a classified precursor locus (<code>genomic_context.classify_he_arms</code> â€” 714)</i>
gctx_mirgenedb ARM BOOLEAN 29%	Is this arm in MirGeneDB â€” the curated high-confidence miRNA set (526 true / 188 false)? AUTHORED values: True (526) Â· False (188) example: <code>hsa-let-7a-5p / let-7-5p/98-5p â†’ True</code>
gctx_n_loci ARM COUNT 29%	How many genomic loci the precursor maps to. Range 1 â€” 3, 29% filled. AUTHORED values: 1.0 (642) Â· 2.0 (62) Â· 3.0 (10) example: <code>hsa-let-7a-5p / let-7-5p/98-5p â†’ 3</code> <i>Defined on: arms with a classified precursor locus (<code>genomic_context.classify_he_arms</code> â€” 714)</i>
gctx_promoter_class ARM CATEGORICAL 29%	Whose promoter drives it â€” <code>host_shared</code> (346) / <code>own</code> (239) / shared. Range 4 levels, 29% filled. AUTHORED values: <code>host_shared</code> (346) Â· <code>own</code> (239) Â· <code>shared_other</code> (128) Â· <code>unknown</code> (1) example: <code>hsa-let-7a-5p / let-7-5p/98-5p â†’ host_shared</code> <i>Defined on: arms with a classified precursor locus (<code>genomic_context.classify_he_arms</code> â€” 714)</i>
gctx_promoter_coord_class ARM CATEGORICAL 29%	The same call from coordinates rather than annotation. Range 4 levels, 29% filled. AUTHORED values: <code>host_shared</code> (480) Â· <code>shared_other</code> (111) Â· <code>unknown</code> (85) Â· <code>intergenic</code> (38) example: <code>hsa-let-7a-5p / let-7-5p/98-5p â†’ host_shared</code> <i>Defined on: arms with a classified precursor locus (<code>genomic_context.classify_he_arms</code> â€” 714)</i>

gctx_promoter_gene ARM IDENTIFIER 24%	The gene whose promoter drives the precursor. Range 307 distinct, 24% filled. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p â†’ MIR100HG <i>Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€” 714)</i>
gctx_promoter_gene_type ARM CATEGORICAL 24%	That gene's biotype. Range 5 levels, 24% filled. AUTHORED values: protein_coding (320) Â· lncRNA (267) Â· misc_RNA (2) Â· transcribed_processed_pseudogene (1) Â· IG_V_gene (1) example: hsa-let-7a-5p / let-7-5p/98-5p â†’ lncRNA <i>Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€” 714)</i>
gctx_promoter_host_tss_kb ARM REAL Â¥ 0 24%	Distance in kb from the precursor to its host's TSS (median 0.5 kb, max 874). AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 376.1
gctx_strand_rel ARM CATEGORICAL 27%	Is the precursor SENSE (633 arms) or ANTISENSE (40) to its host? AUTHORED values: sense (633) Â· antisense (40) example: hsa-let-7a-5p / let-7-5p/98-5p â†’ sense

SURV_ Â· 8

reference â€” context, provenance and rarely-quoted detail

8 columns Â· rung arm Â· coverage 6%â€”6% (median 6%) 4x p-value Â· 3x real â¥ 0 Â· 1x constant
--

surv_dfi_hr_adj ARM REAL Â¥ 0 6%	Adjusted DFI hazard ratio per arm â€” range [0.84, 1.16], median 1.00. The whole distribution sits within Â±16% of no effect. See surv_dfi_q_adj . AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.9371 <i>Defined on: arms in the TCGA outcome panel (145) â€” â” TCGA legs only; the GSE leg is bare-stem matched</i>
surv_dfi_hr_base ARM REAL Â¥ 0 6%	Unadjusted DFI hazard ratio for the arm (0.86 â€” 1.16, median 1.00). AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.9463
surv_dfi_p_adj ARM P-VALUE 6%	Adjusted DFI p. Range 0.0741 â€” 0.997, 6% filled. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.4701 <i>Defined on: arms in the TCGA outcome panel (145) â€” â” TCGA legs only; the GSE leg is bare-stem matched</i>
surv_dfi_p_base ARM P-VALUE 6%	Unadjusted DFI p for this arm. Range 0.0973 â€” 0.998, 6% filled. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.5368 <i>Defined on: arms in the TCGA outcome panel (145) â€” â” TCGA legs only; the GSE leg is bare-stem matched</i>
surv_dfi_q_adj ARM CONSTANT 6%	BH q for the arm's DFI hazard ratio. AUTHORED values: 0.996528491082692 (145) example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.9965
surv_os_hr_adj ARM REAL Â¥ 0 6%	Adjusted OS hazard ratio. Range 0.816 â€” 1.24, 6% filled. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 1.011 <i>Defined on: arms in the TCGA outcome panel (145) â€” â” TCGA legs only; the GSE leg is bare-stem matched</i>
surv_os_p_adj ARM P-VALUE 6%	Adjusted OS p. Range 0.00235 â€” 0.998, 6% filled. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.8856 <i>Defined on: arms in the TCGA outcome panel (145) â€” â” TCGA legs only; the GSE leg is bare-stem matched</i>

surv_os_q_adj

ARM

P-VALUE

6%

BH q for OS

AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p

0.9745

AID_ 4

reference context, provenance and rarely-quoted detail

4 columns

rung

arm

coverage

6%6% (median 6%)

1x fraction

1x real

1x signed

1x count

aid_frac_resolvable

ARM

FRACTION

01

AGG

ARM-IN-FAMILY

6%

Fraction of the arm's multi-arm cells that resolve

median 0.19. Range 0 1, 6% filled.

AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p

0.4118

Defined on: arms inside MULTI-ARM (gene,family) cell the arm rung's 20.3%-of-edges footprint (142)

aid_med_arm_sep_z

ARM

REAL

0

AGG

ARM-IN-FAMILY

6%

Median separation z across those cells. Range

0.00108 5.59, 6% filled.

AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p

2.381

Defined on: arms inside MULTI-ARM (gene,family) cell the arm rung's 20.3%-of-edges footprint (142)

aid_med_oof_drho

ARM

SIGNED

11

AGG

ARM-IN-FAMILY

6%

Median out-of-fold gain from resolving the arm. Range

-0.092 +0.0421, 6% filled.

AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p

-0.004584

Defined on: arms inside MULTI-ARM (gene,family) cell the arm rung's 20.3%-of-edges footprint (142)

aid_n_cells

ARM

COUNT

AGG

ARM-IN-FAMILY

6%

Multi-arm cells this arm participates in fill

5.8% (the arm identifiability block is the sparsest on the card).

AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p

17

WSHIFT_ 6

reference context, provenance and rarely-quoted detail

6 columns

rung

arm

coverage

45%91% (median 52%)

3x real

2x real, signed

1x count

wshift_mean_dGlobalRank

ARM

REAL, SIGNED

52%

Mean within-patient change in its global rank. Range

-36.2 42.7, 52% filled.

AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p

-0.13

Defined on: arms in the WITHIN-PATIENT paired subset (103 matched tumour/NAT pairs; 2,236 arms)

wshift_mean_own_shift

ARM

REAL, SIGNED

52%

Mean within-patient change in the arm's own share. Range

-3.94 5.05, 52% filled.

AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p

-0.223

Defined on: arms in the WITHIN-PATIENT paired subset (103 matched tumour/NAT pairs; 2,236 arms)

wshift_n_pairs

ARM

COUNT

91%

Paired patients behind the within-patient shift. Range

0 103, 91% filled.

AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p

103

Defined on: arms in the WITHIN-PATIENT paired subset (103 matched tumour/NAT pairs; 2,236 arms)

wshift_nat_patient_sd ARM REAL $\hat{\mu}$ 0 74%	Between-patient SD of the arm's NAT level. Range 0 $\hat{\mu}$ 2.73, 74% filled. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p $\hat{\mu}$ 0.556 Defined on: arms in the WITHIN-PATIENT paired subset ($n \hat{\mu} \hat{\mu} 103$ matched tumour/NAT pairs; 2,236 arms)
wshift_own_specific_frac ARM REAL $\hat{\mu}$ 0 45%	Within-patient paired quantities on the ~ 103 matched tumour/NAT participants. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p $\hat{\mu}$ 0.62
wshift_sd_dGlobalRank ARM REAL $\hat{\mu}$ 0 45%	Dispersion of that change. Range 0.03 $\hat{\mu}$ 50.9, 45% filled. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p $\hat{\mu}$ 0.4 Defined on: arms in the WITHIN-PATIENT paired subset ($n \hat{\mu} \hat{\mu} 103$ matched tumour/NAT pairs; 2,236 arms)

CNV_ $\hat{\mu}$ 9

reference $\hat{\mu}$ context, provenance and rarely-quoted detail

9 columns $\hat{\mu}$ rung arm $\hat{\mu}$ coverage 35%$\hat{\mu}$35% (median 35%) 6$\times$ percent $\hat{\mu}$ 3$\times$ real $\hat{\mu}$ 0	
cnv_mean_cn ARM REAL $\hat{\mu}$ 0 35%	Arm-locus copy number: MEAN copy number at this arm's locus, over the TCGA cohort. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p $\hat{\mu}$ 2.403
cnv_median_cn ARM REAL $\hat{\mu}$ 0 35%	Arm-locus copy number: MEDIAN copy number, over the TCGA cohort. AUTHORED values: 2.0 (632) $\hat{\mu}$ 3.0 (175) $\hat{\mu}$ 4.0 (40) $\hat{\mu}$ 2.5 (8) $\hat{\mu}$ 3.5 (1) example: hsa-let-7a-5p / let-7-5p/98-5p $\hat{\mu}$ 2
cnv_pct_altered ARM PERCENT 35%	Arm-locus copy number: percent of patients with ANY alteration, over the TCGA cohort. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p $\hat{\mu}$ 43.66
cnv_pct_amp ARM PERCENT 35%	Arm-locus copy number: percent AMPLIFIED, over the TCGA cohort. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p $\hat{\mu}$ 16.21
cnv_pct_deep_del ARM PERCENT 35%	Percent of samples with a deep deletion at this arm's locus. AUTHORED values: 0.0 (722) $\hat{\mu}$ 0.0953 (75) $\hat{\mu}$ 0.1907 (32) $\hat{\mu}$ 0.286 (10) $\hat{\mu}$ 0.0954 (6) $\hat{\mu}$ 0.3813 (6) $\hat{\mu}$ 1.5253 (4) $\hat{\mu}$ 0.0955 (1) example: hsa-let-7a-5p / let-7-5p/98-5p $\hat{\mu}$ 0
cnv_pct_gain ARM PERCENT 35%	Arm-locus copy number: percent with a low-level GAIN, over the TCGA cohort. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p $\hat{\mu}$ 16.11
cnv_pct_loh ARM PERCENT 35%	Arm-locus copy number: percent with loss of heterozygosity, over the TCGA cohort. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p $\hat{\mu}$ 69.97
cnv_pct_loss ARM PERCENT 35%	Arm-locus copy number: percent with a LOSS, over the TCGA cohort. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p $\hat{\mu}$ 11.34
cnv_sd_cn ARM REAL $\hat{\mu}$ 0 35%	Arm-locus copy number: SD of copy number across patients, over the TCGA cohort. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p $\hat{\mu}$ 0.9746

TIER_ 5

reference “ context, provenance and rarely-quoted detail

5 columns rung arm coverage 91%91% (median 91%) 2x real 0 1x categorical 1x boolean 1x fraction 01		
tier_class ARM CATEGORICAL 91%	Expression tier “ silent on 1,518 of 2,236 arms (67.9%), conditional 476, the rest robust. AUTHORED values: silent (1,518) conditional (476) robust (242) example: hsa-let-7a-5p / let-7-5p/98-5p robust	
tier_expressed ARM BOOLEAN 91%	Is the arm above the expression floor often enough to be usable (718 true / 1,518 false)? AUTHORED values: False (1,518) True (718) example: hsa-let-7a-5p / let-7-5p/98-5p True	
tier_frac_expressed ARM FRACTION 01 91%	Fraction of samples where it is expressed “ median 0.00. Range 0 “ 1, 91% filled. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p 1 Defined on: arms detected in the TCGA miRNA matrix (2,236 of 2,450)	
tier_max ARM REAL 0 91%	The arm's maximum RPM across tumour samples. Range 0.105 “ 19.4, 91% filled. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p 17.48 Defined on: arms detected in the TCGA miRNA matrix (2,236 of 2,450)	
tier_p90 ARM REAL 0 91%	The arm's 90th-percentile RPM across tumour samples (0.10 “ 18.49, median 0.94). AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p 15.76	

FOC_ 7

reference “ context, provenance and rarely-quoted detail

7 columns rung arm coverage 30%30% (median 30%) 2x count 2x signed 11 1x identifier 1x boolean 1x p-value		
foc_focal_locus ARM IDENTIFIER 30%	The focal MIRBase locus used for this arm's CN analysis (437 distinct). AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p MI0000062	
foc_multilocus ARM BOOLEAN 30%	Does this arm map to MORE THAN ONE genomic locus (35 true / 698 false)? AUTHORED values: False (698) True (35) example: hsa-let-7a-5p / let-7-5p/98-5p True	
foc_n ARM COUNT 30%	Patients behind the focal-locus CN correlation. Range 41 “ 1032, 30% filled. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p 1032 Defined on: arms with a paralog-decontaminated FOCAL-locus concordance (733) “ “ DESCRIPTIVE, not an instrument	
foc_n_loci ARM COUNT 30%	How many loci the arm maps to. Range 1 “ 3, 30% filled. AUTHORED values: 1.0 (698) 2.0 (27) 3.0 (8) example: hsa-let-7a-5p / let-7-5p/98-5p 3 Defined on: arms with a paralog-decontaminated FOCAL-locus concordance (733) “ “ DESCRIPTIVE, not an instrument	

foc_partial_p ARM P-VALUE 30%	p for that partial correlation. Range 1.54e-42 â€“ 0.996, 30% filled. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 6.541e-09 <i>Defined on: arms with a paralog-decontaminated FOCAL-locus concordance (733) â€” â†’ DESCRIPTIVE, not an instrument</i>
foc_partial_rho ARM SIGNED Â^'1â€“1 30%	The same, conditioned on covariates. Range -0.269 â€“ +0.424, 30% filled. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.1871 <i>Defined on: arms with a paralog-decontaminated FOCAL-locus concordance (733) â€” â†’ DESCRIPTIVE, not an instrument</i>
foc_spearman_rho ARM SIGNED Â^'1â€“1 30%	Correlation between the arm's dose and its focal locus CN. Range -0.235 â€“ +0.485, 30% filled. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.08216 <i>Defined on: arms with a paralog-decontaminated FOCAL-locus concordance (733) â€” â†’ DESCRIPTIVE, not an instrument</i>

SURROGATE_ **Â** **2**

reference **â€”** context, provenance and rarely-quoted detail

2 columns Â rung arm Â coverage 0%â€“0% (median 0%) 1x fraction 0â€“1 Â 1x categorical	
surrogate_corr ARM FRACTION 0â€“1 0%	Correlation between this arm and its surrogate (0.62 â€“ 0.89) â€” the surrogate's quality gate. See the edge twin. AUTHORED values: 0.621 (1) Â 0.874 (1) Â 0.892 (1) Â 0.842 (1) example: hsa-miR-181a-5p / miR-181-5p â†’ 0.892 <i>Defined on: arms with a same-seed SURROGATE (MH-166) â€” 3.6% of edges</i>
surrogate_instrument ARM CATEGORICAL 0%	Which surrogate stands in for this arm's collapsed GTEx leg (4 levels). AUTHORED values: hsa-miR-93-5p (1) Â hsa-miR-20a-5p (1) Â hsa-miR-181b-5p (1) Â hsa-miR-429 (1) example: hsa-miR-181a-5p / miR-181-5p â†’ hsa-miR-181b-5p

CNVC_ **Â** **5**

reference **â€”** context, provenance and rarely-quoted detail

5 columns Â rung arm Â coverage 31%â€“31% (median 31%) 2x p-value Â 2x signed Â^'1â€“1 Â 1x count	
cnvc_n ARM COUNT 31%	Patients behind the CN-coupling estimate. Range 20 â€“ 1032, 31% filled. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 1032 <i>Defined on: arms with a CNâ†’expression concordance fit (756)</i>
cnvc_partial_p ARM P-VALUE 31%	p for the conditioned CN correlation. Range 1.25e-42 â€“ 1, 31% filled. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.0001062 <i>Defined on: arms with a CNâ†’expression concordance fit (756)</i>
cnvc_partial_q ARM P-VALUE 31%	BH q for the arm's CN-coupling partial correlation. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.0006635
cnvc_partial_rho ARM SIGNED Â^'1â€“1 31%	The same, conditioned â€” barely moves (+0.05 vs +0.06), so the CN association is not carried by the covariates. Range -0.322 â€“ +0.468, 31% filled. AUTHORED example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.1238 <i>Defined on: arms with a CNâ†’expression concordance fit (756)</i>

cncv_spearman_rho

ARM

SIGNED -1

31%

Arm dose vs locus CN, unconditioned. Range **-0.378 to +0.5**, 31% filled.

AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p at 0.0573

Defined on: arms with a CN expression concordance fit (756)

DISC_ 2

reference context, provenance and rarely-quoted detail

2 columns

rung arm

coverage 1 to 100% (median 50%)

1x categorical

1x count

disc_gold_families

ARM

CATEGORICAL

1%

Which seed families this row's discovery-queue edges belong to, semicolon-separated. Usually ONE at the arm rung (an arm has one seed).

AUTHORED

example: hsa-miR-25-3p / miR-25-3p/32-5p/92-3p/363-3p/367-3p at miR-25-3p/32-5p/92-3p/363-3p/367-3p

disc_n_gold_edges

ARM

COUNT

100%

How many of the 157 discovery-queue edges name this arm. Concentrated: miR-106b-5p **39** miR-20a-5p 25 miR-93-5p 25 miR-19a-3p 19 miR-18a-5p 9.

AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p at 0

CLOAD_ 3

reference context, provenance and rarely-quoted detail

3 columns

rung arm

coverage 8 to 60% (median 60%)

1x real at 0

1x fraction at 1

1x categorical

cload_dose_rpm

ARM

REAL at 0

60%

The arm's dose used in the compartment-loading fit. Range **0.279 to 2.4e+05**, 60% filled.

AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p at 3.285e+04

Defined on: arms with a compartment-loading row (1,473) at its source has NO PRODUCER (MH-111 ad-hoc)

cload_lock

ARM

FRACTION at 1

8%

Is this arm's compartment loading LOCKED to one cell type rather than spread across the deconvolution?

AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p at 0.256

cload_top_compartment

ARM

CATEGORICAL

60%

Which compartment the arm's abundance loads on most. Range **8 levels**, 60% filled.

AUTHORED

values: CAFs (1,335) Endothelial (47) PVL (35) Myeloid (31) Normal Epithelial (15) B-cells (4) Plasmablasts (3) T-cells (3)

example: hsa-let-7a-5p / let-7-5p/98-5p at Endothelial

Defined on: arms with a compartment-loading row (1,473) at its source has NO PRODUCER (MH-111 ad-hoc)

MODEL_ 1

reference context, provenance and rarely-quoted detail

1 columns

rung arm

coverage 30 to 30% (median 30%)

1x count

model_n_genes

ARM

COUNT

AGG EDGE

30%

Which model variant produced this row.

AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p at 38

Defined on: arms carrying >=1 curated HE edge in the model design (741)

COMV_ 2

reference " context, provenance and rarely-quoted detail

2 columns rung arm coverage 15%15% (median 15%)
1x count 1x identifier

comv_module

ARMCOUNT15%

Which co-movement module this arm belongs to in the 6b-RETIRED heuristic lane. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p 22

comv_top_partner

ARMIDENTIFIER15%

Co-expression module membership and top co-expressed partner. Pure co-expression " the cross-state class columns are deliberately excluded from this block. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p hsa-let-7f-5p(0.67);hsa-let-7b-5p(0.55);hsa-let-7e-5p(0.43);hsa-let-7b-3p(0.40);hsa-miR-101-3p.1(0.36)

Defined on: arms in a co-expression module (371) " pure co-expression; the state-class columns are excluded

HX_ 1

reference " context, provenance and rarely-quoted detail

1 columns rung arm coverage 28%28% (median 28%)
1x real 0

hx_hly_baseline_imputed

ARMREAL 028%

" IMPUTED healthy baseline (miTED rank-transfer or seed-mate substitution) " LABELLED as imputed, never presented as a measurement. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p 12.32

seed_family_card.tsv

seed_family

One row per the seed family itself, gene-free. 83.7% of families are SINGLETONS “ for those every 'family-level' statistic is trivially that one arm's value. Mask on fam_single_member first.

33 columns 1,959 rows 2 blocks median coverage 92%

Contents “ 2 blocks, most important first

1. seed_ 1 “ CORE
2. fam_ 32 “ REFERENCE

SEED_ “ 1

core “ the columns findings rest on

1 columns “ rung key “ coverage 100%“100% (median 100%)
1x identifier

seed_family

KEY IDENTIFIER 100%

KEY “ the seed family itself, one row per family. AUTHORED
example: miR-4670-3p

FAM_ “ 32

reference “ context, provenance and rarely-quoted detail

32 columns “ rung family “ coverage 4%“100% (median 91%)
12x count “ 8x real “ 0 “ 7x fraction 0“1 “ 2x boolean “ 1x categorical “ 1x identifier “ 1x signed
â`1â€1

fam_ctx_composition

FAMILY CATEGORICAL 29%

The family's genomic-context MIX as a tally string “ sense_coding_host:1 on 251 families, sense_lncRNA_host:1 on 175, antisense_overlap:1 next (29 distinct patterns). AUTHORED
example: miR-34b-5p/449c-5p â†’ sense_lncRNA_host:1,sense_coding_host:1

fam_ctx_context_homogeneous

FAMILY BOOLEAN 29%

Do ALL of this family's members share one genomic context “ True on 524 of 571 (91.8%)? AUTHORED
values: True (524) “ False (47)
example: miR-34b-5p/449c-5p â†’ False

fam_ctx_dose_coding_host_frac

FAMILY FRACTION 0“1
29%

Fraction of the family's DOSE that comes from members with a protein-coding host. AUTHORED
example: miR-34b-5p/449c-5p â†’ 0

fam_ctx_dose_cotranscribed_frac

FAMILY FRACTION 0“1
29%

Fraction of the family's dose from members CO-TRANSCRIBED with their host (sense strand). AUTHORED
example: miR-34b-5p/449c-5p â†’ 0

fam_ctx_n_distinct_coding_hosts FAMILY COUNT 29%	How many DISTINCT coding hosts the family's members sit in. AUTHORED values: 1.0 (288) · 0.0 (264) · 2.0 (17) · 3.0 (2) example: <code>miR-34b-5p/449c-5p</code> · 1
fam_cur_degree_max FAMILY COUNT 36%	The largest curated HE-degree among the family's members. AUTHORED example: <code>miR-3682-3p</code> · 1
fam_dominant_arm FAMILY IDENTIFIER 92%	WHICH member carries the largest share of the family's dose. AUTHORED example: <code>miR-4691-5p/6792-3p</code> · <code>hsa-miR-4691-5p</code>
fam_dominant_share FAMILY FRACTION · 0.5101 92%	The top member's share of the family's total dose. AUTHORED example: <code>miR-4691-5p/6792-3p</code> · 0.5101
fam_dose_hhi FAMILY FRACTION · 0.9346 15%	Concentration of DOSE across the family's members. “FLOOR-CORRECTED” verified 2026-08-19: its minimum sits far below the raw 1/k floor at every member count (0.0000 at k=2 where the raw floor is 0.500), and <code>spearman(n_members, hhi) = +0.1896</code> rather than the strongly negative value a raw index would give. AUTHORED example: <code>miR-378-3p</code> · 0.9346
fam_dose_max_log2 FAMILY REAL · 0.4237 92%	The largest member dose, log2. AUTHORED example: <code>miR-4691-5p/6792-3p</code> · 0.4237
fam_dose_med_log2 FAMILY REAL · 0.4164 92%	The median member dose, log2. AUTHORED example: <code>miR-4691-5p/6792-3p</code> · 0.4164
fam_dose_total_log2 FAMILY REAL · 0.3658 100%	The family's TOTAL dose, log2 · the denominator behind every dose share. AUTHORED example: <code>miR-4670-3p</code> · 0.3658
fam_fame_npmid_sum FAMILY COUNT 100%	Summed publication count over the family's members. AUTHORED example: <code>miR-4670-3p</code> · 13
fam_n_expressed FAMILY COUNT 100%	How many members clear the expression floor. AUTHORED example: <code>miR-4670-3p</code> · 1
fam_n_guide FAMILY COUNT 100%	How many members are classed GUIDE strand by AGO loading (0 · 9). AUTHORED values: 0.0 (1,737) · 1.0 (188) · 2.0 (23) · 3.0 (3) · 6.0 (3) · 5.0 (2) · 4.0 (2) · 9.0 (1) example: <code>miR-4670-3p</code> · 0
fam_n_members FAMILY COUNT 100%	How many arms are in this seed family. AUTHORED example: <code>miR-4670-3p</code> · 1
fam_n_members_context FAMILY COUNT 29%	Arms in the seed family (median 1, max 22). AUTHORED values: 1.0 (486) · 2.0 (61) · 3.0 (14) · 5.0 (4) · 10.0 (2) · 4.0 (2) · 6.0 (1) · 8.0 (1) example: <code>miR-34b-5p/449c-5p</code> · 2
fam_n_model_edges FAMILY COUNT 100%	How many edges the family holds in the LEARNED MODEL's design (0 · 209). AUTHORED example: <code>miR-4670-3p</code> · 0
fam_n_passenger FAMILY COUNT 100%	How many members are classed PASSENGER strand (0 · 3). See <code>fam_n_guide</code>. AUTHORED values: 0.0 (1,784) · 1.0 (167) · 2.0 (5) · 3.0 (3) example: <code>miR-4670-3p</code> · 0 Defined on: seed families in the arm-card universe (1,959; · 83.7% single-member · shares/HHI are 1 or NaN)

fam_n_shift_gt20_gated FAMILY COUNT 100%	How many members show an isomiR seed shift above 20%, among those with enough coverage to judge. Range 0 â€“ 2, 100% filled â€” so this is a rare event, not a distribution. AUTHORED values: 0.0 (1,911) Â· 1.0 (45) Â· 2.0 (3) example: miR-4670-3p â†’ 0 Defined on: seed families in the arm-card universe (1,959; â’” 83.7% single-member â†’ shares/HHI are 1 or NaN)
fam_real_n_c10_sum FAMILY COUNT 100%	Summed count of realized couplings below â^’0.10 over the family's members. AUTHORED example: miR-4670-3p â†’ 0
fam_real_n_scored_sum FAMILY COUNT 100%	Summed count of scored gene-regulator pairs over the family's members â€” the DENOMINATOR for <code>fam_real_n_c10_sum</code>. Range 0 â€“ 169, 100% filled. AUTHORED example: miR-4670-3p â†’ 0 Defined on: seed families in the arm-card universe (1,959; â’” 83.7% single-member â†’ shares/HHI are 1 or NaN)
fam_redund_m_eff FAMILY REAL Â%¥ 0 4%	EFFECTIVE number of independent members after accounting for within-family correlation â€” the family's redundancy-corrected size (mode 2.0, 50 distinct values). AUTHORED example: miR-25-3p/32-5p/92-3p/363-3p/367-3p â†’ 4.82
fam_redund_median_within_rho FAMILY SIGNED Â^’1Â€{1 4%	Median pairwise correlation BETWEEN the family's members (â^’0.236 â€“ +1.00). AUTHORED example: miR-25-3p/32-5p/92-3p/363-3p/367-3p â†’ 0.178
fam_redund_overcount_frac FAMILY FRACTION 0Â€”1 4%	How much the family's naive member count OVERSTATES its independent size, once within-family correlation is accounted for (0 â€“ 0.374). AUTHORED example: miR-25-3p/32-5p/92-3p/363-3p/367-3p â†’ 0.035
fam_seed_shift_max_gated FAMILY FRACTION 0Â€”1 13%	The same, gated to well-covered members. AUTHORED example: miR-27-3p â†’ 0.023
fam_single_member FAMILY BOOLEAN 100%	â€” THE DOMINANT FACT ABOUT THIS ENTIRE CARD: 1,639 of 1,959 families (83.7%) are SINGLETONS. For those rows every 'family-level' statistic is TRIVIALY that one arm's value, and <code>fam_dominant_share</code> is forced to exactly 1.0. AUTHORED values: True (1,639) Â· False (320) example: miR-4670-3p â†’ True
fam_targetome_seed_med FAMILY REAL Â%¥ 0 97%	Median MANE 3'UTR seed-scan targetome over the members. AUTHORED example: miR-4688/6743-5p â†’ 1888
fam_targetome_seq_med FAMILY REAL Â%¥ 0 30%	Median scanMiR K_D targetome over the members. AUTHORED example: miR-3615 â†’ 2353
fam_var_hhi FAMILY FRACTION 0Â€”1 14%	Herfindahl concentration of <code>abund_sd</code> ACROSS the family's members â€” is the family's variance carried by one arm? AUTHORED example: miR-376b-5p/376c-5p â†’ 6.327e-05
fam_var_max FAMILY REAL Â%¥ 0 89%	LARGEST member dispersion in the family. AUTHORED example: miR-4687-3p â†’ 0.2317
fam_var_med FAMILY REAL Â%¥ 0 89%	MEDIAN dispersion (<code>abund_sd</code>) over the family's members. AUTHORED example: miR-4687-3p â†’ 0.2317